1 Groups

We start by recalling the definition of a group and reviewing some examples that you saw last year.

Definition 1.1 (Group).
A group is a pair $(G, .)$ where G is a non-empty set and . is a composition law

$$(.) : G \times G \to G$$

$$(a, b) \mapsto a.b$$

which satisfies the following conditions:

1. $(.)$ is associative: $(a.b).c = a.(b.c)$.
2. $(.)$ admits a neutral element e_G (simply denoted e if there is no risk of confusion) satisfying:

$$a.e_G = e_G.a = a, \ \forall a \in G.$$

3. existence of an inverse: $\forall a \in G, \exists a^{-1} \in G$:

$$a.a^{-1} = a^{-1}.a = e_G,$$

a^{-1} is called the inverse of a.

If the composition law $(.)$ further satisfies $a.b = b.a$ for all $a, b \in G$, we say G is *abelian* or commutative. We say G is *finite* if it has finitely many elements. The number of elements of G is denoted $|G|$ and called the *cardinality* or *order*[1] of G. If G is not finite we say it is *infinite*.

Example 1.1.

1. $(\mathbb{Z}, +)$ the set of integers endowed with the addition law is an abelian, infinite, group. The neutral element is 0 and the inverse of an integer $n \in \mathbb{Z}$ is $-n$.
2. Given $n \geq 1$, recall the integers modulo n:

$$\mathbf{Z}_n = \{[0]_n, [1]_n, \cdots, [n-1]_n\}$$

where $[k]_n$ is the class of the integer k modulo n, $[k]_n = [l]_n$ if and only if $k - l$ is divisible by n.
The addition of integers modulo n is defined by $[i]_n + [j]_n = [i+j]_n$. The set of integers modulo n endowed with the addition law $(\mathbf{Z}_n, +)$ is a finite abelian group of cardinality $|\mathbf{Z}_n| = n$. The neutral element is $[0]_n$, and the inverse of $[k]_n \in \mathbf{Z}_n$ is $[-k]_n$.
3. Given $n \geq 1$, $(S_n, \circ)$ is the group of permutations of the set $A = \{1, \ldots, n\}$: a permutation of A is a bijection $A \to A$. An element $\sigma \in S_n$ is a bijective map $\sigma : A \to A$.
The composition law $\circ$ is the composition of maps: if $\sigma, \tau \in S_n$ then

$$\sigma \circ \tau : A \to A$$

is defined by

$$(\sigma \circ \tau)(a) = \sigma(\tau(a)) \qquad \forall a \in A.$$

[1] Do not confuse this with the order of an element of a group

The neutral element $e = e_{S_n}$ is the identity map $\mathrm{id}_A : A \to A$ (it is a bijection).
S_n is a finite group with $|S_n| = n!$.
Assume $n \geq 2$, and $2 \leq k \leq n$ is an integer. Recall $\sigma = (a_1 a_2 \dots a_k)$ is a *cycle* if $\{a_1, a_2, \dots, a_k\} \subset A$ are k distinct elements of A, $\sigma(a_i) = a_{i+1}$ if $1 \leq i \leq k-1$, $\sigma(a_k) = a_1$, and $\sigma(j) = j$ if $j \notin \{a_1, a_2, \dots, a_k\}$. The integer k is called the *length* of σ.
The number of cycles of length $2 \leq k \leq n$ in S_n equals $\frac{n(n-1)\cdots(n-k+1)}{k}$ (prove this fact).
S_n is not abelian if $n \geq 3$: $(123)(13) \neq (13)(123)$.

4. For $n \geq 3$, $(D_{2n}, .)$ is the group of rotational and reflection symmetries of a regular n-gon Γ_n whose vertices we label by $1, 2, \dots, n$. Let r be the rotation clockwise in the plane of Γ_n with angle $\frac{2\pi}{n}$, and s the rotation through the symmetry axis of Γ_n passing through the vertex 1. Then
$$D_{2n} = \{1, r, \dots, r^{n-1}, s, sr, \cdots, sr^{n-1}\}$$
has $2n$ elements, 1 is the identity rotation, $r^n = s^2 = 1$, $(r^i)^{-1} = r^{-i} = r^{n-i}$, and $sr^i = r^{-i}s = r^{n-i}s$. The group D_{2n} is not abelian if $n \geq 3$: $sr \neq rs = sr^{n-1}$ in this case since $r \neq r^{n-1}$.
Here $r^k = r \circ \dots \circ r$ means r composed with itself k times for $k \geq 1$, it is the rotation clockwise in the plane of Γ_n with angle $\frac{2\pi k}{n}$.
For $n = 2$ we define D_4 as follows
$$D_4 = \{1, s, r, sr\}$$
with 1 the neutral element, $s^2 = r^2 = 1$, and $sr = rs$. So D_4 is abelian. Note that $(sr)^2 = srsr = srrs = sr^2s = ss = s^2 = 1$ in this case.

Remark (Basic Properties).
Let G be a group with neutral element e, and $a, b \in G$.

1. The neutral element e is unique.
2. The inverse a^{-1} of an element a is unique.
3. $(a^{-1})^{-1} = a$.
4. $(ab)^{-1} = b^{-1}a^{-1}$.
5. Cancellation laws: $\{au = av\} \Longrightarrow \{u = v\}$ and $\{ub = vb\} \Longrightarrow \{u = v\}$.
6. For $n \geq 1$, $a^n = a.a \dots a$ is the composite of a with itself n times.
If $n < 0$ we define $a^n = (a^{-n})^{-1}$.
Exponentiation rules: we have $a^n.a^m = a^{n+m}$, and $(a^n)^m = a^{nm}$, for $n, m \in \mathbb{Z}$, and $a \in A$.
7. The associativity of the group law (.) implies that for $a_1, \dots, a_k \in G$ the product $a_1.a_2.\dots.a_k$ is well defined and is independent of the way the expression is bracketed, however the order of the a_i's in the composition process is important unless G is abelian.

Definition 1.2 (Subgroup).
Let $(G, .)$ be a group. A *subgroup* of G is a non-empty subset $H \subset G$ which is closed under the composition law (.) and taking inverses:

- $x, y \in H \quad \Longrightarrow \quad x.y \in H$,
- $x \in H \quad \Longrightarrow \quad x^{-1} \in H$.
In particular if $x \in H$ then $x.x^{-1} = e_G \in H$.
In other words, H is a subgroup of G if H is closed under the composition law (.), and endowed with the latter $(H, .)$ is a group on its own.

Example 1.2.
Check the following are subgroups.

1. Given a group G the subset $\{e_G\}$ is a subgroup called the trivial subgroup.
2. Given $m \in \mathbb{Z}$ the subset $m\mathbb{Z} = \{mk \ : \ k \in \mathbb{Z}\}$ of integers which are multiples of m is a subgroup of $(\mathbb{Z}, +)$, it is infinite if $m \neq 0$.
3. Given $n \geq 1$ the subset $H = \{1, r, \ldots, r^{n-1}\}$ of D_{2n} consisting of all powers of the rotation r is a subgroup of D_{2n} of cardinality n.
4. Given a group G and an element a of G the subset

$$H = \{a^n : n \in \mathbb{Z}\}$$

of G consisting of all integer powers of a is a subgroup, called the cyclic subgroup of G generated by a [note that $a^0 = aa^{-1} = e_G$]. The subgroup H is usually denoted $\langle a \rangle$, and a is called a generator of H.
5. The subset $\{e, (12)(34), (13)(24), (14)(23)\}$ is a subgroup of S_4.

Definition 1.3 (Order of an element).
Let G be a group and $a \in G$. Assume $\exists t \geq 1$ with $a^t = e_G$. We define the *order* of a by

$$\text{ord}(a) = \min \ \{n \geq 1 \ : \ a^n = e_G\}.$$

If $a^n \neq e_G, \forall n \geq 1$, we define $\text{ord}(a) = \infty$ and say a has infinite order.

Example 1.3 (Properties).

1. $\text{ord}(a) = 1$ if and only if $a = e_G$.
2. Let $0 \neq n \in \mathbb{Z}$. Then $\text{ord}(n) = \infty$, for otherwise $\exists t \geq 1$ with $tn = 0$ which is not possible.
3. If G is a group and $a \in G$ then $\text{ord}(a) = \infty$ is equivalent to saying that the elements $\{a^t : t \geq 1\} = \{a, a^2, \ldots, a^i, a^{i+1}, \ldots\}$ of G are all pairwise distinct: for otherwise if $i > j$ and $a^i = a^j$, then $a^{i-j} = e_G$ with $i - j \geq 1$, and hence a would have finite order. In particular, if G is a finite group, every element $a \in G$ has finite order. For otherwise G would contain the infinite set $\{a, a^2, \ldots, a^i, a^{i+1}, \ldots\}$, which is not possible since G is finite.
4. Let a be an element of a group G with finite order $\text{ord}(a) = k$. Then

$$H = \langle a \rangle = \{a^n : n \in \mathbb{Z}\} = \{e_G, a, a^2, \ldots, a^{k-1}\}$$

and $|H| = k$.

Proof. Let $n = tk + r$ with $0 \leq r < k$ (Euclidean division). Then $a^n = a^{tk+r} = a^{tk}.a^r = (a^k)^t a^r = (e_G)^t a^r = e_G a^r = a^r$. Also if $1 \leq j < i \leq k - 1$ and $a^i = a^j$ then $a^{i-j} = e_G$, hence $\text{ord}(a) \leq i - j < k$ which is a contradiction. Thus, the elements $\{e_G, a, a^2, \ldots, a^{k-1}\}$ are all pairwise distinct. □

5. Let $n \geq k \geq 2$ and $\sigma = (a_1 \cdots a_k) \in S_n$ a cycle of length k. Then $\text{ord}(\sigma) = k$. Further if $\sigma \in S_n$ then one can write $\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_m$: a product of disjoint cycles and $\text{ord}(\sigma) = \text{lcm}(\text{ord}(\tau_i))_{i=1}^m$ (will discuss examples in lectures).
6. Let $n \geq 1$ and $[k]_n \in \mathbf{Z}_n$. Then $\text{ord}([k]_n) = \frac{n}{\gcd(n,k)}$.

Proof. Let $d = \gcd(k, n)$. Write $dk' = k$ and $dn' = n$ with $\gcd(n', k') = 1$. We show $n' = \text{ord}([k]_n)$. First $n'[k]_n = [kn']_n = [dn'k']_n = [nk']_n = [0]_n$ so that $\beta = \text{ord}([k]_n) \leq n'$. Assume $\beta < n'$ then $[0] = \beta[k]_n = [\beta k]_n$ hence $n = dn'$ divides $\beta k = \beta dk'$, which implies n' divides $\beta k'$, the latter implies that n' divides β since $\gcd(n', k') = 1$ which contradicts $\beta < n'$. □

Definition 1.4 (Cyclic Groups).
A group G is called *cyclic* if $\exists a \in G$ such that

$$G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$$

equals the (sub)group generated by a. The element a is called a *generator* of G.

A generator of a cyclic group need not be unique, if a generates G then so does a^{-1} since $a^n = (a^{-n})^{-1} = (a^{-1})^{-n}$ for $n \in \mathbb{Z}$.

If $G = \langle a \rangle$ is cyclic, and $\text{ord}(a)$ is infinite, then

$$G = \{e, a, a^{-1}, a^2, a^{-2}, \ldots, a^n, a^{-n}, \ldots\}$$

is infinite, and $a^i \neq a^j$ if $i < j$ are integers.

If $G = \langle a \rangle$ is cyclic, and $\text{ord}(a) = k$ is finite, then $|G| = k$ and

$$G = \{e, a, \ldots, a^{k-1}\}.$$

Example 1.4.

1. $\mathbb{Z} = \langle 1 \rangle$ is an infinite cyclic group generated by 1. Indeed $\forall n \in \mathbb{Z}, n = n.1$. Also $\mathbb{Z} = \langle -1 \rangle$ since $n = (-n)(-1)$.
2. $\mathbf{Z}_n = \langle [1]_n \rangle$ is a cyclic group generated by $[1]_n$. Indeed $[h]_n = h[1]_n$ for all integers h. Further if $k \geq 1$ is an integer with $\gcd(k, n) = 1$ then $\mathbf{Z}_n = \langle [k]_n \rangle$ and $[k]_n$ is also a generator of $\mathbf{Z}_n$. Indeed $\text{ord}([k]_n) = \frac{n}{\gcd(k,n)} = n$ so that the subgroup $H = \langle [k]_n \rangle$ of $\mathbf{Z}_n$ generated by $[k]_n$ must equal $\mathbf{Z}_n$ as $H \subset \mathbf{Z}_n$ and they have the same cardinality n. For example $\mathbf{Z}_8 = \langle [3]_8 \rangle = \langle [5]_8 \rangle$.
3. If $x \in G$ has order n, then $\text{ord}(x^k) = \frac{n}{\gcd(n,k)}$ for each $k \geq 1$ (same proof as in example 2 above).
4. $H = \langle (1234) \rangle = \{e, (1234), (13)(42), (1432)\}$ is a cyclic subgroup of S_4 of cardinality 4. The subgroup $\{e, (12)(34), (13)(24), (14)(23)\}$ is not a cyclic subgroup of S_4 as it has no elements of order 4.
5. $\langle r \rangle = \{1, r, \ldots, r^{n-1}\}$ is a cyclic subgroup of cardinality n of D_{2n}. The subgroup $\{1, s, sr^2, r^2\}$ is a subgroup of D_8 which is not cyclic as it has no elements of order 4: $\text{ord}(r^2) = \text{ord}(s) = \text{ord}(sr^2) = 2$.

Definition 1.5 (Product of Groups).
Let $(G, \circ)$ and $(H, \star)$ be two groups. We define a new group $(G \times H, .)$, called the *product* of G and H, as follows

$$G \times H = \{(g, h) \ : \ g \in G, \ h \in H\}$$

is the set-theoretic product of G and H. The composition law $(.)$ is defined by

$$(g_1, h_1).(g_2, h_2) = (g_1 \circ g_2, h_1 \star h_2),$$

it is a group law on $G \times H$: associativity can be easily checked using associativity of $\circ$ and $\star$ (check this fact). The neutral element is (e_G, e_H):

$$(g,h).(e_G,e_H) = (g \circ e_G, h \star e_H) = (g,h),$$

and the inverse $(g,h)^{-1}$ of (g,h) is (g^{-1},h^{-1}):

$$(g,h).(g^{-1},h^{-1}) = (g \circ g^{-1}, h \star h^{-1}) = (e_G,e_H).$$

Lemma 1.1.
Let $(G,\circ)$ and $(H,\star)$ be groups. If G and H are abelian then so is $G \times H$. If both G and H are finite then so is $G \times H$ and $|G \times H| = |G||H|$.

Proof. Assume G, H are abelian, let $g_1, g_2 \in G$, and $h_1, h_2 \in H$. Then $(g_1,h_1).(g_2,h_2) = (g_1 \circ g_2, h_1 \star h_2) = (g_2 \circ g_1, h_2 \star h_1) = (g_2,h_2).(g_1,h_1)$, where we used the commutativity of $\circ$ and $\star$ for the second equality. If both G and H are finite then the number of elements of $G \times H$ is the number of pairs (g,h) with $g \in G$ and $h \in H$. There are $|G|$ possibilities for g, $|H|$ possibilities for h, which gives $|G||H|$ for the number of such pairs. □

Example 1.5.

1. $\mathbb{Z} \times \mathbb{Z} = \{(m,n) : m,n \in \mathbb{Z}\}$ is a group, we denote the group law by $+$ (and call it addition). Thus
$$(m,n) + (m',n') = (m+m', n+n'),$$
$(0,0)$ is the neutral element, and the inverse of (m,n) is $(-m,-n)$.
2. Given positive integers n and m: $\mathbf{Z}_n \times \mathbf{Z}_m = \{([k]_n,[r]_m) : k,r \in \mathbb{Z}\}$ is a group, we denote the group law by $+$. Thus
$$([k]_n,[r]_m) + ([k']_n,[r']_m) = ([k+k']_n,[r+r']_m),$$
$([0]_n,[0]_m)$ is the neutral element, and the inverse of $([k]_n,[r]_m)$ is $([-k]_n,[-r]_m)$. Further $\mathbf{Z}_n \times \mathbf{Z}_m$ is a finite group of cardinality mn.
3. $\mathbf{Z}_n \times \mathbf{Z} = \{([k]_n, r) : k,r \in \mathbb{Z}\}$ is a group, we denote the group law by $+$. Thus
$$([k]_n,r) + ([k']_n,r') = ([k+k']_n, r+r'),$$
$([0]_n, 0)$ is the neutral element, and the inverse of $([k]_n,r)$ is $([-k]_n,-r)$. Further $\mathbf{Z}_n \times \mathbf{Z}$ is an infinite group.

Definition 1.6 (Group Homomorphism).
Let $(G,\circ)$ and $(H,\star)$ be groups. A map $f : G \to H$ is called a *homomorphism* if

$$f(g_1 \circ g_2) = f(g_1) \star f(g_2), \quad \forall g_1, g_2 \in G.$$

In particular $f(e_G) = e_H$ (check this fact).

If in addition f is a bijection it is called an *isomorphism*.

Example 1.6.
Study examples 3,5 and 6 below.

1. The map $f : \mathbb{Z} \to \mathbf{Z}_n$ defined by $f(k) = [k]_n$ is a group homomorphism since

$$f(r+s) = [r+s]_n = [r]_n + [s]_n = f(r) + f(s).$$

It is surjective as $[k]_n = f(k)$ but is not injective: $f(k) = f(k+n) = [k]_n$.

2. Let $m \geq 0$ be an integer. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(k) = mk$ is a group homomorphism:

$$\phi(k_1 + k_2) = m(k_1 + k_2) = mk_1 + mk_2 = \phi(k_1) + \phi(k_2).$$

This map is injective if $m \neq 0$: $\phi(k_1) = mk_1 = \phi(k_2) = mk_2 \implies m(k_1 - k_2) = 0 \implies k_1 - k_2 = 0 \implies k_1 = k_2$. The map ϕ is not surjective if $m \neq 1$: if m doesn't divide r then $r \neq \phi(k), \forall k \in \mathbb{Z}$.

3. Let $n > 1$ an integer. Recall that a permutation $\sigma \in S_n$ is *even* (resp. *odd*) if σ can be written as the product of an even (resp. odd) number of transpositions (ij). The map

$$\epsilon : S_n \to \mathbf{Z}_2$$

defined by $\epsilon(\sigma) = [0]_2$ if σ is even, and $\epsilon(\sigma) = [1]_2$ if σ is odd, is a group homomorphism (check this fact). It is surjective but is not injective if $n \geq 3$.

4. The map

$$f : D_{2n} \to \mathbf{Z}_2$$

defined by

$$f(s^i r^j) = [i]_2$$

is a group homomorphism. Suppose k is an odd integer then

$$f(s^i r^j . s^k r^l) = f(s^i r^j s r^l) = f(s^i s r^{n-j} r^l) = f(s^{i+1} r^{n-j+l}) = [i+1]_2 = [i+k]_2$$

$$[i]_2 + [k]_2 = f(s^i r^j) f(s^k r^l).$$

if k is even then

$$h(s^i r^j . s^k r^l) = h(s^i r^j r^l) = h(s^i r^{j+l}) = [i]_2 = [i]_2 + [0]_2 = h(s^i r^j) h(s^k r^l).$$

It is surjective: $f(r) = [0]_2$ and $f(s) = [1]_2$, but f is not injective: $f(s) = f(sr)$.

5. The map

$$f : D_4 \to \mathbf{Z}_2 \times \mathbf{Z}_2$$

defined by $f(1) = ([0]_2, [0]_2)$, $f(r) = ([1]_2, [0]_2)$, $f(s) = ([0]_2, [1]_2)$ and $f(sr) = ([1]_2, [1]_2)$ is an isomorphism (check that f is a homomorphism). The homomorphism f is bijective as it is injective and both groups have the same cardinality which equals 4, thus f is an isomorphism.

6. The map

$$h : D_{2n} \to \mathbf{Z}_n$$

defined by

$$h(s^i r^j) = [j]_n$$

is NOT a group homomorphism: if k is odd then

$$\begin{aligned} h(s^i r^j . s^k r^l) &= h(s^i s^k r^{n-j} r^l) = h(s^{i+k} r^{n-j+l}) \\ &= [n - j + l]_n = [-j + l]_n, \end{aligned}$$

which in general is different from $h(s^i r^j) h(s^k r^l) = [j]_n + [l]_n = [j + l]_n$.

Remark (Basic properties of homomorphisms).
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism.

1. Let $n \in \mathbb{Z}$ and $g \in G$ then $f(g^n) = f(g)^n$.
2. Suppose f is an isomorphism. Then the followings hold.
   (i) G is finite $\iff$ H is finite.
   (ii) G is abelian $\iff$ H is abelian.
   (iii) $\mathrm{ord}(g) = \mathrm{ord}(f(g)), \forall g \in G$.

Proof. See Problem Sheet 1, Q.6. □

Definition 1.7 (Image and Kernel of a Homomorphism).
Let

$$f : (G, \circ) \to (H, \star)$$

be a group homomorphism. We define the *image* of f to be

$$\mathrm{Im}(f) = \{h \in H \ : \ \exists g \in G, \ h = f(g)\}.$$

We define the *kernel* of f to be

$$\mathrm{Ker}(f) = \{g \in G \ : \ f(g) = e_H\}.$$

Remark (Basic properties of Image and Kernel).
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism. Then $\mathrm{Im}(f)$ is a subgroup of H and $\mathrm{Ker}(f)$ is a subgroup of G.

Proof. The fact that $\mathrm{Im}(f)$ is a subgroup of H has been proven in Year 1. We show $\mathrm{Ker}(f)$ is a subgroup of G. First, let $g \in \mathrm{Ker}(f)$, i.e. $f(g) = e_H$. Then $e_H = f(e_G) = f(g \circ g^{-1}) = f(g) \star f(g^{-1}) = e_H \star f(g^{-1}) = f(g^{-1})$ so that $g^{-1} \in \mathrm{Ker}(f)$. Second, let $g_1, g_2 \in \mathrm{Ker}(f)$. Then $f(g_1 \circ g_2) = f(g_1) \star f(g_2) = e_H \star e_H = e_H$ and $g_1 \circ g_2 \in \mathrm{Ker}(f)$. □

Example 1.7.

1. The group homomorphism $f : \mathbb{Z} \to \mathbf{Z}_n$ in Example 1.6 has image $\mathrm{Im}(f) = \mathbf{Z}_n$ and kernel $\mathrm{Ker}(f) = n\mathbb{Z}$.
2. The group homomorphism $g : \mathbb{Z} \to \mathbb{Z}$ in Example 1.6 has image $\mathrm{Im}(f) = m\mathbb{Z}$ and kernel $\mathrm{Ker}(f) = \{0\}$ if $m \neq 0$.
3. The group homomorphism $\epsilon : S_n \to \mathbf{Z}_2$ in Example 1.6, for $n > 1$, has image $\mathrm{Im}(f) = \mathbf{Z}_2$ and kernel $\mathrm{Ker}(f) = A_n$: the alternating subgroup of S_n which consists of all even permutations.
4. The group homomorphism $\tau : D_{2n} \to \mathbf{Z}_2$ in Example 1.6 has image $\mathrm{Im}(f) = \mathbf{Z}_2$ and kernel $\mathrm{Ker}(f) = \langle r \rangle = \{1, r, \cdots, r^{n-1}\}$.

Proposition 1.2.
Let $f : (G, \circ) \to (H, \star)$ be a group homomorphism. Then

1. *f is surjective if and only if $Im(f) = H$.*
2. *f is injective if and only if* $\mathrm{Ker}(f) = \{e_G\}$.

Proof. Exercise (compare with the proof of Lemma 6.1 in §6). □

2 Left Cosets and Quotient Groups

In this section whenever we consider a group $(G, \circ)$ we will omit the notation $\circ$, i.e. we will simply write $xy = x \circ y$ if $x, y \in G$.

Definition 2.1 (Subgroup Relation).
Let G be a group and $H \subset G$ a subgroup. We define the relation R among elements of G by $xRy \iff x^{-1}y \in H$.

From the definition, it follows that xRy if and only if there exists some $h \in H$ such that $y = xh$.

Definition.
A relation R on a set S is an *equivalence relation* if:

1. R is *reflexive*: for all $x \in S$, we have xRx.
2. R is *symmetric*: for all $x, y \in S$, if xRy, then yRx.
3. R is *transitive*: for all $x, y, z \in S$, if xRy and yRz, then xRz.

Given an equivalence relation R on a set S and an element x of S, the *equivalence class* of x, denoted $\bar{x}$ is the set $\bar{x} = \{y \in S : xRy\}$.

Lemma 2.1.
The relation R is an equivalence relation. If $x \in G$ then its equivalence class, denoted $\bar{x}$, for the relation R is

$$\bar{x} = xH = \{xh : h \in H\}.$$

Proof. R is reflexive: xRx since $x^{-1}x = e_G \in H$. R is symmetric: assume xRy, i.e. $x^{-1}y \in H$, then $(x^{-1}y)^{-1} = y^{-1}(x^{-1})^{-1} = y^{-1}x \in H$ (since H is a subgroup) which means yRx. R is transitive: assume xRy, i.e. $x^{-1}y \in H$, and yRz, i.e. $y^{-1}z \in H$, then $(x^{-1}y)(y^{-1}z) = x^{-1}yy^{-1}z = x^{-1}z \in H$ (since H is a subgroup) which means xRz.

The equivalence class of $x \in G$ is

$$\bar{x} = \{y \in G : xRy\} = \{y \in G : y = xh, h \in H\} = xH = \{xh : h \in H\}.$$

□

Definition 2.2 (Left Cosets).
The equivalence class $xH = \{xh : h \in H\}$ is called a *left coset* of H; it is a subset of G.

We denote the set of all left cosets of H by

$$(G/H)_{\text{left}} = \{xH : x \in G\}.$$

We usually omit the word left and write G/H instead of $(G/H)_{\text{left}}$.

The cardinality of G/H is denoted $|G : H|$ and is called the *index* of H in G.

As R is an equivalence relation, we have

$$G = \bigcup_{x \in G} \bar{x} = \bigcup_{x \in G} xH$$

where the union is disjoint, i.e. given $x, y \in G$ then either $xH = yH$; which happens if and only if xRy, or $xH \bigcap yH = \emptyset$. The proof of this claim is essentially that given in MTH1001. It generalises to all equivalence relations.

Note that

$$\overline{e_G} = e_G H = \{e_G h : h \in H\} = H \in G/H.$$

If G is finite then G/H is a finite set, its cardinality is denoted

$$|G : H| = |G/H|$$

and is called the *index* of H in G. Clearly $|G : H| \leq |G|$.

Example 2.1.

1. Let $G = (\mathbf{Z}, +)$ and $H = n\mathbf{Z}$ $(n \geq 0)$. In this case

   $$aRb \iff -a + b = b - a \in H = n\mathbf{Z},$$

   i.e. a is congruent to b modulo n. Let $a \in \mathbf{Z}$ with $a = kn + r$, $0 \leq r < n$ (Euclidean division). Then aRr since $-a + r = (-k)n \in n\mathbf{Z}$, and $\bar{a} = \bar{r} = r + n\mathbf{Z}$. Thus

   $$\mathbf{Z}/n\mathbf{Z} = \{\bar{0}, \bar{1}, \cdots, \overline{n-1}\}$$

   is just the set $\mathbf{Z}_n$ of integers modulo n.
2. Let $G = \mathbf{Z}_4 = \{[0]_4, [1]_4, [2]_4, [3]_4\}$ and $H = \langle [2]_4 \rangle = \{[0]_4, [2]_4\}$. In this case

   $$\overline{[0]_4} = \overline{[2]_4} = [0]_4 + H = H$$

   since $[2]_4 - [0]_4 = [2]_4 \in H$, and

   $$\overline{[1]_4} = \overline{[3]_4} = [1]_4 + H = \{[1]_4, [3]_4\}$$

   since $[3]_4 - [1]_4 = [2]_4 \in H$. Here $|G : H| = 2$.
3. Let $D_6 = \{1, r, r^2, s, sr, sr^2\}$ and $H = \langle r \rangle = \{1, r, r^2\}$. Then

   $$\bar{1} = \bar{r} = \overline{r^2} = 1H = H$$

   and

   $$\bar{s} = \overline{sr} = \overline{sr^2} = sH = \{s, sr, sr^2\}$$

   since $s^{-1}(sr) = s(sr) = s^2 r = r \in H$ and $s^{-1}(sr^2) = s(sr^2) = s^2 r^2 = r^2 \in H$. Here $|G : H| = 2$.

Theorem 2.2 (Lagrange's Theorem)**.**
Let G be a finite group and H a subgroup. Then

$$|G| = |H||G/H| = |H||G : H|.$$

Proof. First we show that all left cosets have the same cardinality $|H| = |xH|$. Let $x \in G$ fixed. The map $\sigma : H \to xH$ defined by $\sigma(h) = xh$ is a bijection. Indeed, it is clearly surjective. Assume $\sigma(h_1) = xh_1 = \sigma(h_2) = xh_2$ then $x^{-1}xh_1 = h_1 = x^{-1}xh_2 = h_2$. Since σ is bijective, H and xH have the same cardinality. We know that G is the disjoint union of the left cosets of H and all these cosets have the same cardinality: equal that of H. Thus $|G|$ equals the number of left cosets; i.e. $|G : H|$, times the cardinality of a coset which equals $|H|$. □

Corollary 2.3.

1. *Let G be a finite group and H a subgroup. Then $|H|$ divides $|G|$.*
2. *Let G be a finite group and $x \in G$ then $\mathrm{ord}(x) = |\langle x \rangle|$ divides $|G|$.*

Theorem 2.4 (Cauchy's Theorem).
Let G be a finite group and p a prime number dividing $|G|$. Then $\exists g \in G$ with $\mathrm{ord}(g) = p$, and there exists a subgroup H of G with $|H| = p$.

Definition 2.3 (Normal Subgroup).
A subgroup H of G is called *normal* if

$$xH = \{xh \ : \ h \in H\} = Hx = \{h'x \ : \ h' \in H\}, \quad \forall x \in G.$$

This condition is satisfied if G is abelian since the elements of G commute: $xh = hx$. Thus every subgroup of an abelian group is a normal subgroup.

Example 2.2.

1. Let $D_6 = \{1, r, r^2, s, sr, sr^2\}$ and $H = \langle r \rangle = \{1, r, r^2\}$. Then H has 2 left cosets $\bar{1} = \bar{r} = \overline{r^2} = 1H = H$ and $\bar{s} = \overline{sr} = \overline{sr^2} = sH = \{s, sr, sr^2\}$ (see Examples 2.1). Further H is normal. Indeed $rH = r^2H = Hr = Hr^2 = H$ and $sH = srH = sr^2H = Hs = Hsr = Hsr^2$, since $sr^j = r^{-j}s$ hence
$$sr^k r^j = sr^{k+j} = sr^{j+k} = sr^j r^k = r^{-j} sr^k$$
for $0 \leq j < 3$, and $0 \leq k < 3$.
2. The subgroup $K = \{1, s\} = \langle s \rangle$ of $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ is NOT normal. Indeed $rK = \{r, rs = sr^3\} \neq Kr = \{r, sr\}$ as $sr^3 \neq sr$ (since $r^3 \neq r$).
3. If $f : G \to G'$ is a group homomorphism then $\mathrm{Ker}(f)$ is a normal subgroup of G.

   Proof. Let $x \in G$, we show $x\,\mathrm{Ker}(f) = \mathrm{Ker}(f)x$. First we show $x\,\mathrm{Ker}(f) \subset \mathrm{Ker}(f)x$. Let $h \in \mathrm{Ker}(f)$. We show $xh = h'x$, which is equivalent to $xhx^{-1} = h'$, for some $h' \in \mathrm{Ker}(f)$. We have $f(xhx^{-1}) = f(x)f(h)f(x^{-1}) = f(x)e_{G'}f(x)^{-1} = f(x)f(x)^{-1} = e_{G'}$. Hence $xhx^{-1} = h' \in \mathrm{Ker}(f) = H$ which implies $xh = xhx^{-1}x = h'x$. Similarly one can show $\mathrm{Ker}(f)x \subset x\,\mathrm{Ker}(f)$. □
4. (Prove the following as an exercise) A subgroup H of G is normal if and only if
$$xHx^{-1} = \{xhx^{-1} : h \in H\} = H, \ \forall x \in G.$$
   Indeed this follows easily from the fact that
$$xh = h'x \iff xhx^{-1} = h'.$$

Definition 2.4 (Conjugate Elements).
Two elements g, g' of a group G are called *conjugate* if $\exists x \in G$ with

$$g' = xgx^{-1}.$$

The conjugation relation among elements of G is an equivalence relation (check this fact).

- A subgroup H of G is normal if and only if whenever H contains an element h it contains all its conjugates $\{xhx^{-1} : x \in G\}$ (prove this as an exercise).
- If G is abelian: $xhx^{-1} = xx^{-1}h = h$ and each element equals its own conjugates.

Conjugate elements in S_n

Let $\sigma \in S_n$ and write $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k})$ as a product of k disjoint cycles of lengths $n_1 \leq n_2 \leq \dots \leq n_k$. The k-tuple $(n_1, \dots, n_k)$ is called the *cycle type* of σ.

Example 2.3.
$(12)(345)(6789)$ is of type $(2, 3, 4)$ and $(12)(34)$ is of type $(2, 2)$.

Proposition 2.5.
Two permutations in S_n are conjugates if and only if they have the same cycle type.

Proof. Let $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k}) \in S_n$ be a permutation of type $(n_1, \dots, n_k)$. Let $\tau \in S_n$. We claim

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1) \dots \tau(a_{n_1})) \dots (\tau(b_1) \dots \tau(b_{n_k}))$$

is the decomposition of $\tau \circ \sigma \circ \tau^{-1}$ into the product of disjoint cycles, hence it also has cycle type $(n_1, \dots, n_k)$.

We first demonstrate the equation. For this, we need to show that if $\sigma(i) = j$ then

$$\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(j). \qquad (*)$$

But $\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(\sigma(\tau^{-1}(\tau(i)))) = \tau(\sigma(i)) = \tau(j)$.

Moreover, the cycles $(\tau(a_1) \dots \tau(a_{n_1}))$ $\dots$ $(\tau(b_1) \dots \tau(b_{n_k}))$ are disjoint since the above cycles $(a_1 \dots a_{n_1})$, $\dots$, $(b_1 \dots b_{n_k})$ are disjoint and τ is a bijection.

Conversely let $\sigma = (a_1 \dots a_{n_1}) \dots (b_1 \dots b_{n_k})$ and $\sigma' = (a'_1 \dots a'_{n_1}) \dots (b'_1 \dots b'_{n_k})$ be two permutations with the same cycle type. Let $\alpha_1, \alpha_2, \dots, \alpha_t$ and $\beta_1, \beta_2 \dots, \beta_t$ the the elements of $\{1, 2, \dots, n\}$ such that $\sigma(\alpha_i) = \alpha_i$, $\sigma'(\beta_i) = \beta_i$, respectively, $1 \leq i \leq t$. Define the permutation $\tau \in S_n$ by $\tau(a_i) = a'_i, \dots, \tau(b_i) = b'_i$ and $\tau(\alpha_i) = \beta_i$ (convince yourself that this is a bijection). By the above property $(*)$ we have:

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1) \dots \tau(a_{n_1})) \dots (\tau(b_1) \dots \tau(b_{n_k})) = (a'_1 \dots a'_{n_1}) \dots (b'_1 \dots b'_{n_k}) = \sigma'$$

(check this fact) and σ and σ' are conjugate. □

Example 2.4.

1. Elements of S_n may be conjugate in various distinct ways: let $\sigma_1 = (12)(345)$ and $\sigma_2 = (34)(125)$ then $\sigma_2 = \tau \circ \sigma_1 \circ \tau^{-1}$ where $\tau = (13)(24)$ (see above proof). If we rewrite $\sigma_2 = (43)(251)$ then $\sigma_2 = \tilde{\tau} \circ \sigma_1 \circ \tilde{\tau}^{-1}$ where $\tilde{\tau} = (145)(23)$. Thus σ_1 and σ_2 are conjugate by both τ and $\tilde{\tau}$.
2. In S_3 all 3 cycles are conjugate to each others: $(123) = (23)(132)(23) = (12)(132)(12) = (31)(132)(31)$. Further in S_3 all transpositions are conjugate: $(12) = (213)(23)(231) = (13)(23)(13)$, $(13) = (21)(23)(21) = (231)(23)(213)$, and $(13) = (23)(12)(23)$.
3. In S_4 all products of disjoint transpositions are conjugate: $(12)(34) = (32)[(13)(24)](32)$, $(12)(34) = (423)[(14)(23)](432)$, and $(13)(24) = (43)[(14)(23)](43)$.

4. Being conjugate in S_n doesn't mean necessarily being conjugate in A_n. Permutations $\sigma_1, \sigma_2 \in A_4$ are conjugate in A_4 if $\sigma_2 = \tau\sigma_1\tau^{-1}$ with $\tau \in A_4$. In particular σ_1 and σ_2 must have the same cycle type, but even so they might not be conjugate in A_4: $(132) = \tau(123)\tau^{-1}$ where $\tau \in \{(12), (13), (23)\}$ but
$$\{(12), (13), (23)\} \cap A_4 = \emptyset.$$
5. All product of disjoint transpositions are conjugate in A_4: $(14)(23) = (132)[(13)(24)](123)$, $(12)(34) = (423)[(14)(23)](432)$, $(13)(24) = (134)[(14)(23)](143)$, and
$$\{(132), (423), (134)\} \subset A_4.$$

Definition 2.5 (Quotient Group Law).
Let G be a group and H a **normal** subgroup of G. We define a composition law $(.)$ on the set of left cosets G/H by
$$(.) : G/H \times G/H \to G/H$$
$$(xH, yH) \mapsto xH.yH = xyH.$$

First, we should verify that this composition law is well-defined. That is, we must show that if $x_1H = x_2H$ and $y_1H = y_2H$, then $x_1y_1H = x_2y_2H$. From the definition of the relation R, we know, for all $z_1, z_2 \in G$, that $z_1H = z_2H$ if and only if $z_1^{-1}z_2 \in H$.

Write $x_1^{-1}x_2 = h$ and $y_1^1y_2 = h'$. Then $h \in H$ and $h' \in H$. It suffices to show that $(x_1y_1)^{-1}(x_2y_2) \in H$.

Note
$$(x_1y_1)^{-1}(x_2y_2) = (y_1^{-1}x_1^{-1})(x_2y_2) = y_1^{-1}(x_1^{-1}x_2)y_2 = y_1^{-1}hy_2 = (y_1^{-1}hy_1)(y_1^{-1}y_2) = h''h',$$
where $h'' = (y_1^{-1}hy_1 \in H$ as H is normal in G.

This is well defined since H is normal: if $x'H = xH$, and $y'H = yH$ then $x'y'H = xyH$ (see discussion before Definition 2.3).

Proposition 2.6.
*$(G/H, .)$ is a group, it is called the **quotient group** of G by H.*

Proof. Associativity $(xH.yH).zH = (xy)H.zH = (xy)zH = x(yz)H = xH.yzH = xH.(yH.zH)$. The neutral element is $e_{G/H} = e_GH = H$: $xH.e_GH = xe_GH = xH$. The inverse of xH is $x^{-1}H$: $xH.x^{-1}H = xx^{-1}H = e_GH$. □

Example 2.5.

1. If G is abelian then every subgroup H of G is normal and we can consider the quotient group G/H. In this case the quotient group G/H is abelian (check this fact).
2. Let $G = \mathbf{Z}$, $H = n\mathbf{Z}$ with $n \geq 1$. The quotient group $\mathbf{Z}/n\mathbf{Z}$ equals $\{\bar{0}, \cdots, \overline{n-1}\}$, $\bar{i} = i + n\mathbf{Z}$. Its group law $(.)$ is denoted $+$ and is given by
$$\bar{i} + \bar{j} = (i + n\mathbf{Z}) + (j + n\mathbf{Z}) = (i + j) + n\mathbf{Z} = \overline{i + j}$$
this is the group $(\mathbf{Z}_n, +)$. From now on we will systematically use the notation $\mathbf{Z}/n\mathbf{Z}$ instead of $\mathbf{Z}_n$, as well as the notation $\bar{i} = i + n\mathbf{Z}$ instead of $[i]_n$.

3. $A_3 = \{e, (123), (132)\}$ is a normal subgroup of S_3, it has 2 left cosets:

$$eA_3 = A_3 = (123)A_3 = (132)A_3$$

and

$$(12)A_3 = (13)A_3 = (23)A_3 = \{(12), (23), (13)\}.$$

Thus $S_3/A_3 = \{eA_3, (12)A_3\}$ and the group law is given by: $eA_3.(12)A_3 = (12)A_3$, $eA_3.eA_3 = eA_3$, $(12)A_3.(12)A_3 = (12)^2A_3 = eA_3$. The group S_3/A_3 is an abelian cyclic group of order 2 generated by $(12)A_3$.

4. In $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$ the subgroup $H = \{1, r^2, s, sr^2\}$ is a normal subgroup. Indeed if i is odd

$$s^i r^j r^2 (s^i r^j)^{-1} = s^i r^{j+2-j} s^{-i} = s^i r^2 s^{-i} = r^2 \in H$$

$$s^i r^j s (s^i r^j)^{-1} = s^{i+1} r^{-j-j} s^{-i} = s^{i+1-i} r^{2j} = sr^{2j} \in H$$

and

$$s^i r^j sr^2 (s^i r^j)^{-1} = s^{i+1} r^{2-2j} s^{-i} = s^{2i+1} r^{2j-2} = sr^{2(j-1)} \in H$$

(check by yourself what happens if i is even in which case $s^i = 1$) . The subgroup H has 2 left cosets: $H = 1H = sH = r^2H = sr^2H$ and $rH = r^3H = srH = sr^3H = \{r, r^3, sr, sr^3\}$. Thus $D_8/H = \{1H, rH\}$ is a cyclic group of order 2: $1H.1H = 1H$, $1H.rH = rH$, and $rH.rH = r^2H = 1H$.

Map onto the quotient group. Let H be a **normal** subgroup of G. Consider the map

$$\phi : G \to G/H,$$

defined by

$$g \mapsto \phi(g) = gH.$$

Proposition 2.7.
The map ϕ is a surjective group homomorphism and $\mathrm{Ker}(\phi) = H$.

Proof. The fact that ϕ is surjective is clear $gH = \phi(g)$ for every $g \in G$. It is a homomorphism since

$$\phi(g_1 g_2) = g_1 g_2 H = (g_1 H).(g_2 H) = \phi(g_1).\phi(g_2).$$

We show $\mathrm{Ker}(\phi) = H$. First $H \subset \mathrm{Ker}(\phi)$ since if $g \in H$ then $e_G^{-1} g = g \in H$ and $e_G R g$, hence $\phi(g) = gH = e_G H = e_{G/H}$. Conversely let $g \in \mathrm{Ker}(\phi)$ meaning $\phi(g) = gH = e_G H$, then $e_G R g$ and $e_G^{-1} g = g \in H$. □

Going back to group homomorphisms, we will now state and prove the first isomorphism theorem for groups. Let

$$f : G \to H$$

be a group homomorphism.

The kernel $\mathrm{Ker}(f)$ of f is a normal subgroup of G [see Examples 2.2] and we can consider the quotient group $G/\mathrm{Ker}(f)$. The image $\mathrm{Im}(f) = \{h \in H : h = f(g), g \in G\}$ is a subgroup of H (see discussion after Definition 1.7).

Theorem 2.8 (First Isomorphism Theorem)**.**
The quotient group $G/\operatorname{Ker}(f)$ is isomorphic to $Im(f)$.

Proof. Consider the map

$$\pi : G/\operatorname{Ker}(f) \to \operatorname{Im}(f),$$

defined by

$$\pi(g\operatorname{Ker}(f)) = f(g).$$

We show π is a group isomorphism. First we check that π is well defined. Assume $g\operatorname{Ker}(\pi) = g'\operatorname{Ker}(\pi)$ meaning $g'^{-1}g = \tilde{g} \in \operatorname{Ker}(\pi)$. Then $f(g) = f(g'\tilde{g}) = f(g')f(\tilde{g}) = f(g')e_H = f(g')$ since $\tilde{g} \in \operatorname{Ker}(f)$. Further π is a homomorphism:

$$\pi(g\operatorname{Ker}(f).g'\operatorname{Ker}(f)) = \pi(gg'\operatorname{Ker}(f)) = f(gg') = f(g)f(g') = \pi(g\operatorname{Ker}(f))\pi(g'\operatorname{Ker}(f)).$$

The homomorphism π is surjective: if $f(g) \in \operatorname{Im}(f),\ g \in G$, then

$$f(g) = \pi(g\operatorname{Ker} f)$$

by the definition of π. Finally π is injective: assume $f(g) = \pi(g\operatorname{Ker}(f)) = \pi(g'\operatorname{Ker}(f)) = f(g')$, then $f(g')^{-1}f(g) = f(g'^{-1}g) = e_H$ and $g'^{-1}g \in \operatorname{Ker}(f)$, or equivalently gRg' meaning $g\operatorname{Ker} f = g'\operatorname{Ker} f$. □

Corollary 2.9.
Let G, H be groups with G finite and $f : G \to H$ a group homomorphism. Then $Im(f)$ is finite and

$$|G| = |\operatorname{Ker}(f)||Im(f)|.$$

In particular both $|\operatorname{Ker}(f)|$ and $|Im(f)|$ divide $|G|$.

Proof. The First Isomorphism Theorem states that $G/\operatorname{Ker}(f)$ is isomorphic to $\operatorname{Im}(f)$. Since G is finite, $G/\operatorname{Ker}(f)$ is finite, and thus $\operatorname{Im}(f)$ is finite being isomorphic to $G/\operatorname{Ker}(f)$. Further this isomorphism implies $|G/\operatorname{Ker}(f)| = |\operatorname{Im}(f)|$. On the other hand

$$|G/\operatorname{Ker}(f)| = |G : \operatorname{Ker}(f)| = \frac{|G|}{|\operatorname{Ker}(f)|}$$

by Lagrange's theorem hence the result. □

Note that if $f : G \to H$ is a surjective group homomorphism then $\operatorname{Im}(f) = H$ and the FIT for groups states that $G/\operatorname{Ker}(f)$ is isomorphic to H.

Example 2.6.

1. Consider the map $f : D_6 \to \mathbf{Z}/2\mathbf{Z}$ defined by

   $$f(s^i r^j) = \bar{i}$$

   thus $f(1) = f(r) = f(r^2) = \bar{0}$ and $f(s) = f(sr) = f(sr^2) = \bar{1}$. Then f is surjective and is a homomorphism [see Examples 1.6]:

   $$f(s^i r^j s^t r^e) = f(s^i s^t r^{-j} r^e) = f(s^{i+t} r^{i-e}) = \overline{i+t} = \bar{i} + \bar{t} = f(s^i r^j) + f(s^t r^e).$$

   Further $\operatorname{Ker}(f) = H = \langle r \rangle$ is the subgroup generated by r. The First Isomorphism Theorem asserts that D_6/H is isomorphic to $\operatorname{Im}(f) = \mathbf{Z}/2\mathbf{Z}$.

2. Find, if any, a surjective homomorphism from S_3 to a group of order 2 or 3.

- A surjective homomorphism from S_3 to a group of order 2 would have as kernel a normal subgroup of S_3 of order 3. The subgroup A_3 is a normal subgroup of order 3, and a group of order 2 must be cyclic. We construct a surjective homomorphism

$$f : S_3 \to \mathbf{Z}/2\mathbf{Z} = \{\bar{0}, \bar{1}\}$$

as follows: $f(\sigma) = \bar{0}$ is σ is even and $f(\sigma) = \bar{1}$ if σ is odd, thus

$$f(e) = f((123)) = f((132)) = \bar{0}$$

and

$$f((12)) = f((13)) = f((23)) = \bar{1}$$

(check f is a homomorphism). Further $\mathrm{Ker}(f) = A_3$ and the FIT asserts that S_3/A_3 is isomorphic to $\mathrm{Im}(f) = \mathbf{Z}/2\mathbf{Z}$.

- A surjective homomorphism from S_3 to a group of order 3 would have a kernel a normal subgroup of S_3 of order 2. Such a subgroup would contain an element of order 2, namely a transposition, hence would contain all transpositions as they are all conjugated in S_3 (see Proposition 2.5). Since there are 3 transpositions in S_3 we see that no such a subgroup, hence no such homomorphism, exists.

3 Group Actions

Definition 3.1 (Group Action)**.**
Let $(G, \star)$ be a group and A a non-empty set. A *group action* of G on A is a map:

$$(\cdot) : G \times A \to A$$

$$(g, a) \mapsto g \cdot a$$

satisfying:

(A1) $(g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a), \quad \forall g_1, g_2 \in G, \quad \forall a \in A.$ **(A2)** $e_G \cdot a = a, \quad \forall a \in A.$

Example 3.1.

1. $(\mathbf{Z}, +)$ acts on $\mathbf{R}$ by translations via the action:

   $$(\cdot) : \mathbf{Z} \times \mathbf{R} \to \mathbf{R}$$

   defined by

   $$(n, r) \mapsto n \cdot r = n + r.$$

   Indeed the axiom (A1) is satisfied:

   $$(n_1 + n_2) \cdot r = (n_1 + n_2) + r = n_1 + (n_2 + r) = n_1 \cdot (n_2 \cdot r)$$

   as well as the axiom (A2):

   $$0 + r = r.$$

2. **Action by left multiplication.** A group $(G, \star)$ acts on itself via the action

   $$(\cdot) : G \times G \to G$$

   defined by

   $$(h, g) \mapsto h \cdot g = h \star g.$$

   Indeed the axiom (A1) is satisfied:

   $$(h_1 \star h_2) \cdot g = (h_1 \star h_2) \star g = h_1 \star (h_2 \star g) = h_1 \cdot (h_2 \cdot g)$$

   as well as the axiom (A2):

   $$e_G \cdot g = e_G \star g = g.$$

   This action is called the *left regular action* of G on itself by left multiplication.

3. **Action by conjugation** A group $(G, \star)$ acts on itself via the action

   $$(\cdot) : G \times G \to G$$

   defined by

   $$(h, g) \mapsto h \cdot g = h \star g \star h^{-1}.$$

   Indeed the axiom (A1) is satisfied:

   $$(h_1 \star h_2) \cdot g = (h_1 \star h_2) \star g \star (h_1 \star h_2)^{-1} = (h_1 \star h_2) \star g \star (h_2^{-1} \star h_1^{-1}) = h_1 \star (h_2 \star g \star h_2^{-1}) \star h_1^{-1} = h_1 \cdot (h_2 \cdot g)$$

   as well as the axiom (A2):

   $$e_G \cdot g = e_G \star g \star e_G^{-1} = g.$$

   This action is called the action of G on itself by conjugation.

Definition (Permutation representation).
Let $(\cdot) : G \times A \to A$ be a group action of a group $(G, \star)$ on the set A. For a fixed $g \in G$ the map

$$\tau_g : A \to A$$

defined by

$$\tau_g(a) = g \cdot a$$

is a bijection. Indeed τ_g has an inverse map $\tau_{g^{-1}} : A \to A$ since

$$\tau_{g^{-1}} \circ \tau_g = \tau_g \circ \tau_{g^{-1}} = \mathrm{id}_A$$

as follows from (A1) and (A2). More precisely:

$$(\tau_{g^{-1}} \circ \tau_g)(a) = \tau_{g^{-1}}(\tau_g(a)) = \tau_{g^{-1}}(g \cdot a) = g^{-1} \cdot (g \cdot a) = (g^{-1} \star g) \cdot a = e_G \cdot a = a$$

for all $a \in A$.

Let $(S_A, \circ)$ be the group of bijections from A to itself; S_A is also called the group of symmetries of A, the group law $\circ$ is the composition of bijections. [Check that $(S_A, \circ)$ is indeed a group.] The map

$$\tau : G \to S_A$$

defined by

$$\tau(g) = \tau_g$$

is a group homomorphism since for $g_1, g_2 \in G$ one has $\tau(g_1 \star g_2) = \tau(g_1) \circ \tau(g_2)$:

$$\tau(g_1 \star g_2)(a) = \tau_{g_1 \star g_2}(a) = (g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = \tau_{g_1}(\tau_{g_2}(a)) = (\tau(g_1) \circ \tau(g_2))(a)$$

for all $a \in A$. The homomorphism τ is called the *permutation representation* associated to the given action $(\cdot)$.

Definition 3.2 (Kernel of a Representation).
The kernel of $\tau : G \to S_A$

$$\mathrm{Ker}(\tau) = \{g \in G : \tau(g) = \tau_g = \mathrm{id}_A\} = \{g \in G : g \cdot a = a, \forall a \in A\},$$

is called the *kernel of the representation* τ, it is a normal subgroup of G. If $\mathrm{Ker}(\tau) = \{e_G\}$, or equivalently if τ is injective, we say that the action $(\cdot)$ is *faithful*.

Example 3.2.

1. The left regular representation of G on itself arising from the action of G on itself by left multiplication (see Examples 3.1) is faithful. Indeed let $g \in G$ with $\tau_g = \mathrm{id}_G$, meaning $\tau_g(h) = g \cdot h = h, \forall h \in G$. For $h = e_G$ we obtain $g = g \cdot e_G = e_G$.
2. Consider the action of D_8 on itself by conjugation and the corresponding permutation representation $\tau : D_8 \to S_{D_8}$. This action is not faithful. Indeed $r^2 \in \mathrm{Ker}(\tau)$ since

   $$r^2 s^i r^j (r^2)^{-1} = r^2 s^i r^j r^2 = s^i r^2 r^j r^2 = s^i r^{4+j} = s^i r^j$$

   holds for each $s^i r^j \in D_8$, hence $\tau(r^2) = \mathrm{id}_{D_8}$.

3. Assume G is abelian and consider the action of G on itself by conjugation and the corresponding representation $\tau : G \to S_G$. Then this action is not faithful if $|G| > 1>$. In fact $\mathrm{Ker}(\tau) = G$ as $\tau(g) = \tau_g = \mathrm{id}_G, \forall g \in G$:

$$\tau(g)(h) = \tau_g(h) = g \star h \star g^{-1} = g \star g^{-1} \star h = h$$

for all $h \in G$.

Definition 3.3 (Stabiliser)**.**
Let $G \times A \to A$ be an action of a group $(G, \star)$ on a set A and $a \in A$. The *stabiliser* of a is

$$\mathrm{Stab}(a) = \{g \in G : g \cdot a = a\}.$$

Proposition 3.1.
The stabiliser $\mathrm{Stab}(a)$ *is a subgroup of* G*.*

Proof. First $e_G \in \mathrm{Stab}(a)$ since $e_G \cdot a = a$. Let $g \in \mathrm{Stab}(a)$ meaning $g \cdot a = a$. Then

$$a = e_G \cdot a = (g^{-1} \star g) \cdot a = g^{-1} \cdot (g \cdot a) = g^{-1} \cdot a$$

and $g^{-1} \in \mathrm{Stab}(a)$. Further, let $g_1, g_2 \in \mathrm{Stab}(a)$ meaning $g_1 \cdot a = g_2 \cdot a = a$ then

$$(g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = g_1 \cdot a = a$$

and $g_1 \star g_2 \in \mathrm{Stab}(a)$. □

Example 3.3.

1. In the example 1 of Examples 3.1 let $r \in \mathbf{R}$, then

$$\mathrm{Stab}(r) = \{n \in \mathbf{Z} : n + r = r\} = \{0\}.$$

2. Consider the action of S_3 on the set $\{1, 2, 3\}$ by permutations. Then $\mathrm{Stab}(1) = \{e, (23)\}$, $\mathrm{Stab}(2) = \{e, (13)\}$, and $\mathrm{Stab}(3) = \{e, (12)\}$.

Let $(G, \star)$ be a group acting on a non-empty set A. Define a relation R among the elements of A by:

$$a \ R \ b \iff \exists g \in G : a = g \cdot b$$

for $a, b \in A$.

Proposition 3.2.
The relation R *is an equivalence relation.*

Proof. R is reflexive since $a = e_G \cdot a$ hence $a \ R \ a$, $\forall a \in A$. Let $a, b \in A$ with $a \ R \ b$, i.e. $a = g \cdot b$ for some $g \in G$. Then $g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = (g^{-1} \star g) \cdot b = e_G \cdot b = b$ hence $b \ R \ a$. Finally, let $a, b, c \in A$ with $a \ R \ b$ and $b \ R \ c$. Thus $\exists g, g' \in G$ such that $a = g \cdot b$ and $b = g' \cdot c$. Then $a = g \cdot b = g \cdot (g' \cdot c) = (g \star g') \cdot c$ and $a \ R \ c$. □

Definition 3.4 (Orbit).
Let $a \in A$. The equivalence class of a for the relation R is

$$\bar{a} = \{b \in A : b \ R \ a\} = \{b \in A : \exists g \in G, \ b = g \cdot a\} = \{g \cdot a : g \in G\}.$$

It is called the *orbit* of a, for the given action, and is denoted $\mathrm{orb}(a)$.

Note that A is equal to the disjoint union of its orbits under the given action of G:

$$A = \bigcup_{a \in A} \mathrm{orb}(a).$$

The action of G on A is called *transitive* if there is only one orbit, in which case this orbit necessarily contains all elements of A, i.e. $\forall a, b \in A$, $\exists g \in G$ with $a = g \cdot b$. Thus $A = \mathrm{orb}(a)$ for every $a \in A$ in this case.

Example 3.4.

1. The action
$$S_n \times \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$$
$$(\sigma, i) \mapsto \sigma(i)$$
of S_n on the set $\{1, 2, \ldots, n\}$ is transitive: given $1 \leq i \leq j \leq n$ we can find $\tau \in S_n$ with $\tau(i) = j$. For example take $\tau = \mathrm{id}$ if $i = j$, or $\tau = (ij)$ otherwise.
2. The group D_8 acts transitively on the four vertices of a square and the stabiliser of any vertex is the subgroup of order 2 (and index 4) consisting of identity and the reflection about the line of symmetry passing through that point.

Theorem 3.3 (The Orbit-Stabiliser Theorem).
Assume $(G, \star)$ is a finite group acting on a set A. Then the orbit $\mathrm{orb}(a)$ of an element $a \in A$ is finite and

$$|\mathrm{orb}(a)| = |G : \mathrm{Stab}(a)| = \frac{|G|}{|\mathrm{Stab}(a)|}.$$

Proof. Consider the map

$$f : \mathrm{orb}(a) \to (G/\mathrm{Stab}(a))_{\mathrm{left}}$$

defined by

$$f(g \cdot a) = g.\,\mathrm{Stab}(a).$$

We check this map is well defined, i.e.

$$\{g_1 \cdot a = g_2 \cdot a\} \Rightarrow \{g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)\}.$$

We have $(g_2^{-1} \star g_1) \cdot a = g_2^{-1} \cdot (g_1 \cdot a) = g_2^{-1} \cdot (g_2 \cdot a) = (g_2^{-1} \star g_2) \cdot a = e_G \cdot a = a$ hence $g_2^{-1} \star g_1 \in \mathrm{Stab}(a)$ which implies $g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)$.

Further f is a bijection. Indeed f is clearly surjective, we prove it is injective. Suppose $g_1 \cdot a, g_2 \cdot a \in \mathrm{orb}(a)$ and $f(g_1 \cdot a) = f(g_2 \cdot a)$, i.e. $g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)$ which is equivalent to $g_2^{-1} \star g_1 \in \mathrm{Stab}(a)$. Then $a = (g_2^{-1} \star g_1) \cdot a = g_2^{-1} \cdot (g_1 \cdot a) \Rightarrow g_2 \cdot a = g_2 \cdot (g_2^{-1} \cdot (g_1 \cdot a)) = (g_2 \star g_2^{-1}) \cdot (g_1 \cdot a) = e_G \cdot (g_1 \cdot a) = g_1 \cdot a$ as required.

Now since f is a bijection, $\mathrm{orb}(a)$ and $(G/\operatorname{Stab}(a))_{\text{left}}$; whose cardinality is $|G : \operatorname{Stab}(a)|$, have the same number of elements. The statement follows using Lagrange's theorem which asserts that $|G : \operatorname{Stab}(a)| = \frac{|G|}{|\operatorname{Stab}(a)|}$. □

Example 3.5.

- Consider the action of S_3 on itself by conjugation. Then

$$\operatorname{Stab}(12) = \{e, (12)\}$$

and

$$\mathrm{orb}(12) = \{(23) = (123)(12)(132), (13) = (132)(12)(123), (12) = e(12)e^{-1}\}.$$

Further

$$\operatorname{Stab}(123) = \{1, (123), (132)\}$$

and

$$\mathrm{orb}(123) = \{(123) = e(123)e^{-1}, (132) = (12)(123)(12)\}.$$

Cayley's Theorem. Recall the left regular representation of $(G, \star)$ on itself by left multiplication (see example 2 in Examples 3.1). Assume that G is finite of cardinality n. If we label the elements of G as $\{g_1, \ldots, g_n\}$, the regular representation defines a faithful permutation representation (an injective homomorphism)

$$\tau : G \to S_n$$

called the *regular permutation representation*, defined as follows: if $g \in G$, $\tau(g)$ is the permutation defined for $i, j \in \{1, \ldots, n\}$ by

$$\tau(g)(i) = j \quad \text{if} \quad g \star g_i = g_j.$$

The permutation representation τ depends on the given labelling of the elements of G.

In particular, since $\operatorname{Ker}(\tau) = \{e_G\}$ we obtain by the FIT that G is isomorphic to its image $\tau(G)$; a subgroup of S_n. We thus obtain.

Theorem 3.4 (Cayley's Theorem).
A finite group of cardinality n is isomorphic to a subgroup of S_n.

Next, we define the left action of a group $(G, \star)$ on the set of left cosets of a given subgroup. Let H be a subgroup of G, and $A = (G/H)_{\text{left}}$ the set of left cosets of H. The group G acts on A by

$$g \cdot (g'H) = (g \star g')H$$

(check that the group action axioms are satisfied). This action is called the action of G on the left cosets of H by left multiplication.

If $|G : H| = m$ is finite, and we label the elements of A as $\{g_1H, \ldots, g_mH\}$, then the above representation defines a homomorphism

$$\tau : G \to S_m$$

as follows: if $g \in G$, $\tau(g)$ is the permutation defined for $i, j \in \{1, \ldots, m\}$ by

$$\tau(g)(i) = j \quad \text{if} \quad g \cdot g_iH = (g \star g_i)H = g_jH.$$

The permutation representation τ depends on the given labelling of the elements of A.

Example 3.6.
Let $G = D_8$ and $H = \langle s \rangle = \{1, s\}$. Label the distinct left cosets H, rH, r^2H, and r^3H of H with the integers 1, 2, 3, and 4 respectively. Under this labelling we compute the permutation σ_s induced by the action by multiplication by the element s on the left cosets of H:

$$s \cdot H = \{s, s^2\} = H$$

so $\sigma_s(1) = 1$,

$$s \cdot (rH) = srH = \{sr, srs\} = \{r^3s, r^3\} = r^3H$$

so $\sigma_s(2) = 4$,

$$s \cdot r^2H = sr^2H = \{sr^2, sr^2s\} = \{r^2s, r^2s^2 = r^2\} = \{r^2s, r^2\} = r^2H$$

so $\sigma_s(3) = 3$, and

$$s \cdot r^3H = sr^3H = \{sr^3, sr^3s\} = \{rs, rs^2\} = \{rs, r\} = rH$$

so $\sigma_s(4) = 2$. Hence for this labelling $\sigma_s = (24)$. Similarly one can compute, under this labelling, the permutation σ_r induced by the action by multiplication by the element r on the left cosets of H to find $\sigma_r = (1234)$.

Let

$$\tau_H : G \to S_{G/H}$$
$$g \mapsto \tau_H(g)$$

be the permutation representation associated to the action of G by left multiplication on the left cosets of H.

Thus if $g \in G$,

$$\tau_H(g) : G/H \to G/H$$

is the bijection defined by

$$\tau_H(g)(g'H) = (g \star g')H.$$

Theorem 3.5.
With the notations above, the followings hold.

1. *G acts transitively on G/H.*
2. *The stabiliser of e_GH is the subgroup H.*
3. *$\mathrm{Ker}(\tau_H) = \bigcap_{x \in G} xHx^{-1}$, and $\mathrm{Ker}(\tau_H)$ is the largest normal subgroup of G contained in H.*

Proof.

1. To see that G acts transitively on G/H let $aH, bH \in G/H$ and $g = b \star a^{-1}$. Then $g \cdot (aH) = (b \star a^{-1}) \cdot aH = (b \star a^{-1} \star a)H = bH$ this proves 1.
2. The stabiliser of e_GH is $\{g \in G : g \cdot (e_GH) = gH = H\} = \{g \in G : gH = H\} = H$, this shows 2.
3. Next, we prove 3. By definition $\mathrm{Ker}(\pi_H) = \{g \in G : g \cdot (xH) = xH, \forall x \in G\} = \{g \in G : (g \star x)H = xH, \forall x \in G\} = \{g \in G : (x^{-1} \star g \star x)H = H, \forall x \in G\} = \{g \in G : x^{-1} \star g \star x \in H, \forall x \in G\} = \{g \in G : g \in xHx^{-1}, \forall x \in G\} = \bigcap_{x \in G} xHx^{-1}$. Further $\mathrm{Ker}(\pi_H)$ is contained in $H = e_GHe_G^{-1}$, and $\mathrm{Ker}(\pi_H)$ is a normal subgroup of both G and H. Now let N be a normal subgroup of G contained in H then $N = xNx^{-1} \subset xHx^{-1}$, $\forall x \in G$, hence $N \subset \bigcap_{x \in G} xHx^{-1} = \mathrm{Ker}(\pi_H)$. This shows $\mathrm{Ker}(\pi_H)$ is the largest normal subgroup of G contained in H.

□

Corollary 3.6.
Let G be a finite group of cardinality $n \geq 2$ and p the smallest prime number dividing $n = |G|$. Then any subgroup of G of index p is normal. In particular, if G has a subgroup of index 2 then this subgroup must be normal.

Proof. Suppose $H \leq G$ is a subgroup (the notation $\leq$ is used in the literature to denote a subgroup) and $|G : H| = p$. Let π_H be the permutation representation afforded by multiplication on the set of left cosets of H in G. Let $K = \mathrm{Ker}(\pi_H)$; K is contained in H, and $|H : K| = m$. Then $|G : K| = |G : H||H : K| = pm$ (use Lagrange's theorem for the subgroup H of G and the subgroup K of H to show this).

Since H has p left cosets in G, G/K is isomorphic to a subgroup of S_p (namely the image of G under π_H) by the FIT. By Lagrange's Theorem, $pm = |G/K|$ divides $|S_p| = p!$. Thus m divides $\frac{p!}{p} = (p-1)!$. But all prime factors of $(p-1)!$ are less than p, and by the minimality of p, every possible prime divisor of m is greater or equal to p. This forces $m = 1$ as m divide $|G|$. So $H = K$ is a normal subgroup of G (since K is normal): the equality $|H : K| = 1$ just means that $H = K$. □

4 The Class Equation

Let G be a group and let

$$\mathcal{S}(G) = \{A : A \subset G, A \neq \emptyset\}$$

be the set of non-empty subsets of G. Then G acts on $\mathcal{S}(G)$ by conjugation via

$$\phi : G \times \mathcal{S}(G) \ \to \ \mathcal{S}(G)$$
$$(g, A) \ \mapsto \ gAg^{-1} = \{gag^{-1} \ : \ a \in A\}$$

where $g \in G$ and $A \subset G$ is a nonempty subset of G (check that this definition satisfies the axioms of a group action).

The subset gAg^{-1} of G is called the *conjugate* of A by g. Note that if A is a subgroup of G then gAg^{-1} is also a subgroup of A (check this fact).

Definition 4.1 (Normaliser)**.**
If $A \in \mathcal{S}(G)$, the stabiliser of A under the above action ϕ by conjugation is called the *Normaliser* of A in G and denoted

$$N_G(A) = \{g \in G \ : \ gAg^{-1} = A\}.$$

Thus $N_G(A)$ is a subgroup of G.

The normaliser $N_G(A)$ of a subset A of G acts on A by conjugation

$$\phi_A : N_G(A) \times A \to A$$
$$(g, a) \ \mapsto \ gag^{-1}$$

(check that ϕ_A satisfies the axioms of a group action).

Definition 4.2 (Centraliser-Centre)**.**
If $A \in \mathcal{S}(G)$, the kernel of the action ϕ_A is called the *centraliser* of A in G and denoted

$$C_G(A) = \{g \in G \ : \ gag^{-1} = a \quad \forall a \in A\}.$$

Thus $C_G(A)$ is a normal subgroup of $N_G(A)$ (see Definition 3.2). Note that the equality $gag^{-1} = a$ is equivalent to $ga = ag$, thus the centraliser of A in G

$$C_G(A) = \{g \in G \ : \ ga = ag \quad \forall a \in A\}$$

consists of those elements of G which commute with all elements of A.

In case $A = G$, $N_G(G) = G$ since $gGg^{-1} = G$ for all $g \in G$ (check this fact) and G is equal to its normaliser.

The centraliser $C_G(G)$ of G in G is called the *centre* of G and denoted

$$Z(G) = \{g \in G \ : \ gh = hg \quad \forall h \in G\}.$$

Thus $Z(G)$ is the set of elements of G which commute with every element of G. $Z(G)$ is a normal subgroup of G. Note that $Z(G)$ is contained in $C_G(A)$ for any non-empty subset A of G. Indeed an element of G which commutes with all elements of G commutes, a fortiori, with every element in a subset A of G. In fact

$$Z(G) = \bigcap_{A \subset G} C_G(A).$$

Note also that $Z(G)$ is an abelian group: elements of $Z(G)$ commute with each other.

Example 4.1.
(1) If G is abelian, all elements of G commute with each other: $ab = ba$ for all $a, b \in G$, so $Z(G) = G$. Similarly, $C_G(A) = N_G(A) = G$ for any subset A of G since $gag^{-1} = gg^{-1}a = a$ for every $g \in G$ and $a \in A$.

(2) Let $G = D_8$ be the dihedral group of cardinality 8 with the usual generators r and s satisfying $sr^j = r^{-j}s = r^{4-j}s$, and $A = \{1, r, r^2, r^3\}$ the subgroup of rotations in D_8 generated by r.

- We show $C_G(A) = A$. Since A is abelian: $r^i r^j = r^j r^i = r^{i+j}$, it holds that $A \subset C_G(A)$. The element $s \notin C_G(A)$ since $sr = r^3 s \neq rs$. An element of G not in A is of the form sr^i, and is not an element of $C_G(A)$ since $sr^i r = sr^{i+1} = r^{-1-i}s = r^{4-1-i}s \neq r^{i+1}s$ for $i = 0$ or $i = 2$. For $i = 1$ we have $(sr)r = sr^2 = r^2 s \neq r(sr) = rr^3 s = r^4 s = s$. Thus the only elements of $C_G(A)$ are the elements of A and $A = C_G(A)$.
- We show $N_G(A) = G$. Since $C_G(A) \subset N_G(A)$ [$C_G(A)$ is a subgroup of $N_G(A)$] and $A = C_G(A)$ we have $A \subset N_G(A)$. We have $sAs^{-1} = sAs = \{s1s = 1, srs = ssr^3 = r^3, sr^2 s = ssr^2 = r^2, sr^3 s = ssr = r\} = A$, hence $s \in N_G(A)$. Since $N_G(A)$ is a subgroup of G, $r^i \in N_G(A)$, $s \in N_G(A)$, it holds that $sr^i \in N_G(A)$, hence all elements of $G = D_8$ are in $N_G(A)$ and $N_G(A) = G$.
- We show $Z(G) = \{1, r^2\}$. Since $Z(G) \subset C_G(H)$ for any subset H of G, we have $Z(G) \subset C_G(A) = A$. The elements r, r^3 are not in $Z(G)$ since $rs = sr^3 \neq sr$ and $r^3 s = sr \neq sr^3$. Thus $Z(G) \subset \{1, r^2\}$. The element $r^2 \in Z(G)$ since it commutes with any power r^i of r, it also commutes with s: $sr^2 = r^2 s$, hence r^2 commutes with any element $s^i r^j$ of G. This shows $Z(G) = \{1, r^2\}$.

(3) Let $G = S_3$ and $A = \{1, (12)\}$. We show $C_G(A) = N_G(A) = A$. We have $A \subset C_G(A)$ since A is abelian. In particular A is a subgroup of $C_G(A)$ and 2 divides $|C_G(A)|$. By Lagrange's Theorem $|C_G(A)|$ divides $|G| = 6$, hence $|C_G(A)|$ equals 2 or 6. If $|C_G(A)| = 6$ then $G = C_G(A)$ and (12) would commute with every element of S_3. But this is not the case since (12) doesn't commute with (123): $(12)(123) = (23) \neq (123)(12) = (13)$. Thus, $|C_G(A)| = 2$ and $A = C_G(A)$.

Let $\sigma \in N_G(A)$. Then $\sigma A \sigma = \{\sigma 1 \sigma^{-1} = 1, \sigma(12)\sigma^{-1}\} = A$ holds if and only if $\sigma(12)\sigma^{-1} = (12)$, which implies $\sigma(12) = (12)\sigma$, meaning $\sigma \in C_G(A) = A$. Thus, $N_G(A) \subset C_G(A)$, hence $N_G(A) = C_G(A) = A$ since $C_G(A) \subset N_G(A)$ holds always. Finally $Z(G) = \{1\}$. Indeed $Z(G) \subset C_G(A) = A$. But $(12) \notin Z(G)$ since $(12)(123) = (23) \neq (123)(12) = (13)$. Thus $Z(G) = \{1\}$. We will see later that $Z(S_n) = \{1\}$ holds if $n \geq 3$.

4.1 The Class Equation

Let G be a group and $g \in G$. Consider the subset $\{g\} \subset G$, then

$$N_G(\{g\}) = C_G(\{g\}) = \{h \in G \ : \ hgh^{-1} = g\} = \{h \in G \ : \ hg = gh\}$$

is the subgroup of G of elements which commute with g. We denote $C_G(\{g\})$ simply by $C_G(g)$ and call it the *centraliser of g in G*. Note that $C_G(g) = \mathrm{Stab}(g)$ is precisely the stabiliser of $\{g\}$ under the conjugation action of G on itself.

The orbit of $\{g\}$ under conjugation is

$$\mathrm{orb}(g) = \{hgh^{-1} \ : \ h \in G\}$$

and consists of all elements of G which are conjugate to g; $\operatorname{orb}(g)$ is called the *conjugacy class* of g. Note that $g = e_G g e_G^{-1} \in \operatorname{orb}(g)$. Further $\operatorname{orb}(g) = \{g\}$ if and only if $hgh^{-1} = g$ [which is equivalent to $hg = gh$] for all $h \in G$, and this is equivalent to $g \in Z(G)$. Thus

$$|\operatorname{orb}(g)| = 1 \iff g \in Z(G).$$

Assume now that G is finite. The Orbit-Stabiliser Theorem (see Theorem 3.3) states that

$$|\operatorname{orb}(g)| = \frac{|G|}{|C_G(g)|} = |G : C_G(g)|.$$

The conjugacy classes of elements of G form a partition of G

$$G = \bigcup_{g \in G} \operatorname{orb}(g)$$

where the union is disjoint. By the above discussion, we have

$$Z(G) = \bigcup_{g \in G, |\operatorname{orb}(g)|=1} \operatorname{orb}(g)$$

where the union is over all element of G with $|\operatorname{orb}(g)| = 1$. Let $\{\operatorname{orb}(g_1), \ldots, \operatorname{orb}(g_r)\}$ be the distinct conjugacy classes of G which **are not** contained in $Z(G)$, or equivalently those orbits such that $|\operatorname{orb}(g_i)| \geq 2$ for $1 \leq i \leq r$. Then

$$G = Z(G) \bigcup \Big(\bigcup_{i=1}^{r} \operatorname{orb}(g_i) \Big).$$

Counting the number of elements of G, and bearing in mind the equality $|\operatorname{orb}(g_i)| = |G : C_G(g_i)|$, we find the class equation:

Theorem 4.1 (The Class Equation).
*Let G be a finite group and $\{\operatorname{orb}(g_1), \ldots, \operatorname{orb}(g_r)\}$ the distinct conjugacy classes of G which are **NOT** contained in $Z(G)$, then*

$$|G| = |Z(G)| + \sum_{i=1}^{r} |\operatorname{orb}(g_i)| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

Example 4.2.
(1) If G is abelian then all the elements of G commute, so $Z(G) = G$ and $|G| = |Z(G)|$. The class equation gives no specific information in this case.

(2) In D_8 we have $Z(D_8) = \{1, r^2\}$ [see example 4.1 (2)]. Let $x \in D_8$ and $x \notin Z(D_8)$. We show $|C_{D_8}(x)| = 4$. It holds that $\{1, r^2, x\} \subset C_{D_8}(x)$, since r^2 commutes with x and $C_{D_8}(x)$. Now $|C_{D_8}(x)|$ is a factor of $|D_8| = 8$, by Lagrange's Theorem. Given that $|C_{D_8}| \geq 3$, the only possibilities are 4 and 8. Further, $|C_{D_8}(x)| = 8$ cannot hold for otherwise $C_{D_8}(x) = D_8$ and x commutes with every element of D_8 hence $x \in Z(G)$ which is not the case. Thus $|C_{D_8}(x)| = 4$ necessarily, $|\operatorname{orb}(x)| = \frac{|D_8|}{|C_{D_8}(x)|} = |D_8 : C_{D_8}(x)| = 2$, and x has exactly two distinct conjugates.

The conjugacy classes of D_8 are

$$\{1\},\ \{r^2\},\ \{r, r^3\},\ \{s, sr^2\},\ \{sr, sr^3\}.$$

To see this, note that $r^3 = srs$, $sr^2 = (sr)s(r^3s)$, and $sr^3 = (sr^2)sr(r^2s)$.

The first two classes form $Z(D_8)$ and the class equation for D_8 is

$$|D_8| = 8 = 1 + 1 + 2 + 2 + 2.$$

One important consequence of the Class equation is the following.

Theorem 4.2.
If p is a prime number and G is a finite group of cardinality $|G| = p^s$ for some $s \geq 1$, then G has a nontrivial centre meaning $|Z(G)| > 1$.

Proof. By the class equation

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

where $g_1, \ldots, g_r$ are representative of the distinct conjugacy classes not contained in $Z(G)$. By definition $C_G(g_i) \neq G$ since $g_i \notin Z(G)$, hence $|G : C_G(g_i)| = \frac{|G|}{|C_G(g_i)|} > 1$, and p divides $|G : C_G(g_i)|$ (since the latter is > 1 and divides $|G| = p^s$). Since p divides $|G|$ it must also divide

$$|Z(G)| = |G| - \sum_{i=1}^{r} |G : C_G(g_i)|,$$

in particular $|Z(G)| > 1$ and $Z(G) \neq \{e_G\}$. □

4.2 Cauchy's Theorem

Using among others the class number formula we now prove Cauchy's Theorem.

Theorem 4.3 (Cauchy's Theorem)**.**
Let G be a finite group and p a prime number which divides $|G|$. Then there exists an element of G of order p, and a subgroup of G of cardinality p.

Proof. The proof is by strong induction on $|G|/p$. If $|G| = p$, then G is cyclic of order p and every element $x \in G$ different from e_G has order p. So assume $|G| > p$, and p divides $|G|$.

First, we treat the case where G is **abelian**. Let $x \in G$ be an element distinct from e_G and $n = \text{ord}(x) > 1$. If p divides n, and $m = \frac{n}{p}$, then x^m is an element of order p. Otherwise, if $\gcd(p, n) = 1$, let $H = \langle x \rangle$ be the subgroup generated by x; which is normal since G is abelian. Consider the factor group G/H whose cardinality is smaller than that of G, and is divisible by p since p doesn't divide $|H| = n$ and divides $|G| = |H||G/H|$. By the induction hypothesis G/H has an element yH of order p. But yH is the image of y under the natural homomorphism $G \to G/H$. Thus, $p = \text{ord}(yH)$ divides $r = \text{ord}(y)$ and $y^{\frac{r}{p}}$ is an element of G of order p.

Next, we treat the general case where G need not be abelian. If p divides $|Z(G)|$ then, since $Z(G)$ is abelian, we deduce from the above that there exists an element x of $Z(G)$, hence an element of G, such that $\text{ord}(x) = p$. Otherwise consider the class equation

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

where $\{\mathrm{orb}(g_1), \ldots, \mathrm{orb}(g_r)\}$ are the distinct conjugacy classes of G which are not contained in $Z(G)$. Now p divides $|G|$, and doesn't divide $|Z(G)|$, hence there exists $1 \leq i \leq r$ such that p doesn't divide $|G : C_G(g_i)|$ (otherwise p would divide $|G| - \sum_{i=1}^{r} |G : C_G(g_i)| = |Z(G)|$). This implies that p divides $|C_G(g_i)| = \frac{|G|}{|G:C_G(g_i)|}$. Now $C_G(g_i)$ is different from G, since g_i is not in $Z(G)$, and p divides $|C_G(g_i)| < |G|$. By the induction hypothesis there exists an element $x \in C_G(g_i)$; hence a fortiori an element of G, such that $\mathrm{ord}(g) = p$.

Finally, if $x \in G$ has order p then the subgroup $\langle x \rangle$ generated by x is a subgroup of G of cardinality p. □

4.3 Conjugacy Classes in S_n

We have already seen that two permutations $\sigma, \tau \in S_n$ are conjugate if and only if they have the same cycle type (see Proposition 2.5).

Let $n \geq m \geq 2$ be integers and $\sigma = (a_1 \ldots a_m)$ an m-cycle in S_n. There are n-possibilities for a_1. Once a_1 is fixed there are $n-1$ possibilities for a_2, and so on. Once $a_1, \ldots, a_{m-1}$ are fixed there are $n - (m-1) = n - m + 1$ choices for a_m. Shifting the position of $a_1, \ldots, a_m$ cyclically to obtain $(a_i a_{i+1} \ldots a_m a_1 a_2 \ldots a_{i-1})$ doesn't change the permutation σ. Counting in this way the m-cycles we find that their number is

$$\frac{n(n-1)\ldots(n-m+1)}{m}.$$

The number $\frac{n(n-1)\ldots(n-m+1)}{m}$ is thus the cardinality of the orbit $\mathrm{orb}(\sigma)$ of σ under conjugation by S_n, which is also the index of the centraliser of σ in S_n:

$$|\,\mathrm{orb}_{S_n}(\sigma)| = \frac{|S_n|}{|C_{S_n}(\sigma)|} = |S_n : C_{S_n}(\sigma)| = \frac{n(n-1)\ldots(n-m+1)}{m}.$$

Since $|S_n| = n!$ we deduce that

$$|C_{S_n}(\sigma)| = \frac{|S_n|}{\frac{n(n-1)\ldots(n-m+1)}{m}} = \frac{n!}{\frac{n(n-1)\ldots(n-m+1)}{m}} = m(n-m)!.$$

The elements $\{1, \sigma, \ldots, \sigma^{m-1}\}$, which form the subgroup of S_n generated by σ, all commute with σ. Also all permutations in S_n which are disjoint from σ commute with σ and there are $(n-m)!$ permutations of this type: the full symmetric group on the digits in $\{1, 2, \ldots, n\}$ not appearing in σ, i.e. distinct from $\{a_1 \ldots a_m\}$.

The product of elements of these two types already account for $m(n-m)!$ permutations commuting with σ. By the above computation of $|C_{S_n}(\sigma)|$ these form the full centraliser of σ in S_n. Summarising, if $\sigma \in S_n$ is an m-cycle, then

$$C_{S_n}(\sigma) = \left\{\sigma^i \tau \ : \ 0 \leq i \leq m-1, \ \tau \in S_{n-m}\right\}$$

where S_{n-m} denotes the subgroup of S_n which fixes all integers appearing in the m-cycle σ [this subgroup is isomorphic to the symmetric group on $n-m$ letters]: these are precisely the permutations of S_n which are disjoint from σ.

For example the centraliser of $\sigma = (135)$ in S_7 is the subgroup

$$\left\{(135)^i \tau \ : \ 0 \leq i \leq 2 \quad \text{and } \tau \text{ fixes } 1, 3, 5\right\}.$$

Note that $\tau \in S_A$ where $A = \{2, 4, 6, 7\}$, so there are $4!$ choices for τ and S_A is isomorphic to S_4. In this case the centraliser $C_{S_7}(\sigma)$ has cardinality $3.4! = 72$.

4.4 Simple Groups

A finite group G is called **simple** if $|G| > 1$ and the only normal subgroups of G are $\{e_G\}$ and G itself.

By Lagrange's Theorem, if $|G| = p$ is a prime number, then the only subgroups of G are G and $\{e_G\}$, so G is necessarily simple in this case. Thus all cyclic groups with cardinality a prime number are simple. In fact every abelian simple group G is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ for some prime number p (think how to prove this fact as an exercise).

An important example of a finite simple non abelian group is the following.

Theorem 4.4.
The alternating group A_5 is simple.

Proof. First, note that a normal subgroup H of a group G is the union of the conjugacy classes of its elements. Indeed, whenever H contains an element g of G it contains its orbit $\mathrm{orb}(g)$ under conjugation. Hence if H is normal it holds that

$$H = \bigcup_{g \in H} \mathrm{orb}(g)$$

where the union is disjoint.

We work out the conjugacy classes of elements of A_5 and their cardinalities. Remember that permutations which are conjugate in S_n need not be conjugate in A_n [see example 2.3 (4)] so we need to be careful.

Representatives of the conjugacy classes of even permutations in S_5 can be taken to be

$$1 \qquad (123) \qquad (12345) \qquad \text{and} \qquad (12)(34).$$

The centralisers of 3-cycles and 5-cycles in S_5 have been determined above. We have

$$C_{S_5}((123)) = \{(123)^i \tau \ : \ 0 \leq i \leq 2 \quad \text{and } \tau \text{ fixes } 1, 2, 3\},$$

thus τ is either identity or the transposition (45), and

$$C_{S_5}((12345)) = \{(12345)^i \ : \ 0 \leq i \leq 4\}.$$

Checking which of these conjugates are even we find

$$C_{A_5}((123)) = C_{S_5}((123)) \cap A_5 = \langle (123) \rangle = \{(123)^i \ : \ 0 \leq i \leq 2\}$$

and

$$C_{A_5}((12345)) = C_{S_5}((12345)) \cap A_5 = \{(12345)^i \ : \ 0 \leq i \leq 4\}.$$

These groups have cardinalities 3 and 5, hence have index 20 and 12 in A_5, respectively. Thus there are 20 distinct conjugates of (123) in A_5 and 12 distinct conjugates of (12345) in A_5:

$$|\,\mathrm{orb}_{A_n}((123))| = 20$$

$$|\,\mathrm{orb}_{A_n}((12345))| = 12.$$

Since there are twenty 3-cycles in S_5, and all of them are in A_5, we see that all twenty 3-cycles are conjugate in A_5.

The number of 5-cycles in A_5 is 24, but there are only 12 distinct conjugates of the 5-cycle (12345) in A_5. Thus some 5-cycle σ is not conjugate to (12345) in A_5 [in fact (13245) is such a σ]. Hence the 5-cycles in A_5 form two distinct conjugacy classes in A_5, each of these classes has 12 elements.

Since the 3 and 5-cycles account for all the nonidentity elements of A_5 with odd order, the 15 remaining nonidentity elements of A_5 must have even order and therefore have cycle type $(2, 2)$.

Certainly, the conjugacy class of $(12)(34)$ in S_5 consists of all permutations of cycle type $(2, 2)$. We count these. Let σ be a permutation of cycle type $(2, 2)$. There are exactly five choices for i such that $\sigma(i) = i$. Suppose σ acts non-identically on $\{j_1, \ldots, j_4\}$. Then there are 3 choices for $\sigma(j_1)$. This fixes σ. So there are $5 \cdot 3 = 15$ choices for σ.

Now I'll show that every $(2, 2)$ cycle is conjugate to $\tau = (12)(34)$ in A_5. Suppose σ is a permutation of cycle type $(2, 2)$. Then, $\sigma = g\tau g^{-1}$, for some $g \in S_5$. If g is even, then σ is conjugate to τ in A_5. Otherwise, noting that (12) commutes with τ, we have

$$(g(12))\tau(g(12))^{-1} = g\tau g^{-1} = \sigma.$$

But $g(12)$ is even, so σ is conjugate to τ in A_5. Thus, $|\operatorname{orb}_{A_n}((12)(34))| = 15$.

In summary the conjugacy classes of A_5 have orders 1, 15, 20, 12 and 12.

Now suppose H is a normal subgroup of A_5, it must be the union of conjugacy classes of A_5. Thus the cardinality of H must be both a divisor of 60 (the cardinality of A_5) and the sum of some collection of the integers $\{1, 12, 12, 15, 20\}$ (the sizes of the conjugacy classes in A_5). A quick check shows that the only possibilities are $|H| = 1$ and $|H| = 60$, so that H is either equal to $\{1\}$ or A_5 itself. □

A more general result holds, namely the alternating group A_n is simple for every $n \geq 5$. We omit the proof of this fact.

5 The Sylow Theorems

The Sylow Theorems provide a partial converse to Lagrange's Theorem. We start with a definition.

Definition 5.1.
Let p be a prime number. A finite group G of cardinality p^t for some $t \geq 1$ is called a (finite) *p-group*.

Let G be a finite group.

1. Subgroups of G which are p-groups are called *p-subgroups*.
2. If $|G| = p^r m$ with $\gcd(p, m) = 1$, then a p-subgroup of G has cardinality p^i with $1 \leq i \leq r$. A subgroup of G of cardinality p^r is called a *Sylow p-subgroup of G* or a *p-Sylow subgroup of G*.
3. The set of Sylow p-subgroups of G will be denoted $\mathrm{Syl}_p(G)$ and the cardinality of $\mathrm{Syl}_p(G)$, which is the number of Sylow p-subgroups of G, will be denoted

$$n_p(G) = |\operatorname{Syl}_p(G)|.$$

Example 5.1.

1. The group S_3 has $6 = 2 \times 3$ elements

$$S_3 = \{1, (123), (132), (12), (13), (23)\}.$$

The unique Sylow 3-subgroup of S_3 is

$$\langle (123) \rangle = \{1, (123), (132)\}$$

[this is the only subgroup of S_3 of cardinality 3], and the Sylow 2-subgroups of S_3 are

$$\{1, (12)\} \qquad \{1, (13)\} \qquad \text{and} \qquad \{1, (23)\}.$$

Thus $n_3(S_3) = 1$ and $n_2(S_3) = 3$.

2. The group D_{12} has $12 = 2^2 \times 3$ elements. It has three Sylow 2-subgroups, namely $\{1, r^3, s, sr^3\}$, $\{1, r^3, sr, sr^4\}$, and $\{1, r^3, sr^2, sr^5\}$, each of cardinality 2^2. It has many 2-subgroups: the Sylow 2-subgroups above, five subgroups $\langle sr^i \rangle$ for $i = 0, \ldots, 5$ and the subgroup $\langle r^3 \rangle$.
It has exactly one Sylow 3-subgroup, namely $\{1, r^2, r^4\}$. This is its only 3-subgroup.
None of the other subgroups of D_{12} are p-subgroups of D_{12}. For instance, $\langle r \rangle$, of cardinality 6, is not a p-subgroup of D_{12}, for any prime p.

Theorem 5.1 (The Sylow Theorems).
Let G be a finite group of cardinality $|G| = p^r m$ with $r \geq 1$ and $\gcd(p, m) = 1$.

(Sylow I) *Sylow p-subgroups of G exist, i.e. $\mathrm{Syl}_p(G) \neq \emptyset$.*

(Sylow II) *The Sylow p-subgroups of G are conjugate: if P_1 and P_2 are Sylow p-subgroups of G, then there exists $g \in G$ such that $P_1 = gP_2g^{-1}$.*

(Sylow III)

- *The number $n_p(G)$ of Sylow p-subgroups of G satisfies $n_p(G) \equiv 1 \mod (p)$.*
- *The number $n_p(G)$ of Sylow p-subgroups of G divides $m = \frac{|G|}{p^r}$.*
- *Moreover, $n_p(G) = |G : N_G(P)|$.*

We will extend Sylow I with Theorem 5.5, which states that every p-subgroup of G and every element of order a power of p is contained in a Sylow p-subgroup of G.

In order to prove the Sylow Theorems we add some complements to group actions in the case of an action of a p-group.

5.1 Action of finite p-groups

Let $(\cdot) : H \times X \to X$ be a group action where H is a p-group and X is a finite set. The Orbit-Stabiliser Theorem gives

$$|\operatorname{orb}(x)| = \frac{|H|}{|\operatorname{Stab}(x)|},$$

for any $x \in X$. In particular the following cases occur:

- either $|\operatorname{orb}(x)| > 1$ in which case $|\operatorname{orb}(x)|$ is divisible by p, since $|H|$ is a power of p and $|\operatorname{orb}(x)|$ divides $|H|$;
- or $|\operatorname{orb}(x)| = 1$. In this case $\operatorname{orb}(x) = \{x\}$ and we call x a *fixed point* of the given action of H.

We know

$$X = \bigcup_{1 \le i \le l} \operatorname{orb}(x_i)$$

is a finite union of disjoint orbits. Counting the number of elements of X as the sum of the cardinalities of the orbits $\operatorname{orb}(x_i)$ we have:

$$|X| = \sum_{i=1}^{l} |\operatorname{orb}(x_i)|.$$

Each $\operatorname{orb}(x_i)$ contributes to $|X|$ either by a power of p if x_i is not a fixed point of the action, or by 1 if x_i is a fixed point. In particular reducing modulo p, $|\operatorname{orb}(x_i)|$ reduces to 0 if x_i is not a fixed point, and we find that $|X|$ is congruent modulo p to the sum of 1's over the set of all fixed points of the action which is precisely the number of fixed points. This leads to the following important congruence modulo p for an action of a finite p-group.

Theorem 5.2 (Fixed Point Congruence)**.**
Let H be a finite p-group acting on a finite set X and $\operatorname{Fix}_H(X)$ the subset of fixed points of X under this action. Then

$$|X| \equiv |\operatorname{Fix}_H(X)| \mod (p).$$

Corollary 5.3.
Let H be a finite p-group acting on a finite set X. If $|X|$ is not divisible by p then there is at least one fixed point in X. If $|X|$ is divisible by p, then the number of fixed points is divisible by p (possibly equals 0).

Before embarking on the proof of Sylow I, we record the following useful lemma.

Lemma 5.4.
Let G be a group of cardinality $p^r m$. Let H be a subgroup of G of cardinality p^i, for some $1 \le i < r$. Then there exists a subgroup H' of G such that H is a subgroup of H' and $|H' : H| = p$ and hence $|H'| = p^{i+1}$.

Proof. Consider the left multiplication action

$$H \times G/H \to G/H$$

$$(h, gH) \mapsto hgH.$$

By the Fixed Point Congruence theorem, we have

$$|G/H| \equiv |\operatorname{Fix}_H(G/H)| \mod (p). \tag{*}$$

Now $gH \in \operatorname{Fix}_H(G/H)$ if and only if for all $h \in H$, we have $hgH = gH$. Equally, for all $h \in H$, $g^{-1}hg \in H$. Equally, $g^{-1}Hg \subset H$. Equally, $g^{-1}Hg = H$ (as $|g^{-1}Hg| = |H|$). Now,

$$g^{-1}Hg = H \iff gHg^{-1} = H \iff g \in N_G(H).$$

Thus,

$$\operatorname{Fix}_H(G/H) = \{gH : g \in N_G(H)\} = N_G(H)/H$$

and the above congruence (*) becomes

$$|G/H| \equiv |N_G(H)/H| \mod (p). \tag{**}$$

Now if $|H| = p^i$ with $1 \leq i < r$ then p divides $|G/H| = \frac{|G|}{|H|} = \frac{p^r m}{p^i} = p^{r-i}m$ hence divides $|N_G(H)/H|$ by (**). By Cauchy's Theorem $N_G(H)/H$ has a subgroup of order p which is of the form H'/H (as are all subgroups of $N_G(H)/H$) where H' is a subgroup of $N_G(H)$ containing H. Thus $|H' : H| = |H'/H| = \frac{|H'|}{|H|} = p$ which implies $|H'| = p|H| = pp^i = p^{i+1}$. □

Proof of (Sylow I). We show a stronger result namely that given any $1 \leq i \leq r$ there exists a subgroup of G of cardinality p^i. We prove this by induction on i. The case $i = 1$ is already established by Cauchy's Theorem.

Suppose that H is a subgroup of G of cardinality p^i. By Lemma 5.4, there exists a subgroup H' of G of cardinality p^{i+1}. The result follows by induction. □

Theorem 5.5 (Extension of Sylow I)**.**
Using Lemma 5.4, we may easily prove the following:

- *If H is a p-subgroup of G, then there exists a Sylow p-subgroup P of G such that $H \subset P$.*
- *An element of G whose order is a power of p is contained in a Sylow p-subgroup of G.*

Proof. For the first result, follow the proof of Sylow I above, but starting with H rather than a subgroup of cardinality p. For the second result, let x be an element of order p^i, for some $1 \leq i < r$. Then $x \in \langle x \rangle$. By the first part of this result, $\langle x \rangle$ belongs to some Sylow p-subgroup of G. □

Proof of (Sylow II). Let P and Q be Sylow subgroups of G, we show there exists $g \in G$ such that $Q = gPg^{-1}$.

Consider the action

$$Q \times G/P \to G/P$$

$$(g', gP) \mapsto g'gP.$$

Since Q is a finite p-group we have the Fixed Point Congruence

$$|G/P| \equiv |\operatorname{Fix}_Q(G/P)| \mod (p). \qquad (*)$$

As P is a Sylow p-subgroup of G, $|G/P| = m$ is not 0 modulo p, hence $|\operatorname{Fix}_Q(G/P)|$ is not congruent to 0 modulo p by (*). In particular, $|\operatorname{Fix}_Q(G/P)| \geq 1$ and $\operatorname{Fix}_Q(G/P) \neq \emptyset$.

Let $gP \in \operatorname{Fix}_Q(G/P)$, then $g'gP = gP$ for all $g' \in Q$. The latter implies $g'g \in gP$ for all $g' \in Q$, which implies $g' \in gPg^{-1}$ for all $g' \in Q$ hence $Q \subset gPg^{-1}$. This implies the equality $Q = gPg^{-1}$ since both Q and gPg^{-1} have the same cardinality p^r. □

Proof of (Sylow III). First, we prove $n_p(G) \equiv 1 \mod (p)$. Consider the action of a Sylow p-subgroup Q of G on $\operatorname{Syl}_p(G)$ by conjugation:

$$Q \times \operatorname{Syl}_p(G) \to \operatorname{Syl}_p(G)$$

$$(g, P) \mapsto gPg^{-1}.$$

The cardinality of $\operatorname{Syl}_p(G)$ is $n_p(G)$. Since Q is a finite p-group we have the Fixed Point Congruence

$$|\operatorname{Syl}_p(G)| \equiv |\operatorname{Fix}_Q(\operatorname{Syl}_p(G))| \mod (p). \qquad (*)$$

Fixed points of $\operatorname{Syl}_p(G)$ under the action of Q are Sylow p-subgroups P such that $gPg^{-1} = P$ for all $g \in Q$, one such P is Q itself. For all such P we have $Q \subset N_G(P)$ by the previous equality, and also $P \subset N_G(P)$. Thus both Q and P are Sylow p-subgroups of $N_G(P)$. Applying (Sylow II) to the group $N_G(P)$ we find that P and Q are conjugate in $N_G(P)$. Since P is normal in $N_G(P)$ the only subgroup of $N_G(P)$ that is conjugate to P is P itself and we obtain $Q = P$. Thus $\operatorname{Fix}_Q(\operatorname{Syl}_p(G)) = \{Q\}$ and $|\operatorname{Fix}_Q(\operatorname{Syl}_p(G))| = 1$, the congruence (*) above shows that $n_p(G) = |\operatorname{Syl}_p(G)| \equiv 1 \mod (p)$.

To show $n_p(G) = |G : N_G(P)|$, consider the action of G on $\operatorname{Syl}_p(G)$ by conjugation:

$$G \times \operatorname{Syl}_p(G) \to \operatorname{Syl}_p(G)$$

$$(g, P) \mapsto gPg^{-1}.$$

Let P be a chosen Sylow p-subgroup of G (this exists by Sylow I). Since all Sylow p-subgroups of G are conjugate by (Sylow II), this group action has only one orbit which is necessarily equal to $\operatorname{Syl}_p(G) = \operatorname{orb}(P)$. By the Orbit–Stabiliser Theorem,

$$n_p(G) = |\operatorname{Syl}_p(G)| = |\operatorname{orb}(P)| = \frac{|G|}{|\operatorname{Stab}_G(\{P\})|} = \frac{|G|}{|N_G(P)|} = |G : N_G(P)|.$$

Now P is a subgroup of $N_G(P)$, so

$$p^r m = |G| = |G : N_G(P)|\,|N_G(P)| = n_p(G)|N_G(P) : P|\,|P| = n_p(G)\,|N_G(P) : P|\,p^r,$$

whence $m = |N_G(P) : P| n_p(G)$, so $n_p(G)$ divides m. □

Corollary 5.6.
Let P be a Sylow p-subgroup of a finite group G. Then P is the unique Sylow p-subgroup of G, i.e. $n_p(G) = 1$, if and only if P is a normal subgroup of G.

Proof. Note that every conjugate gPg^{-1} of P is a Sylow p-subgroup, and the converse holds by Sylow II. Thus

$$\mathrm{Syl}_p(G) = \{gPg^{-1} : g \in G\}.$$

Now $n_p(G) = 1$, i.e. $\mathrm{Syl}_p(G) = \{P\}$ if and only if $gPg^{-1} = P$ for every $g \in G$, which is equivalent to P being a normal subgroup. □

Example 5.2.
(1) A finite abelian group A has a unique Sylow p-subgroup $A_p \subset A$ for each prime number p dividing its cardinality since all subgroups of A are normal. The subgroup A_p consists of all elements $x \in A$ whose order is a power of p.

If $G = \mathbb{Z}/p^r\mathbb{Z} \times \mathbb{Z}/q^s\mathbb{Z}$ with p and q distinct primes then $\mathbb{Z}/p^r\mathbb{Z} \times \{0\}$ is the Sylow p-subgroup of G and $\{0\} \times \mathbb{Z}/q^s\mathbb{Z}$ is the Sylow q-subgroup of G.

(2) $|A_4| = 12 = 3 \times 2^2$. The alternating group A_4 has a unique Sylow 2-subgroup: $V_4 = \{1, (12)(34), (13)(24), (14)(23)\}$, it is a normal subgroup of A_4. A Sylow 3-subgroup of A_4 has cardinality 3 and is generated by a 3-cycle, there are four Sylow 3-subgroups: $\langle(123)\rangle$, $\langle(134)\rangle$, $\langle(124)\rangle$, $\langle(234)\rangle$ in A_4. Note that $4 \equiv 1 \mod (3)$.

(3) $|S_4| = 24 = 3 \times 2^3$. Prove that $n_2(S_4) = 3$, $n_3(S_4) = 4$, and find all the Sylow 3-and Sylow 2-subgroups of S_4.

The Sylow Theorems are useful in classifying groups of small cardinality and groups whose cardinality has few prime factors.

Groups of order pq, where p and q are distinct primes

Suppose $|G| = pq$ for primes p and q with $p < q$. Let $P \in \mathrm{Syl}_p(G)$ and $Q \in \mathrm{Syl}_q(G)$. We show that Q is normal in G; i.e. $n_q(G) = 1$, and P is normal in G if p doesn't divide $q - 1$. We have $n_q(G) = 1 + kq$ for some $k \geq 0$, $n_q(G)$ divides p and $p < q$. This forces $k = 0$. Since $n_q(G) = 1$, Q is the unique Sylow q-subgroup of G and is a normal subgroup of G. Since $n_p(G)$ divides q we have $n_p(G) = 1$ or q. Further $n_p(G) \equiv 1 \mod (p)$. So if q is not congruent to 1 modulo p; meaning p doesn't divide $q - 1$, then $n_p(G) = 1$ necessarily and P is a normal subgroup of G.

For example if $p = 5$ and $q = 13$ then a finite group with cardinality $65 = 5 \times 13$ has a unique Sylow 5-subgroup and a unique Sylow 13-subgroup since 13 is not congruent to 1 modulo 5.

Suppose $p = 3$ and $q = 7$. A finite group G of cardinality $21 = 3 \times 7$ has a unique Sylow 7-subgroup H which is normal in G. Further $n_3(G)$ is congruent to 1 modulo 3 and divides 7 hence equals either 1 or 7.

Groups of order 12

Let G be a finite group of order $12 = 3 \times 2^2$. We show that either G has a normal Sylow 3-subgroup or G is isomorphic to A_4 (in the latter case G has a normal Sylow 2-subgroup). Suppose $n_3(G) \neq 1$ and let $P \in \mathrm{Syl}_3(G)$. Since $n_3(G)$ divides 4, and $n_3(G)$ is congruent to 1 modulo 3, we must have $n_3(G) = 4$. As $|G| = 3 \times 2^2$, the Sylow 3-subgroups have cardinality 3, so the intersection of distinct Sylow 3-subgroups is the trivial group. Each Sylow 3-subgroup contains two elements of order 3. So G contains 2×4 elements of order 3. Since $|G : N_G(P)| = n_3(G) = 4$, by (Sylow III), we must have $N_G(P) = P$ as $P \subset N_G(P)$ and they have the same cardinality: $|N_G(P)| = \frac{|G|}{|G:N_G(P)|}$.

Now G acts on its 4 Sylow 3-subgroups $\{P_1, P_2, P_3, P_4\}$ by conjugation, so this action affords a representation

$$\phi : G \to S_4.$$

The kernel K of this action is the subgroup of G which normalises all Sylow 3-subgroups of G, i.e.

$$K = \bigcap_{i=1}^{4} N_G(P_i) = \bigcap_{i=1}^{4} P_i.$$

In particular $K = \{1\}$, since $P_i \cap P_j = \{1\}$ if $i \neq j$, i.e. ϕ is injective and G is isomorphic to its image $\phi(G) \subset S_4$. Since G has 8 elements of order 3 and there are precisely 8 elements of order 3 in S_4, all contained in A_4, it follows that $\phi(G)$ intersects A_4 in a subgroup of order at least 8. Since both groups have order 12 it follows that $\phi(G) = A_4$, so that G is isomorphic to A_4.

Groups of order p^2q, where p and q are distinct primes

Suppose $|G| = p^2q$ for distinct primes p and q. We show G has a normal Sylow subgroup for either p or q. Let $P \in \mathrm{Syl}_p(G)$ and $Q \in \mathrm{Syl}_q(G)$. Consider the case $p > q$. Since $n_p(G)$ divides q and $n_p(G) = 1 + kp$ for some $k \geq 0$, we must have $n_p(G) = 1$ and P is a normal subgroup of G in this case. Consider now the case $p < q$. If $n_q(G) = 1$, Q is normal in G. Assume therefore that $n_q(G) > 1$, i.e. $n_q(G) = 1 + tq$ for some $t > 0$. Now $n_q(G)$ divides p^2 so $n_q(G)$ equals p or p^2. Since $q > p$ we can not have $n_q(G) = p$ so $n_q(G) = 1 + tq = p^2$. Thus

$$tq = p^2 - 1 = (p-1)(p+1).$$

Since q is prime, either q divides $p - 1$ or q divides $p + 1$. The former is impossible since $q > p$ so the latter holds. Since $q > p$ and q divides $p + 1$ we must have $q = p + 1$ which only holds if $p = 2$, $q = 3$, and $|G| = 12$. The result now follows from the previous example which shows G is isomorphic to A_4 in this case and the Sylow 2-subgroup of G is normal in G.

6 Polynomials

Definition 6.1.
A *polynomial* with coefficients in $\mathbb{Q}$ is an infinite sequence

$$(a_0, a_1, \ldots, a_n, \ldots) \qquad a_i \in \mathbb{Q}$$

satisfying the following: $\exists N \geq 0$ such that $a_i = 0, \forall i \geq N$.

Further

$$(a_0, a_1, \ldots, a_n, \ldots) = (b_0, b_1, \ldots, b_n, \ldots) \iff a_i = b_i \qquad \forall i \geq 0.$$

We define addition and multiplication of polynomials by the following rules:

$$(a_0, a_1, \ldots, a_n, \ldots) + (b_0, b_1, \ldots, b_n, \ldots) = (c_0, c_1, \ldots, c_n, \ldots)$$

where

$$c_i = a_i + b_i \qquad \forall i \geq 0$$

and

$$(a_0, a_1, \ldots, a_n, \ldots) \times (b_0, b_1, \ldots, b_n, \ldots) = (d_0, d_1, \ldots, d_n, \ldots)$$

where

$$d_i = \sum_{s+t=i} a_s b_t \qquad \forall i \geq 0.$$

Example 6.1.

$$(1, \frac{1}{2}, -\frac{1}{3}, 0, \ldots) + (0, 2, \frac{1}{4}, 1, 0, \ldots) = (1, \frac{5}{2}, -\frac{1}{12}, 1, 0, \ldots),$$

and

$$(1, 2, -1, 0, \ldots) \times (1, 3, 0, 0, \ldots) = (1, 5, 5, -3, \ldots).$$

Remark (Notations).

- Let $X = (0, 1, 0, \ldots)$ and $n \geq 1$ then

  $$X^n = X \times \ldots \times X = (0, 0, \ldots, 1, 0, \ldots)$$

  where the 1 is shifted to the n-th digit.
  Further for all $i, j \geq 0$

  $$X^i X^j = X^{i+j}$$

  (prove the above facts, for the first use induction on n).
- For $a \in \mathbb{Q}$ the polynomial

  $$a = (a, 0, \ldots, 0, \ldots)$$

  is called a *constant* polynomial and denoted simply a.
- Every polynomial $(a_0, a_1, \ldots, a_n, 0, \ldots)$, with $a_i = 0$ for all $i \geq n + 1$, can be written as

  $$(a_0, 0, \ldots) + (0, a_1, 0, \ldots) + (0, 0, a_2, 0, \ldots) + \ldots + (0, \ldots, 0, a_n, 0, \ldots)$$

  which; since

  $$(0, 0, \ldots, 0, a, 0 \ldots) = (a, 0, \ldots, 0, \ldots)(0, 0, \ldots, 0, 1, 0, \ldots),$$

equals

$$(a_0, 0, \ldots) + (a_1, 0, \ldots)(0, 1, 0, \ldots) + (a_2, 0, \ldots)(0, 0, 1, 0, \ldots) + \ldots + (a_n, 0, \ldots)(0, \ldots, 0, 1, 0, \ldots)$$
$$= a_0 + a_1(0, 1, 0, \ldots) + a_2(0, 0, 1, 0, \ldots) + \ldots + a_n(0, \ldots, 0, 1, 0, \ldots)$$

which we denote

$$f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n.$$

The $\{a_i\}_{0\leq i\leq n}$ are called the *coefficients* of f. If $a_n \neq 0$, then n is called the *degree* of the polynomial f and denoted $\deg(f)$. The coefficient a_n is called the *leading coefficient* of f, and a_0 is called the *constant coefficient* of f.

- We write

$$\mathbb{Q}[X] = \left\{ f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n \ : \ n \geq 0, \quad a_i \in \mathbb{Q} \right\}$$

and call it the set of polynomials with rational coefficients. Identifying the set of constant polynomials with $\mathbb{Q}$ we can view $\mathbb{Q}$ as a subset of $\mathbb{Q}[X]$.

- The constant polynomial $a_0 = 0$ is called the *zero* polynomial.

Note that $\mathbb{Q}[X]$ is endowed with two operations $+$ and $\times$ as defined above which are the addition and multiplication of polynomials.

Lemma 6.1.
Given polynomials $f, g \in \mathbb{Q}[X]$ we have

$$\deg(f + g) \leq \max(\deg(f), \deg(g)),$$

with equality if $\deg(f) \neq \deg(g)$, and

$$\deg(f \times g) = \deg(f) + \deg(g)$$

if f and g are both non zero polynomials.

Proof. Exercise. □

Proposition 6.2.
$(\mathbb{Q}[X], +)$ is an abelian group.

Proof. First it follows from the definition of addition of polynomials that it is commutative since the addition of rationals is. Associativity of addition of polynomials also follows easily from the associativity of addition of rationals. The neutral element is the constant polynomial $0 = (0, 0, \ldots, 0, \ldots)$, also called the zero polynomial: $f(X) + 0 = f(X)$, $\forall f \in \mathbb{Q}[X]$. Further, given $f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n$, then $-f(X) = (-a_0) + (-a_1)X + (-a_2)X^2 + \ldots + (-a_n)X^n$ satisfies $f(X) + (-f(X)) = 0$ hence is the inverse of $f(X)$. □

Definition 6.2.
Let $f, g \in \mathbb{Q}[X]$. We say that g divides f if $\exists h \in \mathbb{Q}[X]$ such that

$$f(X) = g(X)h(X).$$

Example 6.2.
The polynomial $1+X+X^2$ divides $2+3X+3X^2+X^3$ since $(2+X)(1+X+X^2) = 2+3X+3X^2+X^3$.

Lemma 6.3.
Let $u \in \mathbb{Q}[X]$. Then u divides the constant polynomial 1 if and only if u is a non zero constant polynomial, i.e. $u \in \mathbb{Q} \setminus \{0\}$. Such u is called a unit of $\mathbb{Q}[X]$.

Proof. Assume $0 \neq u = a_0 \in \mathbb{Q}$ is a non zero constant polynomial. Let $v = \frac{1}{a_0}$ then $uv = 1$ and u divides 1.

Conversely let $f(X) = a_0 + a_1X + \ldots a_nX^n$ be a unit with $n = \deg(f)$, i.e. $a_n \neq 0$. There exists a polynomial $g(X) = b_0 + b_1X + \ldots b_mX^m$ of degree m; thus $b_m \neq 0$, such that $f(X)g(X) = a_0b_0 + \ldots + a_nb_mX^{n+m} = 1$.

In particular $a_nb_m \neq 0$ and $\deg(f(X)g(X)) = n + m = \deg(1) = 0$. Hence $n = m = 0$ and $u = a_0 \neq 0$ since $a_0b_0 = 1 \neq 0$. □

Definition 6.3.
Let $f \in \mathbb{Q}[X]$ be a non constant polynomial, i.e. $f \in \mathbb{Q}[X] \setminus \mathbb{Q}$ hence $\deg(f) \geq 1$. The notation $\mathbb{Q}[X] \setminus \mathbb{Q}$ means those elements in $\mathbb{Q}[X]$ which are not in $\mathbb{Q}$.

We say that f is *irreducible* if whenever $f(X) = g(X)h(X)$, where $g(X)$ and $h(X)$ are polynomials in $\mathbb{Q}[X]$, then either g or h is a unit, i.e. either g or h is a non zero constant polynomial.

In other words, f is irreducible if $f(X)$ doesn't factor as the product $g(X)h(X)$ of two polynomials in $\mathbb{Q}[X]$ with $\deg(g(X)) \geq 1$ and $\deg(h(X)) \geq 1$.

Example 6.3.

1. A polynomial $f = a_0 + a_1X \in \mathbb{Q}[X]$ of degree 1 (thus $a_1 \neq 0$) is necessarily irreducible. Indeed $f = gh$ implies $\deg(g) = 0$ or $\deg(h) = 0$, as $\deg(f) = 1 = \deg(g) + \deg(h)$. Thus either g, or h, is a constant polynomial, and is non zero (otherwise $f = gh = 0$). Hence either g or h is a unit.
2. The polynomial $X^2 - 3$ is irreducible. Indeed assume

$$X^2 - 3 = (aX + b)(cX + d) = acX^2 + (ad + bc)X + bd$$

with a and c both non zero.
Equating we get $ac = 1$, $ad + bc = 0$, and $bd = -3$. Hence $\{ad + bc = \frac{1}{c}d - \frac{3}{d}c = 0\} \Longrightarrow \{d^2 - 3c^2 = 0\} \Longrightarrow \{d^2 = 3c^2\} \Longrightarrow \{d = c = 0\}$, where the last implication follows from the fact that 3 is not a square of a rational number. But this contradicts $ac = 1$.

Theorem 6.4 (Euclidean division. The division algorithm).
Let $f, g \in \mathbb{Q}[X]$ with $g \neq 0$. Then $\exists h, r \in \mathbb{Q}[X]$ unique such that

$$f = hg + r$$

where either $r = 0$ (in which case g divides f), or $r \neq 0$ and $\deg(r) < \deg(g)$.

Proof. Write $f(X) = a_0 + a_1X + \ldots + a_mX^m$, $g(X) = b_0 + b_1X + \ldots + b_nX^n$ with $a_m \neq 0$ and $b_n \neq 0$. Assume for simplicity that $m \geq n$ and set $d = m - n \geq 0$ [if $m < n$ take $h(X) = 0$ and $r(X) = f(X)$)]. We argue by induction on d.

If $d = 0$, i.e. $m = n$, then $f(X) = \frac{a_m}{b_n} g(X) + r(X)$ where $r(X) = f(X) - \frac{a_m}{b_n} g(X) = a_0 + \dots a_m X^m - b_0 \frac{a_m}{b_n} - \dots - \frac{a_m b_n}{b_n} X^n$ which has degree $< n$.

Assume now that $k = d > 0$ and the theorem holds for $0 \leq d \leq k-1$. Set $f_1(X) = f(X) - \frac{a_m}{b_n} X^{m-n} g$. then $\deg(f_1) < \deg(f)$ (check this fact). If $f_1 = 0$ then $f(X) = \frac{a_m}{b_n} X^{m-n} g(X)$ and the theorem holds with $h(X) = \frac{a_m}{b_n} X^{m-n}$ and $r(X) = 0$. If $f_1(X) \neq 0$ then by the induction hypothesis $f_1(X) = h_1(X)g(X) + r_1(X)$ where either $r_1(X) = 0$ or $r_1 \neq 0$ and $\deg(r_1) < \deg(g)$. Thus $f(X) = h_1(X)g(X) + r_1(X) + \frac{a_m}{b_n} X^{m-n} g(X) = (h_1(X) + \frac{a_m}{b_n} X^{m-n}) g(X) + r_1(X)$ and the theorem holds with $h(X) = h_1(X) + \frac{a_m}{b_n} X^{m-n}$ and $r(X) = r_1(X)$.

Next, we show the unicity of h and r. Assume $f(X) = g(X)h(X) + r(X) = g(X)h'(X) + r'(X)$ with (h, r) and (h', r') satisfying the conclusion of the Theorem. If $r = 0$ then $f = gh = g'h' + r'$ hence $g(h - h') = r'$ which implies $r' = 0$ [otherwise $\deg(r') \geq \deg(g)$. Likewise $r' = 0$ implies $r = 0$]. Assume both r and r' are non zero, in which case $r = f - gh$ and $r' = f - gh'$ have both degree less than $\deg(g)$. The difference of these two polynomials $r - r' = gh' - gh = g(h' - h)$ has also degree less than that of g. If $h' - h = 0$ then $r' - r = 0$ and we have $h = h'$, $r = r'$. On the other hand $\deg(r - r') = \deg(g) + \deg(h - h')$ if $h - h'$ is a non zero polynomial, which contradicts $\deg(r - r') < \deg(g)$. Thus $h - h' = 0$ and $r - r' = 0$ necessarily which implies $h = h'$ and $r = r'$. □

Example 6.4.
Let $f(X) = X^6 + 2X^5 + X^4 + 3X$ and $g(X) = X^2 + 1$. Then $X^6 + 2X^5 + X^4 + 3X = (X^4 + 2X^3 - 2X)(X^2 + 1) + 5X$, thus $h(X) = X^4 + 2X^3 - 2X$ and $r(X) = 5X$ in this case.

Theorem 6.5 (Unique Factorisation of Polynomials).
Let $f(X) \in \mathbb{Q}[X]$ be a non zero polynomial which is not a unit, i.e. $f(X) \in \mathbb{Q}[X] \setminus \mathbb{Q}$, or equivalently $\deg(f(X)) \geq 1$.

Then it holds

$$f = g_1 g_2 \dots g_r$$

(product of polynomials) with g_i irreducible for all $1 \leq i \leq r$. Further if

$$f = h_1 h_2 \dots h_s$$

is another such factorisation, with h_i irreducible for $1 \leq i \leq s$, then $r = s$ and after rearranging the $\{g_i\}$ one has $g_i = u_i h_i$ where u_i is a unit.

Proof. The existence of a factorisation $f = g_1 g_2 \dots g_r$ as above follows by induction on the degree of f: if f is irreducible then we are done. Otherwise $f = qm$ where $q, m \in \mathbb{Q}[X]$ are non zero, non units, with $1 \leq \deg(q) < \deg(f)$ and $1 \leq \deg(m) < \deg(f)$. By induction both q and m have a factorisation as a product of irreducible polynomials which induces a similar factorisation for f.

The second assertion follows from the fact that if an irreducible polynomial h divides the product $f_1 f_2$ of two polynomials then either h divides f_1 or h divides f_2 (more details on the uniqueness of factorisation will be seen in §10). □

Definition 6.4.

1. Let $f(X) = a_0 + a_1 X + \dots + a_n X^n \in \mathbb{Q}[X]$ and $c \in \mathbb{Q}$. We define the *value* $f(c)$ of f at c by

$$f(c) = a_0 + a_1 c + \dots + a_n c^n \in \mathbb{Q}.$$

Thus f can be viewed as a function

$$f : \mathbb{Q} \to \mathbb{Q}$$

defined by

$$c \mapsto f(c).$$

If $g \in \mathbb{Q}[X]$ is another polynomial then (check this fact)

$$(f \times g)(c) = f(c)g(c), \quad (f + g)(c) = f(c) + g(c).$$

2. We say $c \in \mathbb{Q}$ is a *root* of f if $f(c) = 0$. In this case we say f has a root in $\mathbb{Q}$.

Example 6.5.

1. A polynomial $f(X) = a_0 + a_1X \in \mathbb{Q}[X]$ of degree 1 (thus $a_1 \neq 0$) has necessarily a root in $\mathbb{Q}$. Indeed $f(-\frac{a_0}{a_1}) = 0$ and $-\frac{a_0}{a_1}$ is a root of f.
2. The polynomial $X^2 - 2$ has no root in $\mathbb{Q}$: a root would be equal either to $\sqrt{2}$ or $-\sqrt{2}$ which are both not rational numbers.

Lemma 6.6.
Let $f \in \mathbb{Q}[X]$, $\deg(f) \geq 1$. If $c \in \mathbb{Q}$ is a root of f then the polynomial $X - c$ divides f.

Proof. By the division algorithm $\exists h, r \in \mathbb{Q}[X]$, unique, with

$$f(X) = h(X)(X - c) + r(X)$$

and either $r = 0$ or $\deg(r) < \deg(X - c) = 1$. Thus $r \in \mathbb{Q}$ is a constant.

Further

$$f(c) = h(c)(c - c) + r = 0 + r = 0$$

hence $r = 0$, and $f(X) = h(X)(X - c)$. □

Lemma 6.7.
Let $f \in \mathbb{Q}[X]$ with $\deg(f) = n \geq 1$. Then f has at most n roots in $\mathbb{Q}$.

Proof. We argue by induction on $n = \deg(f)$. If $n = 1$ then $f(X) = a_0 + a_1X$ and $-\frac{a_0}{a_1}$ is the only root of f.

Suppose the lemma is true for polynomials of degree $< n$. Assume $c \in \mathbb{Q}$ is a root of f. Then by Lemma 6.6 there exists $g \in \mathbb{Q}[X]$ with

$$f(X) = (X - c)g(X)$$

and $\deg g = n - 1$. Assume $c' \neq c$, $c' \in \mathbb{Q}$, is another root of f. Then

$$f(c') = (c' - c)g(c') = 0,$$

hence $g(c') = 0$ necessarily and c' is a root of g since $c - c' \neq 0$. By induction there are at most $n - 1$ such c', hence at most n roots of f. □

In a similar way we could have defined the set of polynomials $\mathbb{C}[X]$ with coefficients in $\mathbb{C}$ (and indeed in any field, see §7 for the definition of a field)

$$\mathbb{C}[X] = \{f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n \ : \ n \geq 0, \ a_i \in \mathbb{C}\}.$$

Polynomials with coefficients in $\mathbb{C}$ share several of the properties of polynomials in $\mathbb{Q}[X]$: addition, multiplication, Euclidean division, unique factorisation. The following result is fundamental in Algebra.

Theorem 6.8 (Fundamental Theorem of Algebra)**.**
Let $f \in \mathbb{C}[X]$ *be a polynomial with* $\deg(f) \geq 1$. *Then* $\exists c \in \mathbb{C}$ *with* $f(c) = 0$.

7 Rings

Definition 7.1.
A (commutative) ring $(R, +, \times)$ consists of a nonempty set R endowed with two operations: addition $+$, and multiplication $\times$, satisfying the following axioms.

(A1) $(R, +)$ is an abelian group. We denote the neutral element of $+$ by 0_R or simply 0, and the inverse of an element a for addition by $-a$. Thus

$$a + (-a) = 0.$$

(A2) $a \times b = b \times a \quad \forall a, b \in R$ (multiplication is commutative).

(A3) $(a \times b) \times c = a \times (b \times c) \quad \forall a, b, c \in R$ (associativity of multiplication).

(A4) $a \times (b + c) = a \times b + a \times c \quad \forall a, b, c \in R$ (distributivity of $\times$ with respect to $+$).

We say that R has an identity element (called a ring with identity) if $\exists$ an element $1_R \in R$ (or simply denoted 1) such that

$$a \times 1_R = a, \ \forall a \in R.$$

In the notations we often omit the sign $\times$ and write $ab = a \times b$. Also we denote

$$a - b = a + (-b).$$

Example 7.1.

1. $(\mathbb{Z}, +, \times)$ is a ring with identity 1.
2. $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$ are rings with identity 1.
3. For $m \geq 1$, $(\mathbb{Z}/m\mathbb{Z}, +, \times)$ is a ring with identity the class $\bar{1}$ of 1.
4. $(\mathbb{Q}[X], +, \times)$ is a ring with identity the constant polynomial $f(X) = 1$.

Lemma 7.1.
Let R be a ring. Then the followings hold.

1. $0_R a = a 0_R = 0_R, \ \forall a \in R.$
2. $(-a)b = a(-b) = -(ab), \ \forall a, b \in R.$
3. $(-a)(-b) = ab, \ \forall a, b \in R.$
4. *If R has an identity element 1_R then* $-a = (-1_R)a, \ \forall a \in R.$

Proof. See Problem Sheet 4 exercise 1. □

Definition 7.2.
Let R be a ring. A nonzero element $a \in R \setminus \{0_R\}$ is called a *zero divisor* if $\exists b \in R \setminus \{0_R\}$ such that

$$ab = 0_R.$$

The notation $R \setminus \{0_R\}$ denotes the set of nonzero elements of R.

Example 7.2.

1. $\bar{2} \in \mathbb{Z}/4\mathbb{Z}$ is a zero divisor since $\bar{2} \times \bar{2} = \bar{4} = \bar{0}$.
2. $\mathbb{Z}$ has no zero divisors since
$$\{ab = 0, a, b \in \mathbb{Z}\} \implies \{a = 0 \quad \text{or} \quad b = 0\}.$$
3. $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is a prime integer.

   Proof. Indeed if n is not a prime: $n = n_1 n_2$ with $1 < n_i < n$, hence $\bar{n} = 0 = \bar{n}_1 \bar{n}_2$ and $0 \neq \bar{n}_i$ is a zero divisor, $1 \leq i \leq 2$. Conversely assume $n = p$ is a prime integer and $\bar{a}\bar{b} = \bar{0} = \bar{p} = \overline{ab}$, then p divides ab and hence either p divides a (which is equivalent to $\bar{a} = \bar{0}$) or p divides b (which is equivalent to $\bar{b} = \bar{0}$). □
4. $\mathbb{Q}[X]$ has no zero divisors.

   Proof. We prove that the product of two nonzero polynomials in $\mathbb{Q}[X]$ is a nonzero polynomial. Suppose $f(X) = a_0 + a_1 X + \ldots + a_n X^n \neq 0$, thus $\deg(f) = n \geq 0$ and $a_n \neq 0$; and $g(X) = b_0 + b_1 X + \ldots + b_m X^m \neq 0$, thus $\deg(g) = m \geq 0$ and $b_m \neq 0$. Then
   $$f(X)g(X) = a_0 b_0 + (a_0 b_1 + a_1 b_0)X + \ldots + a_n b_m X^{n+m} \neq 0,$$
   since $a_n b_m \neq 0$. □

From now on when we assume that a ring R has an identity element 1_R, we suppose $1_R \neq 0_R$ [if $0_R = 1_R$ then all elements of R equal 0_R: $a = (1_R)a = (0_R)a = 0_R$].

Definition 7.3.
Let R be a ring with an identity element 1. An element $u \in R$ is called a *unit* if $\exists v \in R$ such that
$$uv = 1.$$
We denote v by u^{-1} and call it the *inverse* of u for multiplication.

Lemma 7.2.
Let R be a ring with identity 1. The set of units of R is denoted
$$R^\times = \{u \in R : u \text{ unit}\}.$$
The set $R^\times$ is closed under multiplication, and $(R^\times, \times)$ is an abelian group.

Proof. First $1 \in R^\times$ since $1 \times 1 = 1$, hence 1 is a unit. Let $u_1, u_2 \in R^\times$, there exists $v_1, v_2 \in R$ such that
$$u_1 v_1 = u_2 v_2 = 1.$$
Then
$$1 = (u_1 v_1)(u_2 v_2) = (u_1 u_2)(v_1 v_2)$$
hence $u_1 u_2$ is a unit and
$$(u_1 u_2)^{-1} = v_1 v_2 = u_1^{-1} u_2^{-1}.$$
Further let $u \in R^\times$, there exists $v \in R$ such that $uv = 1$ hence $v \in R^\times$ is a unit and $v = u^{-1}$ is the inverse of u. Note that the group law $\times$ of $R^\times$ is associative by the axiom (A3) of a ring. □

Example 7.3.

1. Let $u \in \mathbb{Z}$ be a unit, then $\exists v \in \mathbb{Z}$ such that $uv = 1$ which implies $u = 1$ or $u = -1$. Hence $\mathbb{Z}^\times = \{1, -1\}$.
2. $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \ : \ \gcd(a, n) = 1\}$.

   Proof. Indeed $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \exists \bar{b} \in \mathbb{Z}/n\mathbb{Z} : \bar{a}\bar{b} = \overline{ab} = \bar{1} \iff \exists k \in \mathbb{Z} : ab = nk + 1 \iff \exists k \in \mathbb{Z} : ab - nk = 1 \iff \gcd(a, n) = 1$. □
3. $(\mathbb{Q}[X])^\times = \mathbb{Q}^\times = (\mathbb{Q} \setminus \{0\}, \times)$ (see Lemma 6.3).

Definition 7.4.
A ring is called an *integral domain* if it has no zero divisors.

Example 7.4.

1. $\mathbb{Z}$ is an integral domain
2. $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is a prime integer.

Definition 7.5.
A ring F with identity is called a *field* if $F^\times = F \setminus \{0\}$, in other words F is a field if every nonzero element of F is a unit.

Example 7.5.

1. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are fields.
2. $\mathbb{Z}$ and $\mathbb{Q}[X]$ are not fields: 2 has no inverse in $\mathbb{Z}$ and X has no inverse in $\mathbb{Q}[X]$.
3. $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if $n = p$ is a prime integer.

   Proof. Indeed if $\bar{0} \neq \bar{k} \in \mathbb{Z}/n\mathbb{Z}$ then $\bar{k} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \gcd(k, n) = 1$ (see example 2 in Examples 7.3). Hence every non zero element of $\mathbb{Z}/n\mathbb{Z}$ is a unit if and only if n has no positive divisor k with $1 < k < n$, which is precisely the definition of n being a prime. □

Definition 7.6.
Let p be a prime integer. The field $\mathbb{Z}/p\mathbb{Z}$ is denoted $\mathbb{F}_p$ and called the *finite field with p element*.

Lemma 7.3.
A field F is necessarily an integral domain.

Proof. Let $a, b \in F$ with $ab = 0$. We show either $a = 0$ or $b = 0$. Suppose for example $a \neq 0$, we show that $b = 0$. Since $a \neq 0$, a is a unit of F and $\exists a^{-1} \in F$ such that $a^{-1}a = 1$. Thus $a^{-1}(ab) = (a^{-1}a)b = 1b = b = 0$. □

Definition 7.7.
A subset S of a ring R is called a *subring* if $(S, +)$ is a subgroup of $(R, +)$ and S is closed under multiplication, i.e. $a, b \in S$ implies $ab \in S$.

Example 7.6.

1. $\mathbb{Q}$ is a subring of $\mathbb{R}$, $\mathbb{R}$ is a subring of $\mathbb{C}$, $\mathbb{Q}$ is a subring of $\mathbb{Q}[X]$.
2. For $m \in \mathbb{Z}$ set $m\mathbb{Z} = \{ma : a \in \mathbb{Z}\}$. Then $m\mathbb{Z}$ is a subring of $\mathbb{Z}$ (without identity if $|m| > 1$).
3. Let $D \in \mathbb{Z}$ be a non zero square free integer, i.e. D is not divisible by a square. Equivalently, $D = p_1 p_2 \dots p_t$, $D = -1$, or $D = -p_1 p_2 \dots p_t$, where $p_1, p_2, \dots, p_t$ are distinct primes.

Set
$$\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} : a, b \in \mathbb{Q}\} \subset \mathbb{C}.$$
Then $\mathbb{Q}(\sqrt{D})$ is a subring of $\mathbb{C}$. Indeed, $0 = 0 + 0\sqrt{D} \in \mathbb{Q}(\sqrt{D})$,
$$(a_1 + b_1\sqrt{D}) + (a_2 + b_2\sqrt{D}) = (a_1 + a_2) + (b_1 + b_2)\sqrt{D} \in \mathbb{Q}(\sqrt{D}),$$
$$-(a_1 + b_1\sqrt{D}) = -a_1 + (-b_1)\sqrt{D} \in \mathbb{Q}(\sqrt{D})$$
and
$$(a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D}) = (a_1a_2 + b_1b_2D) + (a_2b_1 + a_1b_2)\sqrt{D} \in \mathbb{Q}(\sqrt{D})$$
if $a_1 + b_1\sqrt{D}, a_2 + b_2\sqrt{D} \in \mathbb{Q}(\sqrt{D})$.

- In fact $\mathbb{Q}(\sqrt{D})$ is a field. Indeed let $0 \neq a + b\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ which implies $a \neq 0$ or $b \neq 0$: if $a = b = 0$ then $a + b\sqrt{D} = 0 + 0\sqrt{D} = 0$. We have
$$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \neq 0$$
[if $a^2 - Db^2 = 0$; which is equivalent to $a^2 = Db^2$, then both a and b are non zero (if one of them is zero then the other is also zero which contradicts $a \neq 0$ or $b \neq 0$), and $D = \frac{a^2}{b^2} = (\frac{a}{b})^2$ would be a square], and
$$(a + b\sqrt{D})\frac{a - b\sqrt{D}}{a^2 - Db^2} = (a + b\sqrt{D})(\frac{a}{a^2 - Db^2} - \frac{b}{a^2 - Db^2}\sqrt{D}) = 1$$
hence $a + b\sqrt{D}$ is a unit since $(\frac{a}{a^2-Db^2} - \frac{b}{a^2-Db^2}\sqrt{D}) \in \mathbb{Q}(\sqrt{D})$.

4. With the notations in (3) above set
$$\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} : a, b \in \mathbb{Z}\} \subset \mathbb{Q}(\sqrt{D}).$$
Then $\mathbb{Z}[\sqrt{D}]$ is a subring of $\mathbb{Q}(\sqrt{D})$ (check this fact).
$$\mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$$
is called the *ring of Gaussian integers*.
5. Define
$$\mathbb{Z}[X] = \{a_0 + a_1X + \ldots + a_nX^n : a_i \in \mathbb{Z}, n \geq 0\} \subset \mathbb{Q}[X].$$
Then $\mathbb{Z}[X]$ is a subring of $\mathbb{Q}[X]$ (check this fact).

Example 7.7.

1. Show that $(\mathbb{Z}[\sqrt{-1}])^\times = \{1, -1, i, -i\}$. [Hint: Show and use the fact that if $a + ib \in \mathbb{Z}[i]$ is a unit then $|a + ib| = a^2 + b^2 = 1$.]
Show that the group $((\mathbb{Z}[\sqrt{-1}])^\times, \times)$ is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.
2. $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$.
The group of units $(\mathbb{Z}[\sqrt{2}])^\times$ is infinite. Indeed $1 + \sqrt{2}$ is a unit since
$$(\sqrt{2} - 1)(1 + \sqrt{2}) = 1.$$

Further $(1+\sqrt{2})^n$ is a unit $\forall n \geq 1$ since

$$(\sqrt{2}-1)^n(1+\sqrt{2})^n = [(\sqrt{2}-1)(1+\sqrt{2})]^n = 1^n = 1.$$

The set

$$\{(1+\sqrt{2})^n : n \geq 1\}$$

is infinite as

$$(1+\sqrt{2})^{n_1} \neq (1+\sqrt{2})^{n_2} \quad \text{if} \quad n_1 > n_2.$$

Indeed otherwise $(1+\sqrt{2})^{n_1-n_2} = 1$ which is not possible since $1+\sqrt{2} > 1$ and $n_1 - n_2 > 0$.

8 Ring Homomorphisms and Ideals

Definition 8.1.
Let R and S be rings. A map $\phi : R \to S$ is called a ring homomorphism if it satisfies

1. $\phi(a+b) = \phi(a) + \phi(b) \qquad \forall a, b \in R,$
   i.e. ϕ is a group homomorphism $\phi : (R, +) \to (S, +)$.
2. $\phi(ab) = \phi(a)\phi(b) \qquad \forall a, b \in R.$
3. $\phi(1_R) = 1_S$ if both R and S have identity.
   In particular $\phi(0_R) = 0_S$ and $\phi(-a) = -\phi(a)$ for all $a \in A$ since ϕ is a group homomorphism for $+$.
   A ring homomorphism which is a bijection is called an *isomorphism*.

Example 8.1.

1. The map
$$\phi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$$
$$a \ \mapsto \ \phi(a) = \bar{a}$$
is a ring homomorphism: $\phi(a+b) = \overline{a+b} = \bar{a} + \bar{b} = \phi(a) + \phi(b)$, $\phi(ab) = \overline{ab} = \bar{a}\bar{b} = \phi(a)\phi(b)$, and $\phi(1) = \bar{1}$.
2. The map $\phi : \mathbb{Z} \to \mathbb{Z}$ defined by $\phi(a) = 2a$ is NOT a ring homomorphism: $\phi(ab) = 2ab \neq \phi(a)\phi(b) = 2a2b = 4ab$ if a and b are both non-zero.
3. The map
$$\phi : \mathbb{Q}[X] \to \mathbb{Q}$$
$$f(X) = a_0 + a_1X + \ldots + a_nX^n \ \mapsto \ \phi(f(X)) = f(0) = a_0$$
is a ring homomorphism (check this fact).
4. The map
$$\phi : \mathbb{Q}[X] \to \mathbb{Q}(\sqrt{2})$$
$$f(X) = a_0 + a_1X + \ldots + a_nX^n \ \mapsto \ \phi(f(X)) = f(\sqrt{2}) = a_0 + a_1\sqrt{2} + \ldots + a_n(\sqrt{2})^n$$
is a ring homomorphism.

Definition 8.2.
Let $\phi : R \to S$ be a ring homomorphism. We define
$$\mathrm{Ker}(\phi) = \{r \in R : \phi(r) = 0_S\}$$
called the *kernel* of ϕ, and
$$\mathrm{Im}(\phi) = \{s \in S : \exists r \in R \text{ with } \phi(r) = s\}$$
called the image of ϕ.

Example 8.2.

1. Consider the ring homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ defined by $\phi(a) = \bar{a}$. Then $a \in \mathrm{Ker}(\phi) \iff \phi(a) = \bar{a} = \bar{0} \iff n$ divides a, thus $\mathrm{Ker}(\phi) = n\mathbb{Z}$. Further $\mathrm{Im}(\phi) = \mathbb{Z}/n\mathbb{Z}$.

2. Consider the homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}$ defined by $\phi(f(X)) = f(0)$, i.e. if $f(X) = a_0 + a_1X + \ldots + a_nX^n$ then $\phi(f(X)) = a_0$ (see Examples 8.1).
Then
$$f(X) \in \mathrm{Ker}(\phi) \iff f(0) = 0 \iff a_0 = 0.$$
Thus
$$\mathrm{Ker}(\phi) = \{f(X) = a_1X + a_2X + \ldots + a_nX^n : a_i \in \mathbb{Q}, n \geq 1\}.$$
Further $\mathrm{Im}(\phi) = \mathbb{Q}$ since $a = \phi(a + X)$ for all $a \in \mathbb{Q}$.

Lemma 8.1.
Let $\phi : R \to S$ be a ring homomorphism. Then

1. $\mathrm{Ker}(\phi)$ *is a subring of* R.
2. $Im(\phi)$ *is a subring of* S.
3. ϕ *is surjective* $\iff Im(\phi) = S$.
4. ϕ *is injective* $\iff \mathrm{Ker}(\phi) = \{0_R\}$.

Proof.

1. Let $a, b \in \mathrm{Ker}(\phi)$. Then
$$\phi(a + b) = \phi(a) + \phi(b) = 0_S + 0_S = 0_S$$
and
$$\phi(ab) = \phi(a)\phi(b) = 0_S \times 0_S = 0_S,$$
hence $a + b, ab \in \mathrm{Ker}(\phi)$.
2. If $s_1 = \phi(r_1), s_2 = \phi(r_2)$ then
$$\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2) = s_1 + s_2$$
and
$$\phi(r_1r_2) = \phi(r_1)\phi(r_2) = s_1s_2,$$
hence $s_1 + s_2, s_1s_2 \in \mathrm{Im}(\phi)$.
3. Is clear.
4. Assume $\mathrm{Ker}(\phi) = \{0_R\}$. Let $r_1, r_2 \in R$ with $\phi(r_1) = \phi(r_2) \implies \phi(r_1) + (-\phi(r_2)) = \phi(r_1) + \phi(-r_2) = \phi(r_1 + (-r_2)) = 0_S \implies 0_R = r_1 + (-r_2) \in \mathrm{Ker}(\phi) \implies 0_R + r_2 = r_2 = r_1 + (-r_2) + r_2 = r_1 + ((-r_2) + r_2) = r_1 + 0_R = r_1$, hence $r_1 = r_2$.
Conversely suppose ϕ is injective. Let $r \in \ker(\phi)$. Then $\phi(r) = \phi(0_R) = 0_S$ hence $r = 0_R$ since ϕ is injective and $\mathrm{Ker}(\phi) = \{0_R\}$.

□

Definition 8.3.
Let R be a ring. A subset $I \subset R$ is called an *ideal* if the followings hold:

1. $(I, +)$ is a subgroup of $(R, +)$.
2. I is closed under multiplication by arbitrary elements of R:
$$ab \in I \qquad \forall a \in I \qquad \forall b \in R.$$

Example 8.3.

1. Let $m \in \mathbb{Z}$ then $m\mathbb{Z} = \{ma : a \in \mathbb{Z}\}$ is an ideal of $\mathbb{Z}$: if $k \in \mathbb{Z}, ma \in m\mathbb{Z}$ then $k(ma) = m(ka) \in m\mathbb{Z}$.
2. Let R be a ring and $a \in R$. The subset

$$(a)_R = \{ab : b \in R\} \subset R$$

is an ideal of R, called the *ideal generated by* a. Indeed, $0_R = a0_R \in (a)_R$, $ab + ac = a(b + c) \in (a)_R$, and $-(ab) = a(-b) \in (a)_R$. Further if $d \in R$ is arbitrary and $ac \in (a)_R$ then $d(ac) = dac = adc = a(dc) \in (a)_R$.

Lemma 8.2.
The kernel $\mathrm{Ker}(\phi)$ *of a ring homomorphism* $\phi : R \to S$ *is an ideal of* R.

Proof. We already know that $\mathrm{Ker}\,\phi$ is a subgroup of $(R, +)$. Let $a \in \mathrm{Ker}\,\phi$ and $b \in R$. Then $\phi(ab) = \phi(a)\phi(b) = 0_S\phi(b) = 0_S$ hence $ab \in \mathrm{Ker}(\phi)$. □

- Let $\phi : R \to S$ be a ring homomorphism and I and ideal of S then the inverse image

$$\phi^{-1}(I) = \{a \in R : \phi(a) \in I\}$$

is an ideal of R. Indeed, $\phi(0_R) = 0_S \in I \implies 0_R \in \phi^{-1}(I)$. If $c = \phi(a) \in I, d = \phi(b) \in I$ then $c + d = \phi(a) + \phi(b) = \phi(a + b) \in I \implies a + b \in \phi^{-1}(I)$. If $c = \phi(a) \in I$ then $-c = \phi(-a) \in I \implies -a \in \phi^{-1}(I)$ since $-c \in I$. Finally let $a \in \phi^{-1}(I)$ and $b \in R$. Then $\phi(a) \in I$ and $\phi(ab) = \phi(a)\phi(b) \in I$ since I is an ideal.

Definition (Construction of the quotient ring R/I).

Let R be a ring and I an ideal of R. Recall the relation $\mathcal{H}$ among elements of R defined by (see Definition 2.1):

$$a \mathcal{H} b \iff b - a \in I.$$

The relation $\mathcal{H}$ is an equivalence relation (see Lemma 2.1). The equivalence class of $a \in R$ for the relation $\mathcal{H}$ is $\bar{a} = \{b \in R : b = a + c, c \in I\}$. Thus

$$\bar{a} = a + I = \{a + c : c \in I\}$$

is called the *class of* a *modulo* I [this is also the left coset $a + I$ of the subgroup $(I, +)$ of $(R, +)$]. Set

$$R/I = \{\bar{a} : a \in A\} = \{a + I : a \in R\}$$

for the set of equivalence classes of elements of R modulo I, this is the set of left cosets of the subgroup $(I, +)$ in R. Note that

$$\bar{a} = \bar{b} \iff a - b \in I.$$

Any element in $\bar{a}$ is called a *representative* of the class of a modulo I.

We define addition and multiplication among elements of R/I by the formulas

$$\bar{a} + \bar{b} = \overline{a + b} \qquad (a + I) + (b + I) = a + b + I,$$

and

$$\bar{a} \times \bar{b} = \overline{ab} \qquad (a+I)(b+I) = ab + I.$$

(Check these are well defined and independent of the choice of a representative of a given class, for $+$ see Definition 2.5.)

$(R/I, +, \times)$ is a ring (check this fact) called the *quotient ring* of R by the ideal I. The neutral element for addition is

$$0_{R/I} = \overline{0_R} = 0_R + I = I.$$

If R has an identity 1_R then R/I has an identity as well and

$$1_{R/I} = \overline{1_R} = 1_R + I$$

since $\overline{1_R} \times \bar{a} = \bar{a}$ for all $a \in A$.

Lemma 8.3.
Consider the map $\phi : R \to R/I$ defined by $\phi(a) = \bar{a}$ for $a \in R$. Then

1. *ϕ is a surjective ring homomorphism.*
2. $\mathrm{Ker}(\phi) = I$.

Proof. 1) follows immediately from the definition of ϕ (see also Proposition 2.7 where we proved ϕ is a homomorphism for $+$). For 2): $a \in \mathrm{Ker}(\phi) \iff \bar{a} = \overline{0_R} \iff a - 0_R = a \in I$ (see Proposition 2.7). □

Example 8.4.

1. Let $n \geq 1$ an integer and $I = n\mathbb{Z} = \{na : a \in \mathbb{Z}\}$ the principal ideal of $\mathbb{Z}$ generated by n. The relation $\mathcal{H}$ in this case is the relation of congruence modulo n and $\mathbb{Z}/I = \mathbb{Z}/n\mathbb{Z}$ endowed with $+$ and $\times$: $\bar{i} + \bar{j} = \overline{i+j}$, $\bar{i} \times \bar{j} = \overline{ij}$, is a ring.
2. Let $R = \mathbb{Z}[X]$ and consider the principal ideal of R generated by X^2:

$$I = (X^2)_R = \{h(X) = X^2(b_2 + b_3X + \ldots b_nX^{n-2}) \ : \ b_i \in \mathbb{Z}, n \geq 2\}.$$

$$I = \{h(X) = b_2X^2 + b_3X^3 + \ldots b_nX^n \ : \ b_i \in \mathbb{Z}, n \geq 2\}$$

Let $f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_mX^m \in \mathbb{Z}[X]$. Then $f(X) - (a_0 + a_1X) = a_2X^2 + \ldots + a_mX^m \in I$ which implies $\overline{f(X)} = \overline{a_0 + a_1X}$. Hence every class of a polynomial in $\mathbb{Z}[X]$ modulo I is represented by the class of a polynomial of degree ≤ 1 and

$$\mathbb{Z}[X]/I = \{\overline{a_0 + a_1X} : a_0, a_1 \in \mathbb{Z}\}.$$

In $\mathbb{Z}[X]/I$ we have

$$\overline{3 + 5X + X^4} = \overline{3 + 5X + X^3 - X^5},$$
$$(\overline{1 + 3X}) + (\overline{-4 + 5X}) = \overline{-3 + 8X},$$

and

$$(\overline{1 + 3X})(\overline{-4 + 5X}) = \overline{-4 - 7X + 15X^2} = \overline{-4 - 7X}.$$

3. Let $R = \mathbb{Q}[X]$ and

$$I = (X^2 - 2)_{\mathbb{Q}[X]} = \{(X^2 - 2)f(X) : f(X) \in \mathbb{Q}[X]\}.$$

Let $g(X) \in \mathbb{Q}[X]$. Then $g(X) = (X^2 - 2)f(X) + (a_0 + a_1X)$ (Euclidean division), and $g(X) - (a_0 + a_1X) \in I \iff \overline{g(X)} = \overline{a_0 + a_1X}$ where $a_0 + a_1X$ is the rest of the division of $g(X)$ by $X^2 - 2$. Hence

$$\mathbb{Q}[X]/I = \{\overline{a_0 + a_1X} : a_0, a_1 \in \mathbb{Q}\}.$$

Theorem 8.4 (First Isomorphism Theorem for Rings).
Let $f : R \to S$ be a surjective ring homomorphism. Then $R/\operatorname{Ker}(f)$ is isomorphic to S.

Proof. We consider the map $\pi : R/\operatorname{Ker}(f) \to S$ defined by $\pi(\bar{a}) = f(a)$ for $a \in R$. First we check π is well defined. Assume $\bar{a} = \bar{b} \iff a-b \in \operatorname{Ker}(f) \iff f(a-b) = 0_S \iff f(a) = f(b) \iff \pi(\bar{a}) = \pi(\bar{b})$.

Next we show that π is a homomorphism: $\pi(\bar{a}+\bar{b}) = \pi(\overline{a+b}) = f(a+b) = f(a)+f(b) = \pi(\bar{a})+\pi(\bar{b})$ and $\pi(\bar{a}\bar{b}) = \pi(\overline{ab}) = f(ab) = f(a)f(b) = \pi(\bar{a})\pi(\bar{b})$.

We check π is bijective. First π is surjective since given $s \in S$ there exists $a \in R$ with $s = f(a)$ (as f is surjective) hence $s = \pi(\bar{a})$. To show π is injective we show $\operatorname{Ker}(\pi) = \{\bar{0}_R\}$. Let $a \in R$ with $\pi(\bar{a}) = f(a) = 0_S$. Then $a \in \operatorname{Ker}(f)$ which implies $\bar{a} = \bar{0}_R$.

Alternatively, using the First Isomorphism Theorem for groups, see 2.10, we constructed $\pi : R/\operatorname{Ker}(f) \to S$ and showed it is an isomorphism of groups for $+$. Thus we just need to show π is a ring homomorphism by showing $\pi(\bar{a}\bar{b}) = \pi(\bar{a})\pi(\bar{b})$ which we did above. □

Example 8.5.

1. The homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}$ defined by
$$\phi(f(X)) = f(0)$$
is surjective (see Examples 8.1). We compute $\operatorname{Ker}(\phi)$. We have $\operatorname{Ker}(\phi) = \{f(X) = a_0 + a_1X + \ldots + a_nX^n \in \mathbb{Q}[X] : f(0) = a_0 = 0\} = \{f(X) = a_1X + \ldots + a_nX^n \in \mathbb{Q}[X]\} = \{X(a_1 + a_2X + \ldots + a_nX^{n-1}) \in \mathbb{Q}[X]\} = (X)_{\mathbb{Q}[X]}$ is the principal ideal of $\mathbb{Q}[X]$ generated by X. The first isomorphism theorem for rings implies that $\mathbb{Q}[X]/(X)$ is isomorphic to $\mathbb{Q}$.
2. The homomorphism $\phi : \mathbb{Q}[X] \to \mathbb{Q}(\sqrt{2})$ defined by
$$\phi(f(X)) = f(\sqrt{2})$$
[if $f(X) = a_0 + a_1X + \ldots + a_nX^n$ then $f(\sqrt{2}) = a_0 + a_1\sqrt{2} + \ldots + a_n(\sqrt{2})^n$] is a surjective homomorphism. Indeed if $x = a + b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ then $x = \phi(a + bX)$.
We claim $\operatorname{Ker}(\phi) = I = (X^2 - 2)_{\mathbb{Q}[X]}$ is the principal ideal I generated by $X^2 - 2$. Let $f(X) = (X^2-2)g(X) \in I = (X^2-2)_{\mathbb{Q}[X]}$, then $\phi(f) = \phi(X^2-2)\phi(g) = (\sqrt{2}^2-2)g(\sqrt{2}) = 0$ and $f \in \operatorname{Ker}(\phi)$. Conversely let $f \in \operatorname{Ker}(\phi)$. Then $f(X) = (X^2-2)g(X)+r(X)$ with $r(X) = a+bX$ [$\deg r < \deg(X^2 - 2) = 2$]. In particular, $0 = f(\sqrt{2}) = (\sqrt{2}^2 - 2)g(\sqrt{2}) + a + b\sqrt{2} = 0 + a + b\sqrt{2} \implies a = -b\sqrt{2} \implies a = b = 0$ and f is divisible by $X^2 - 2$ hence $f \in I$.
The first isomorphism theorem for rings implies that $\mathbb{Q}[X]/(X^2 - 2)$ is isomorphic to $\mathbb{Q}(\sqrt{2})$

Definition 8.4.
Let R be a ring and I, J, ideals of R. Set
$$I + J = \{a + b : a \in I, b \in J\}$$
and
$$IJ = \{\sum_{i=1}^{m} a_ib_i : a_i \in I, b_i \in J, m \geq 1\}.$$

Lemma 8.5.
Let R be a ring and I, J ideals of R. Then $I + J$ and IJ are ideals of R.

Proof. will do in lecture. □

Definition 8.5.
Let R be a ring.

1. An ideal I of R is called *principal* if $I = (a)_R$ is generated by one element $a \in R$, called a *generator* of I. Recall $(a)_R = \{ab : b \in R\}$.
2. Let $a, b \in R$ and define $(a, b)_R = \{ac + bd : c, d \in R\} = (a)_R + (b)_R$. Then $(a, b)_R$ is an ideal (being the sum of two ideals) called the ideal *generated* by a and b.

Lemma 8.6.
Let R be a ring.

1. *$(b)_R \subset (a)_R \iff b = ac$ for some $c \in R$.*
2. *If R is an integral domain with identity, then*

$$(a)_R = (b)_R \iff a = bu : u \in R^\times.$$

Proof. Will do in lecture. □

Example 8.6 (Example of a non principal ideal)**.**
Let

$$R = \mathbb{Z}[X]$$

and

$$I = (2, X)_{\mathbb{Z}[X]} = \{2h(X) + Xg(X), h, g \in \mathbb{Z}[X]\}$$

the ideal of $\mathbb{Z}[X]$ generated by 2 and X. I claim I is NOT principal.

We argue by contradiction. Indeed assume the contrary

$$I = (f(X))_{\mathbb{Z}[X]}$$

is principal generated by some $f(X) \in \mathbb{Z}[X]$.

Then

$$2 = 2 \times 1 + X \times 0 \ \in \ I$$

hence $\exists p(X) \in R$ such that

$$2 = f(X)p(X).$$

In particular $0 = \deg 2 = \deg(f) + \deg(p)$ which implies $\deg(f) = \deg(p) = 0$ and both f and p are constant polynomials.

Further since f divides 2 it holds that $f \in \{1, -1, 2, -2\}$. If $f = 1$ or $f = -1$ then

$$1 = 2h(X) + Xg(X)$$

for some polynomials $h, g \in \mathbb{Z}[X]$ which is not possible (the constant coefficient of $2h(X) + Xg(X)$ is even). So $f = 2$ or $f = -2$. But

$$X = 0 \times 2 + 1 \times X \ \in \ I$$

hence $X = 2q(X)$ for some $q(X) \in \mathbb{Z}[X]$, which again is not possible.

Example 8.7 (Construction of a surjective homomorphism with a given kernel)**.**
Let

$$R = \mathbb{Z}[\sqrt{-3}] = \{a + b\sqrt{-3} : a, b \in \mathbb{Z}\}$$

and

$$J = (2, 1 + \sqrt{-3})_R = \{2x + (1 + \sqrt{-3})y : x, y \in \mathbb{Z}[\sqrt{-3}]\}.$$

Construct a surjective homomorphism from R to a finite field with kernel J.

Consider the map

$$f : \mathbb{Z}[\sqrt{-3}] \ \to \ \mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z} = \{\bar{0}, \bar{1}\}$$

defined by

$$a + b\sqrt{-3} \quad \mapsto \quad f(a + b\sqrt{-3}) = \bar{a} + \bar{b}.$$

This map is surjective as $\bar{1} = f(1)$ and $\bar{0} = f(2)$.

- We show f is a ring homomorphism.
  Indeed
  $f(a + b\sqrt{-3} + c + d\sqrt{-3}) = f(a + c + (b + d)\sqrt{-3}) = \overline{a + c} + \overline{b + d} = \bar{a} + \bar{b} + \bar{c} + \bar{d} = f(a + b\sqrt{-3}) + f(c + d\sqrt{-3})$
  and
  $f((a+b\sqrt{-3})(c+d\sqrt{-3})) = f(ac-3bd+(ad+bc)\sqrt{-3}) = \overline{ac - 3bd}+\overline{ad + bc} = \overline{ac + bd}+\overline{ad + bc} = \overline{ac} + \overline{bd} + \overline{ad} + \overline{bc} = \bar{a}\bar{c} + \bar{b}\bar{d} + \bar{a}\bar{d} + \bar{b}\bar{c} = (\bar{a} + \bar{b})(\bar{c} + \bar{d}) = f(a + b\sqrt{-3})f(c + d\sqrt{-3})$ (here we used for the third equality the fact that $\overline{-3} = \bar{1}$ in $\mathbb{F}_2$).
- We show $\mathrm{Ker}(f) = (2, 1 + \sqrt{-3})_R$.
  Clearly $J = (2, 1 + \sqrt{-3})_R \subset \mathrm{Ker}(f)$ since $f(2) = f(1 + \sqrt{-3}) = \bar{0}$ hence $f(2x + (1 + \sqrt{-3})y) = f(2)f(x) + f(1 + \sqrt{-3})f(y) = 0f(x) + 0f(y) = 0$ for $x, y \in R$.
  We show that $\mathrm{Ker}(f) \subset (2, 1 + \sqrt{-3})_R$. Let $a + b\sqrt{-3} \in \mathrm{Ker}(f)$ so that $f(a + b\sqrt{-3}) = \bar{a} + \bar{b} = \overline{a - b} = \bar{0}$. Hence $a - b = 2k$ and $a = b + 2k$ for some $k \in \mathbb{Z}$. Thus $a + b\sqrt{-3} = 2k + b + b\sqrt{-3} = 2k + b(1 + \sqrt{-3}) \in J$ as required.

9 Prime and Maximal Ideals

In this section all rings are **commutative** with an identity element $1 \neq 0$.

The sign $\subsetneq$ means contained and not equal.

Let R be a ring. An ideal I of R is called **proper** if $I \subsetneq R$.

Definition 9.1 (Maximal Ideal)**.**
A proper ideal $M \subsetneq R$ is called a *maximal* ideal if the only ideals of R containing M are M and R, i.e. if $M \subset I$ and I is an ideal then $M = I$ or $I = R$.

Maximal ideals are maximal elements within the set of proper ideals for the relation of inclusion.

Proposition 9.1.
Let $J \subsetneq R$ be a proper ideal. There exists a maximal ideal M of R containing J: $J \subset M$.

Proof. Let $\mathcal{S}$ be the set of all proper ideals of R containing J. The set $\mathcal{S}$ is nonempty (as it contains J) and (partially) ordered by the relation of inclusion. Let $J \subset \ldots I_k \subset I_{k'} \subset \ldots$, with $k, k', \ldots \in K$: an index set, be a chain of proper ideals containing J. By Zorn's Lemma we need to show that $\{I_k\}_{k\in K}$ has an upper bound. Let $\mathcal{I} = \bigcup_{k\in K} I_k$. Then $\mathcal{I}$ is an ideal of R: if $a, b \in \mathcal{I}$ then $a \in I_k, b \in I_{k'}$ for some $k, k' \in K$. Since the family $\{I_k\}_{k\in K}$ is a chain either $I_k \subset I_{k'}$ or $I_{k'} \subset I_k$, say $I_k \subset I_{k'}$, hence $a + b, ab \in I_{k'} \in \mathcal{I}$. Similarly if $a \in \mathcal{I}$, $r \in R$, $\exists k \in K$ with $a \in I_k$, hence $ra \in I_k \subset \mathcal{I}$ since I_k is an ideal. Further $1 \notin \mathcal{I}$, for otherwise $\exists k \in K$ with $1 \in I_k$ hence $I_k = R$ which is not the case since I_k is a proper ideal. Clearly $\mathcal{I}$ is an upper bound for $\{I_k\}_{k\in K}$. □

In particular, applying Proposition 9.1 to the zero ideal $\{0_R\}$ of R we find that maximal ideals always exist.

Theorem 9.2 (Lattice Isomorphism Theorem)**.**
Let $J \subset R$ be an ideal. There is a one to one correspondence between the set of ideals I of R containing J and the set of ideals of the factor ring R/J.

Proof. Let $\mathcal{S}$ be the set of ideals I of R such that $J \subset I$. Let $\mathcal{T}$ be the set of ideals of R/J. It suffices to construct a bijection from $\mathcal{S}$ to $\mathcal{T}$.

For $I \in \mathcal{S}$, define I/J to be the set $\{\bar{b} \in R/J : \bar{b} = \bar{a}, \text{for some } a \in I\}$. If $\bar{b} \in I/J$, then $\bar{b} = \bar{a}$, for some $a \in I$. So $b - a \in J \subset I$. As $a \in I$, $b \in I$. Thus, $\bar{b} \in I/J$ if and only if $b \in I$.

Moreover, $I/J \in \mathcal{T}$. Indeed, $0 \in I$, so $\bar{0} \in I/J$. Suppose $\bar{a}, \bar{b} \in I/J$, so $a \in I$ and $b \in I$. Then $\bar{a} + \bar{b} = \overline{a + b}$ and $a + b \in I$, so $\bar{a} + \bar{b} \in I/J$. Likewise, $-a \in I$ so $-\bar{a} = \overline{-a} \in I/J$, as $-a \in I$. If $\bar{r} \in R/J$, then $ar \in I$, whence $\bar{a}\bar{r} \in I/J$.

Let $\phi : R \to R/J$ be the homomorphism $\phi(a) = \bar{a}$. The preceding comments show that $\phi(I) = I/J \in \mathcal{T}$ for each $I \in \mathcal{S}$. Define $f : \mathcal{S} \to \mathcal{T}$ by $f(I) = \phi(I)$. It remains to construct a map $g : \mathcal{T} \to \mathcal{S}$ that is an inverse of f.

Let $\widetilde{I} \in \mathcal{T}$ be an ideal of R/J. Then, $\phi^{-1}(\widetilde{I})$ is an ideal of R. Further, $J \subset \phi^{-1}(\widetilde{I})$ since if $a \in J$ then $\phi(a) = \bar{a} = \bar{0} \in \widetilde{I}$. Thus there is map $g : \mathcal{T} \to \mathcal{S}$ given by $g(\widetilde{I}) = \phi^{-1}(\widetilde{I})$.

It is left as an exercise to show that $g(f(I)) = I$ for all $I \in \mathcal{S}$ and that $f(g(\widetilde{I})) = \widetilde{I}$, for all $\widetilde{I} \in \mathcal{T}$. From this, it follows that f is a one-to-one correspondence between the ideals of R that contain J and the ideals of R/J. □

Theorem 9.3.
The ideal $M \subsetneq R$ is a maximal ideal if and only if the quotient ring R/M is a field.

Proof. By the Lattice Isomorphism Theorem there is a one to one correspondence between the set of ideals I of R containing M and the ideals of the quotient ring R/M. Now R/M is a field if and only if the only ideals of R/M are R/M and $\{\bar{0}\}$ [this is essentially the content of CW4], in which case the corresponding ideals of R containing M are: $\phi^{-1}(R/M) = R$ and $\phi^{-1}(0_{R/M}) = \mathrm{Ker}(\phi) = M$; where $\phi : R \to R/M$ is the map $a \mapsto \phi(a) = \bar{a}$, i.e. M is a maximal ideal. □

Example 9.1.

1. Let p be a prime number. The ideal $p\mathbb{Z} = (p)_{\mathbb{Z}} = \{pk : k \in \mathbb{Z}\}$ is a maximal ideal of $\mathbb{Z}$ since the factor ring $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field (see Examples 7.5).
2. The ideal $I = (X)_R = \{Xf(X) : f \in \mathbb{Z}[X]\}$ is **NOT** a maximal ideal of $R = \mathbb{Z}[X]$ since $(X)_R \subsetneq (2, X)_R \subsetneq \mathbb{Z}[X]$.
3. Let $R = \{f : [0, 1] \to \mathbb{R}\}$ be the set of all functions from $[0, 1]$ to $\mathbb{R}$. then R is a ring for the operations $+$, $\times$, defined by $(f + g)(x) = f(x) + g(x)$ and $(f \times g)(x) = f(x)g(x)$. Let $a \in \mathbb{R}$ and $M_a = \{f \in R : f(a) = 0\}$, then M_a is an ideal of R (check this fact). The map $\phi : R \to \mathbb{R}$ defined by $f \mapsto \phi(f) = f(a)$ is a surjective ring homomorphism (check this fact) and $\mathrm{Ker}(\phi) = M_a$. The FIT implies R/M_a is isomorphic to $\mathbb{R}$ which is a field hence R/M_a is a field and M_a is a maximal ideal.
4. If $\phi : R \to F$ is a surjective homomorphism from a ring R to a field F with kernel I then I is a maximal ideal. Indeed by the FIT: R/I is isomorphic to the field F hence R/I itself is a field (check this fact) and I is a maximal ideal.
5. The ideal
$$M = (2, X) = \{2f(X) + Xg(X) : f, g \in \mathbb{Z}[X]\}$$
is a maximal ideal of $\mathbb{Z}[X]$. Indeed the map
$$\phi : \mathbb{Z}[X] \to \mathbb{F}_2 = \{\bar{0}, \bar{1}\}$$
defined by
$$\phi(a_0 + a_1X + \ldots + a_nX^n) = \bar{a}_0$$
is a surjective homomorphism (check this fact) with kernel M. Indeed $\phi(2) = \phi(X) = \bar{0}$ hence $\phi(2f(X) + Xg(X)) = \phi(2)\phi(f(X)) + \phi(X)\phi(g(X)) = 0$ and $M \subset \mathrm{Ker}\,\phi$.
If $f(X) = a_0 + a_1X + \ldots + a_nX^n \in \mathbb{Z}[X]$ with $\phi(f(X)) = \bar{a}_0 = \bar{0}$ [i.e. $f(X) \in \mathrm{Ker}(\phi)$] then $a_0 = 2b$ and $f(X) = 2b + X(a_1 + a_2X + \ldots + a_nX^{n-1}) \in M$.
The FIT implies $\mathbb{Z}[X]/M$ is isomorphic to $\mathbb{F}_2$: a field, hence $\mathbb{Z}[X]/M$ is a field and M is a maximal ideal.
6. Let $R = \mathbb{Z}[i]$, we show the ideal
$$I = (1 + i)_R = \{(1 + i)(a + bi) : a, b \in \mathbb{Z}\}$$
is a maximal ideal of R by showing the existence of a surjective homomorphism from R to a field with kernel I. Let
$$\phi : \mathbb{Z}[i] \to \mathbb{F}_2 = \{\bar{0}, \bar{1}\}$$
be the map defined by
$$\phi(a + ib) = \overline{a + b}$$

The map ϕ is surjective as $\bar{0} = \phi(1+i)$ and $\bar{1} = \phi(1+2i)$. Further ϕ is a ring homomorphism:

$$\begin{aligned}\phi[(a+ib)+(c+id)] &= \phi[(a+c)+i(b+d)] = \overline{(a+c)+(b+d)} \\ &= \overline{(a+b)+(c+d)} = \overline{(a+b)} + \overline{(c+d)} \\ &= \phi(a+bi)+\phi(c+di).\end{aligned}$$

and

$$\begin{aligned}\phi[(a+ib)(c+id)] &= \phi[(ac-bd)+i(ad+bc)] = \overline{ac-bd+ad+bc} \\ &= \overline{ac+bd+ad+bc} = \overline{(a+b)(c+d)} \\ &= \overline{(a+b)}\,\overline{(c+d)} = \phi(a+ib)\phi(c+id).\end{aligned}$$

[for the third equality we used the fact that $\overline{-1} = \bar{1}$ in $\mathbb{F}_2$]. We show $\mathrm{Ker}(\phi) = (1+i)_R$. Clearly $(1+i)_R \subset \mathrm{Ker}(\phi)$ since $\phi(1+i) = \overline{1+1} = \bar{0}$, hence $\phi((1+i)(a+ib)) = \phi(1+i)\phi(a+ib) = \bar{0}\phi(a+ib) = \bar{0}$ for every element $(1+i)(a+ib) \in I$.
Reciprocally let $a+ib \in \mathrm{Ker}(\phi)$. Then $\phi(a+ib) = \overline{a+b} = \overline{a-b} = \bar{0}$ hence $a = 2k+b$ for some integer k and $a+ib = 2k+b+ib = (1+i)(1-i)k + b(1+i) = (1+i)(b+k-ki) \in I$ as required, here we used $2 = (1+i)(1-i)$ in $\mathbb{Z}[i]$.
The FIT implies $\mathbb{Z}[i]/\mathrm{Ker}(\phi) = \mathbb{Z}[i]/I$ is isomorphic to $\mathbb{F}_2$: a field, hence $\mathbb{Z}[X]/I$ is a field and I is a maximal ideal.

Definition 9.2 (Prime Ideal).
A proper ideal $P \subsetneq R$ is called a *prime* ideal if whenever the product ab of two elements $a, b \in R$ is an element of P then either a or b is an element of P:

$$\{ab \in P\} \Longrightarrow \{a \in P \text{ or } b \in P\}.$$

Example 9.2.
If p is a prime number the ideal $p\mathbb{Z} = (p)_{\mathbb{Z}} = \{pk : k \in \mathbb{Z}\}$ is a prime ideal. indeed let $a, b \in \mathbb{Z}$ with $ab \in p\mathbb{Z}$. Thus $ab = pk$ for some $k \in \mathbb{Z}$ and p divides ab. Since p is prime either p divides a which implies $a \in p\mathbb{Z}$ or p divides b which implies $b \in p\mathbb{Z}$.

Theorem 9.4.
The ideal $P \subsetneq R$ is a prime ideal if and only if the factor ring R/P is an integral domain.

Proof. Let $a, b \in R$ with $ab \in P$ and consider the homomorphism

$$\phi : R \to R/P$$

defined by

$$r \mapsto \phi(r) = \bar{r}.$$

Then $\overline{ab} = \bar{a}\bar{b} = \bar{0}_R$ since $ab \in P$. Now P is a prime ideal if and only if whenever $ab \in P$, which is equivalent $\overline{ab} = \bar{a}\bar{b} = \bar{0}_R$, then either $a \in P$ which is equivalent $\bar{a} = \bar{0}_R$, or $b \in P$ which is equivalent $\bar{b} = \bar{0}_R$, this latter condition is equivalent to R/P being an integral domain. □

Example 9.3.

1. In an integral domain R the ideal $\{0_R\}$ is a prime ideal since whenever $ab = 0_R$ then either $a = 0_R$ or $b = 0_R$.
2. In $\mathbb{Z}$ the ideals $\{0\}$ and $\{p\mathbb{Z}\}_p$ for p a prime number are prime ideals. The ideal $\{0\}$ is not a maximal ideal since $\{0\} \subsetneq m\mathbb{Z} \subsetneq \mathbb{Z}$ for any nonzero integer m which is different from both 1 and -1.
3. If $\phi : R \to D$ is a surjective homomorphism from a ring R to an integral domain D with kernel J then J is a prime ideal. Indeed by the FIT: R/J is isomorphic to the integral domain D hence R/J itself is an integral domain (prove this) and J is a prime ideal.
4. The ideal $(X)_{\mathbb{Z}[X]} = \{Xf(X) : f(X) \in \mathbb{Z}[X]\}$ of $\mathbb{Z}[X]$ is a prime ideal. Indeed consider the map
$$\phi : \mathbb{Z}[X] \to \mathbb{Z}$$
defined by
$$\phi(a_0 + a_1X + \ldots + a_nX^n) = a_0.$$
Then ϕ is a surjective homomorphism (check this fact) with kernel
$$\mathrm{Ker}(\phi) = \{f(X) = a_0 + a_1X + \ldots + a_nX^n : a_0 = 0\} = (X)_{\mathbb{Z}[X]}$$
and $(X)_{\mathbb{Z}[X]}$ is a prime ideal since $\mathbb{Z}$ is an integral domain. Note that the ideal $(X)_{\mathbb{Z}[X]}$ is prime but not maximal (see Examples 9.1).

A prime ideal need not be a maximal ideal [see Examples 9.3 and (4)]. However the converse holds.

Proposition 9.5.
A maximal ideal is a prime ideal.

Proof. Indeed if M is a maximal ideal of R then R/M is a field hence is an integral domain (since every field is an integral domain, see §7), and M is a prime ideal by Theorem 9.4. □

9.1 Field of Fractions

Let R be an integral domain and
$$\widetilde{F} = \{(a, b) : a, b \in R, b \neq 0_R\}.$$
Define the relation $\mathcal{H}$ among elements of F by
$$(a, b)\ \mathcal{H}\ (c, d) \iff ad = bc.$$
The relation $\mathcal{H}$ is an equivalence relation (check this fact). Let $F = \widetilde{F}/\mathcal{H}$ be the set of equivalence relations of $\mathcal{H}$. We will denote the equivalence class (a, b) by the fraction $\frac{a}{b}$. Thus
$$F = \left\{\frac{a}{b} :\ a, b \in R,\ b \neq 0_R\right\},$$
and
$$\frac{a}{b} = \frac{c}{d} \iff ad = bc.$$
Define the operations $+$ and $\times$ on the set F by
$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

Note that $bd \neq 0_R$ since $b \neq 0_R$, $d \neq 0_R$, and R is integral. Hence both $\frac{ad+bc}{bd}$ and $\frac{ac}{bd}$ are elements of F.

These operations are well defined and do not depend on the choice of a representative of the class of equivalence of (a, b).

Indeed assume $(a, b)\ \mathcal{H}\ (a', b') \iff ab' = ba'$, and $(c, d)\ \mathcal{H}\ (c', d') \iff cd' = dc'$, then

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} = \frac{a'd' + b'c'}{b'd'} = \frac{a'}{b'} + \frac{c'}{d'}$$

since $(ad + bc)b'd' = adb'd' + bcb'd' = ba'db' + bb'c'd = bd(a'd' + b'c')$, and similarly (check)

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} = \frac{a'c'}{b'd'} = \frac{a'}{b'} \times \frac{c'}{d'}.$$

The set F endowed with the operations $+$ and $\times$ is a commutative ring (check this fact) with identity

$$1_F = \frac{1_R}{1_R} \qquad 0_F = \frac{0_R}{1_R} \qquad \text{and} \qquad -(\frac{a}{b}) = \frac{-a}{b}.$$

Further F is a field: if $\frac{a}{b} \neq 0_F$ then $a \neq 0_R$ $[\frac{a}{b} = 0_F = \frac{0_R}{1_R} \iff a = a1_R = b0_R = 0_R]$

$$\frac{a}{b} \times \frac{b}{a} = \frac{ab}{ab} = \frac{1_R}{1_R} = 1_F,$$

and $\frac{b}{a}$ is the inverse of $\frac{a}{b}$. The field F is called the **field of fraction** of R and denoted $\mathrm{Fr}(R)$.

Example 9.4.

1. $\mathbb{Z}$ is an integral domain and $\mathrm{Fr}(\mathbb{Z}) = \mathbb{Q}$.
2. Let K be a field. The ring $K[X]$ is an integral domain, its field of fraction is denoted $K(X) = \mathrm{Fr}(K[X])$ and called the ring of rational functions in the variable X. Thus

$$K(X) = \Big\{\frac{f(X)}{g(X)} : f, g \in K[X], g \neq 0\Big\}$$

with

$$\frac{f(X)}{g(X)} + \frac{h(X)}{q(X)} = \frac{f(X)q(X) + g(X)h(X)}{g(X)q(X)} \qquad \text{and} \qquad \frac{f(X)}{g(X)} \times \frac{h(X)}{q(X)} = \frac{f(X)h(X)}{g(X)q(X)}.$$

If $\frac{f(X)}{g(X)} \in K(X)$ is nonzero then $f(X) \neq 0$ and the inverse of $\frac{f(X)}{g(X)}$ is $\frac{g(X)}{f(X)}$:

$$\frac{f(X)}{g(X)} \times \frac{g(X)}{f(X)} = \frac{f(X)g(X)}{f(X)g(X)} = \frac{1_K}{1_K} = 1_{K(X)}.$$

Proposition 9.6.
Let R be an integral domain. Then the followings hold.

(i) *The map $\psi : R \to \mathrm{Fr}(R)$ defined by $r \mapsto \psi(r) = \frac{r}{1_R}$ is an injective ring homomorphism.* ***(ii)*** *Let $\phi : R \to K$ be an injective homomorphism into a field K. There exists a unique injective homomorphism $\tau : \mathrm{Fr}(R) \to K$ with the property*

$$\tau(\psi(r)) = \tau(\frac{r}{1_R}) = \phi(r)$$

for every $r \in R$.

Proof. **(i)**

$$\psi(r+s) = \frac{r+s}{1_R} = \frac{r}{1_R} + \frac{s}{1_R} = \psi(r) + \psi(s)$$

and

$$\psi(rs) = \frac{rs}{1_R} = \frac{r}{1_R} \times \frac{s}{1_R} = \psi(r)\psi(s).$$

Further if $\psi(r) = \frac{r}{1_R} = \frac{0_R}{1_R}$ then $r = r \times 1_R = 0_R \times 1_R = 0_R$ and ψ is injective.

(ii) Define $\tau : \mathrm{Fr}(R) \to K$ by

$$\tau(\frac{r}{s}) = \phi(r)\phi(s)^{-1}$$

[which makes sense since $s \neq 0_R$, $\phi(s) \neq 0_K$ as ϕ is injective, and the inverse $\phi(s)^{-1}$ of $\phi(s)$ exists in K]. Then

$\tau(\frac{r}{s} + \frac{t}{h}) = \tau(\frac{rh+st}{sh}) = \phi(rh+st)\phi(sh)^{-1} = [\phi(r)\phi(h) + \phi(s)\phi(t)]\phi(s)^{-1}\phi(h)^{-1} = \phi(r)\phi(s)^{-1} + \phi(t)\phi(h)^{-1} = \tau(\frac{r}{s}) + \tau(\frac{t}{h})$,

further

$\tau(\frac{r}{s}\frac{t}{h}) = \tau(\frac{rt}{sh}) = \phi(rt)\phi(sh)^{-1} = [\phi(r)\phi(t)]\phi(s)^{-1}\phi(h)^{-1} = \phi(r)\phi(s)^{-1}\phi(t)\phi(h)^{-1} = \tau(\frac{r}{s})\tau(\frac{t}{h})$, hence τ is a homomorphism.

and

$\tau(1_{\mathrm{Fr}(R)}) = \tau(\frac{1_R}{1_R}) = \phi(1_R)\phi(1_R)^{-1} = 1_K(1_K)^{-1} = 1_K$.

If $\tau(\frac{r}{s}) = \phi(r)\phi(s)^{-1} = 0_K$ then $\phi(r) = 0_K$ [since K is integral and $\phi(s)^{-1} \neq 0_K$] hence $r = 0_R$ since ϕ is injective. This shows τ is injective.

The property $\tau(\psi(r)) = \tau(\frac{r}{1_R}) = \phi(r)$ determines τ uniquely since $\tau(\frac{r}{s}) = \tau(\psi(r)\psi(s)^{-1}) = \tau(\psi(r))\tau(\psi(s)^{-1}) = \tau(\psi(r))\tau(\psi(s))^{-1} = \phi(r)\phi(s)^{-1}$. □

9.2 The Chinese Remainder Theorem

Given two rings R_1 and R_2 we define their **product**

$$R_1 \times R_2 = \{(r_1, r_2) : r_1 \in R_1, r_2 \in R_2\}$$

and endow it with the operations + and × defined by

$$(r_1, r_2) + (s_1, s_2) = (r_1 + s_1, r_2 + s_2) \qquad (r_1, r_2) \times (s_1, s_2) = (r_1 s_1, r_2 s_2).$$

The set $R_1 \times R_2$ endowed with + and × is a commutative ring (check this fact) with identity and zero elements:

$$1_{R_1 \times R_2} = (1_{R_1}, 1_{R_2}) \qquad \text{and} \qquad 0 = (0_{R_1}, 0_{R_2}) \qquad \text{respectively.}$$

Similarly as above we can define the product of any given set of rings with addition and multiplication defined component-wise.

- Two ideals of a ring R are called **co-maximal** if $I + J = R$. If I, J are co-maximal ideals of R then $IJ = I \bigcap J$ [see Exercise 7(d) in Problem Sheet 4]

Theorem 9.7 (Chinese Remainder Theorem)**.**
Let I, J be ideals of R. The map

$$\phi : R \to R/I \times R/J$$

defined by

$$r \mapsto (r + I, r + J)$$

is a ring homomorphism with kernel $I \bigcap J$.

If I and J are co-maximal then this map is surjective and $I \bigcap J = IJ$, in this case $R/IJ = R/(I \bigcap J)$ is isomorphic to $R/I \times R/J$.

Proof. We have

$$\phi(r+s) = (r+s+I, r+s+J) = (r+I+s+I, r+J+s+J) = (r+I, r+J)+(s+I, s+J) = \phi(r)+\phi(s)$$

$$\phi(rs) = (rs + I, rs + J) = ((r + I)(s + I), (r + J)(s + J)) = (r + I, r + J)(s + I, s + J) = \phi(r)\phi(s)$$

and

$$\phi(1_R) = (1_R + I, 1_R + J) = 1_{(R/I\times R/J)}$$

hence ϕ is a homomorphism.

$\mathrm{Ker}(\phi) = \{r \in R : (r + I, r + J) = (0_{R/I}, 0_{R/J})\} = \{r \in R : r \in I, r \in J\} = I \bigcap J.$

Suppose $I + J = R$, we show ϕ is surjective in this case.

Let $(r_1 + I, r_2 + J) \in R/I \times R/J$. There exists $a \in I, b \in J$ with

$$1 = a + b$$

(since $I + J = R$). In particular $a = 1 - b$ and

$$a + J = (1 - b) + J = 1 + J - (b + J) = 1 + J,$$

$b = 1 - a$ and

$$b + I = (1 - a) + I = 1 + I - (a + I) = 1 + I.$$

Consider the element

$$r = r_2a + r_1b \in R$$

then

$$\phi(r) = ((r_2a + r_1b) + I, (r_2a + r_1b) + J) = (r_2a + I + r_1b + I, r_2a + J + r_1b + J) = (r_1b + I, r_2a + J)$$

$$= (r_1 + I, r_2 + J)(b + I, a + J) = (r_1 + I, r_2 + J)(1 + I, 1 + J) = (r_1 + I, r_2 + J).$$

Further $\mathrm{Ker}(\phi) = I \bigcap J = IJ$ [see Exercise 8(d) in Problem Sheet 4] and the FIT for rings implies that $R/\mathrm{Ker}(\phi) = R/(I \bigcap J) = R/IJ$ is isomorphic to $R/I \times R/J$. □

Example 9.5.

- Let $n = p_1^{a_1} p_2^{a_2}$ be an integer with $p_1 \neq p_2$ primes. Then $\mathbb{Z} = p_1^{a_1}\mathbb{Z} + p_2^{a_2}\mathbb{Z}$, and the ideals $(p_1^{a_1})$ and $(p_1^{a_1})$ of $\mathbb{Z}$ are co-maximal. The Chinese Remainder Theorem implies that $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to the the product $\mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \mathbb{Z}/p_2^{a_2}\mathbb{Z}$

- If $I_1, I_1, \ldots, I_n$ are ideals of R which are pairwise co-maximal: $I_i + I_j = R$ for $1 \leq i < j \leq n$, then the Chinese Remainder Theorem generalises as follows.
  The map
  $$\phi : R \to R/I_1 \times R/I_2 \times \ldots \times R/I_n$$
  defined by
  $$r \mapsto (r + I_1, r + I_2, \ldots, , r + I_n)$$
  is a surjective ring homomorphism with kernel
  $$I_1 \bigcap I_2 \bigcap \ldots \bigcap I_n = I_1 I_2 \ldots I_n$$
  and $R/(I_1 I_2 \ldots I_n) = R/(I_1 \bigcap \ldots \bigcap I_n)$ is isomorphic to $R/I_1 \times R/I_2 \times \ldots \times R/I_n$. Here the ideal $I_1 I_2 \ldots I_n$ consists of elements of R that are finite sums
  $$\sum a_1 a_2 \ldots a_n$$
  where $a_i \in I_i$ for $1 \leq i \leq n$.
  In particular if $m = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}$ is an integer with $p_1, p_2, \ldots, p_n$ distinct primes, then the ideals $(p_i^{a_i})$ and $(p_j^{a_j})$ of $\mathbb{Z}$ are co-maximal for $1 \leq i < j \leq n$, and the Chinese Remainder Theorem states that $\mathbb{Z}/m\mathbb{Z}$ is isomorphic to the product $\mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \mathbb{Z}/p_2^{a_2}\mathbb{Z} \times \ldots \times \mathbb{Z}/p_n^{a_n}\mathbb{Z}$.

10 Divisibility and Factorisation in Rings

In this section all rings are **commutative** with an identity element $1 \neq 0$.

10.1 Euclidean Domains

Let R be an integral domain (ring without zero divisors). A function

$$N : R \setminus \{0_R\} \to \mathbb{N}$$

is called a **norm**.

A norm N is called multiplicative if :

$$N(ab) = N(a)N(b)$$

for all $a, b \in R \setminus \{0_R\}$.

R is called an **Euclidean domain (ED)** if there exists a norm N on R such that for all $a, b \in R$ with $b \neq 0$ there exists $q, r \in R$ so that

$$a = bq + r$$

with $r = 0$, or $N(r) < N(b)$ [r is the remainder of the division of a by b].

Example 10.1.
(1) $\mathbb{Z}$ is an Euclidean domain with multiplicative norm

$$N : \mathbb{Z} \setminus \{0\} \to \mathbb{N}$$

$$a \mapsto |a|.$$

(2) $\mathbb{Q}[X]$, or more generally $F[X]$ where F is a field, is an Euclidean domain with norm

$$N : \mathbb{Q}[X] \setminus \{0\} \to \mathbb{N}$$

$$f(X) \mapsto \deg(f).$$

(3) $\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$ is an Euclidean domain with norm

$$N : \mathbb{Z}[i] \setminus \{0\} \to \mathbb{N}$$

$$a + ib \ \mapsto \ N(a + ib) = a^2 + b^2.$$

First we show N is a multiplicative norm. Let $a + ib, c + id \in \mathbb{Z}[i]$ then

$$N[(a + ib)(c + id)] = N[(ac - bd) + i(ad + bc)] = (ac - bd)^2 + (ad + bc)^2$$
$$= (ac)^2 - 2acbd + (bd)^2 + (ad)^2 + 2adbc + (bc)^2 = (ac)^2 + (bd)^2 + (ad)^2 + (bc)^2$$
$$a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 = (a^2 + b^2)(c^2 + d^2) = N(a + ib)N(c + id).$$

Next we show the Euclidean division property. Let $\alpha = a + ib$, and $0 \neq \beta = c + id \in \mathbb{Z}[i]$. Then

$$\frac{\alpha}{\beta} = \frac{(a + ib)(c - id)}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + i\frac{bc - ad}{c^2 + d^2} \in \mathbb{C}.$$

Write

$$r = \frac{ac+bd}{c^2+d^2} \in \mathbb{Q} \qquad s = \frac{bc-ad}{c^2+d^2} \in \mathbb{Q}.$$

Let $p, q \in \mathbb{Z}$ such that $|r-p| \leq \frac{1}{2}$, $|s-q| \leq \frac{1}{2}$, and let

$$\theta = \frac{\alpha}{\beta} - (p+iq) = (r+is) - (p+iq) = r - p + i(s-q) \in \mathbb{Q}(i)$$

and

$$\gamma = \beta\theta = \alpha - (p+iq)\beta \in \mathbb{Z}[i].$$

Thus

$$\alpha = (p+iq)\beta + \gamma$$

with $\gamma \in \mathbb{Z}[i]$.

We show $N(\gamma) < N(\beta)$. We have $N(\gamma) = N(\beta)N(\theta)$ and

$$N(\theta) = (r-p)^2 + (s-q)^2 \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

which implies $N(\gamma) \leq \frac{N(\beta)}{2} < N(\beta)$ as required.

A **principal ideal domain (PID)** is an integral domain R with the property that every ideal of R is principal generated by one element: if $I \subset R$ is an ideal then

$$I = (a)_R = \{ab : b \in R\}$$

for some element a of R.

Theorem 10.1.
Let R be an Euclidean domain then R is a principal ideal domain (PID)

$$\textbf{ED} \implies \textbf{PID}.$$

Proof. Let $\{0_R\} \neq I$ be an ideal of R, we show it is principal. Let

$$n = \min\{N(\tilde{a}) : \tilde{a} \in I, \tilde{a} \neq 0_R\},$$

and $a \in I$ with $N(a) = n$. We show $I = (a)_R$.

Clearly $(a)_R \subset I$ since $a \in I$, we show $I \subset (a)_R$.

Let $b \in I$ then $b = \alpha a + r$ with either $r = 0$ or $N(r) < N(a)$. Also $r = b - \alpha a \in I$ since $a, b \in I$, $\alpha a \in I$ since I is an ideal, and $(I, +)$ is a subgroup of $(R, +)$. This implies $r = 0$ necessarily by the minimality of a [a non zero $r \in I$ with $N(r) < N(a)$ would contradict the minimality of a] hence $b = \alpha a \in (a)_R$ and $I \subset (a)_R$. □

Example 10.2.
(1) $\mathbb{Z}$, $\mathbb{Q}[X]$ (or more generally $F[X]$ where F is a field), and $\mathbb{Z}[i]$ are PID since they are Euclidean domains.

(2) $R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$ is NOT a PID, in particular R is NOT an ED. Note that

$$N : \mathbb{Z}[\sqrt{-5}] \to \mathbb{N}$$

$$a+b\sqrt{-5} \ \mapsto \ N(a+b\sqrt{-5}) = a^2+5b^2$$

is a multiplicative norm (check this fact). Further $N(a+b\sqrt{-5}) = a^2+5b^2 = 1 \iff a^2 = 1, b = 0$ hence $a+ib \in \{1,-1\}$.

The ideal

$$I = (3, 2+\sqrt{-5})_R = \{3x + (2+\sqrt{-5})y : x, y \in R\}$$

is not principal. Indeed suppose the contrary $I = (c+d\sqrt{-5})_R$ is principal. Then $3 = \alpha(c+d\sqrt{-5})$ $(*)$, and $2+\sqrt{-5} = \beta(c+d\sqrt{-5})$ $(**)$, for some $\alpha, \beta \in R$. Now $(*)$ implies $N(3) = 9 = N(\alpha)N(c+d\sqrt{-5}) = N(\alpha)(c^2+5d^2)$ hence c^2+5d^2 equals 1, 3, or 9. We have the following possibilities:

$N(c+d\sqrt{-5}) = 9 \Longrightarrow N(\alpha) = 1 \Longrightarrow \alpha \in \{-1,1\} \Longrightarrow c+d\sqrt{-5} \in \{-3,3\}$ which is impossible by $(**)$ since β would be 1 or -1.

$N(c+\sqrt{-5}d) = 3 = c^2+5d^2$ which is not possible.

$N(c+\sqrt{-5}d) = 1 \Longrightarrow c+d\sqrt{-5} \in \{-1,1\} \Longrightarrow I = R$ which is not possible. Indeed if $1 = 3x+(2+\sqrt{-5})y$ for some $x, y \in R$ then $2-\sqrt{-5} = 3(2-\sqrt{-5})x+(2+\sqrt{-5})(2-\sqrt{-5})y = 3(2-\sqrt{-5})x+9y$ which implies 3 divides $2-\sqrt{-5}$: $2-\sqrt{-5} = 3v \Longrightarrow N(v) = 1$ hence $v \in \{1,-1\}$ which is not possible.

(3) $\mathbb{Z}[X]$ is NOT a PID since the ideal $(2, X)$ is not principal. In particular $\mathbb{Z}[X]$ is NOT an ED.

Definition 10.1.
Let R be a ring and $a, b \in R$. **(1)** We say that b *divides* a (or a is a *multiple* of b), and write b/a, if $a = bc$ for some $c \in R$. **(2)** A *greatest common divisor* of a and b is an element $d \in R$ such that:

(i) d/a and d/b.

(ii) If $d' \in R$, d'/a and d'/b then d'/d.

In this case we write $d = \gcd(a, b)$.

Remark 10.1.
$\gcd(a, b)$ is not unique in general, if $d = \gcd(a, b)$ then $ud = \gcd(a, b)$ for any unit $u \in R^\times$ (check this fact).

Proposition 10.2.
Let R be a PID and $a, b \in R$. Then $\gcd(a, b)$ *exists and*

$$(a)_R + (b)_R = (d)_R$$

where $d = \gcd(a, b)$*. Further,* $d = \gcd(a, b)$ *is unique up to multiplication by a unit, and we have a Bézout identity:*

$$d = sa + rb$$

for some $s, r \in R$.

Proof. The ideal $I = (a)_R + (b)_R$ is principal hence $(a)_R + (b)_R = (d)_R$ for some $d \in R$, we show $d = \gcd(a, b)$. We have $a = 1a+0_R b \in (d)_R \Longrightarrow a = \alpha d$ for some $\alpha \in R$ which implies d/a. Similarly $b = 1b+0_R a \in (d)_R \Longrightarrow b = \beta d$ for some $\beta \in R$ which implies d/b.

Also $d = d1_R \in (d)_R = (a)_R + (b)_R$, hence

$$d = ra + sb \qquad (*)$$

for some $s, r \in R$.

We show $d = \gcd(a, b)$. Assume $d' \in R$, $d'/a \iff a = d'x$, and $d'/b \iff b = d'y$ for some $x, y \in R$. The relation $(*)$ above implies $d = sd'x + rd'y = d'(sx + ry)$ and d'/d which shows $d = \gcd(a, b)$. □

Remark 10.2.
In an Euclidean domain R the $\gcd(a, b)$ exists for all $a, b \in R$ and can be computed by the Euclidean algorithm (as for integers):

$$a = sb + r \qquad N(r) < N(b)$$
$$b = s_1 r + r_1 \qquad N(r_1) < N(r)$$

. . . . . .

$$r_{k-2} = s_k r_{k-1} + r_k \qquad N(r_k) < N(r_{k-1})$$
$$r_{k-1} = s_{k+1} r_k$$

This algorithm has to stop after finitely many steps since $0 \leq N(r_k) < N(r_{k-1}) < \ldots < N(r_1) < N(r)$, and $\gcd(a, b) = r_k$: the last nonzero remainder in the above algorithm.

Proposition 10.3.
Every nonzero prime ideal in a PID is a maximal ideal.

Proof. Let R be a PID and $P = (p)_R \subset R$ a nonzero prime ideal. Hence $p \neq 0_R$ for otherwise $P = \{0_R\}$ would be the zero ideal. By Proposition 9.1 there exists a maximal ideal $M \subsetneq R$ containing P. We show $M = P$. Since R is a PID the ideal $M = (m)_R$ must be principal say generated by $m \in R$. Now $p = p1_R \in P \subset M = (m)_R$ hence $p = mx$ for some $x \in R$. Since P is a prime ideal and $p = mx \in P$ either $m \in P$ or $x \in P$. If $m \in P$ then $M = (m)_R \subset P$, hence $M = (m)_R = P = (p)_R$ and the proof is complete. If $x \in P$ write $x = py$ for some $y \in R$. Thus $p = xm = pym \Longrightarrow p(1 - ym) = 0 \Longrightarrow 1 - ym = 0 \Longrightarrow ym = 1$, m is a unit, and $M = (m)_R = R$, but this contradicts $M \subsetneq R$. We conclude $P = M$ is a maximal ideal. □

Definition 10.2.
Let R be an integral domain.

(1) An element $r \in R \setminus \{0_R, \text{units}\}$ (nonzero and non unit) is called *irreducible* if

$$r = ab \quad \Longrightarrow \quad a \in R^\times \quad \text{or} \quad b \in R^\times.$$

(2) An element $p \in R \setminus \{0_R, \text{units}\}$ (nonzero and non unit) is called *prime* if

$$p/ab \quad \Longrightarrow \quad p/a \quad \text{or} \quad p/b.$$

(3) Two elements $a, b \in R$ are called *associate*; we write $a \sim b$, if

$$a = bu \qquad \text{where} \qquad u \in R^\times.$$

Example 10.3.
(1) A prime number $p \in \mathbb{Z}$ is both irreducible and prime in the above sense.

(2) An irreducible polynomial $P(X) \in \mathbb{Q}[X]$ is both irreducible and prime in the above sense.

(3) Let $x = a + ib \in \mathbb{Z}[i]$. Then $N(x) = a^2 + b^2 = 1$ implies either $a^2 = 0$, $b^2 = 1$, hence $x \in \{i, -i\}$, or $b^2 = 0$, $a^2 = 1$, hence $x = \{-1, 1\}$. In particular, $N(x) = 1 \Longrightarrow x \in (\mathbb{Z}[i])^\times$.

The element $1+i \in \mathbb{Z}[i]$ is irreducible: if $1+i = xy$ with $x, y \in \mathbb{Z}[i]$ then $N(1+i) = 1^2+1^2 = 2 = N(x)N(y)$ which implies $N(x) = 1$ or $N(y) = 1$ hence either x is a unit or y is a unit.

(4)

- $\mathbb{Z}^\times = \{-1, 1\}$ and $m, n \in \mathbb{Z}$ are associate if and only if $m = n$ or $m = -n$.
- In $\mathbb{Z}[i]$: $(2-i) \sim 1+2i$ since $(1+2i) = i(2-i)$ and $i \in \mathbb{Z}[i]^\times$.
- $f, g \in \mathbb{Q}[X]$ are associate if and only if $f(X) = \alpha g(X)$ with $\alpha \in \mathbb{Q}^\times$.

Proposition 10.4.
In an integral domain R a prime element is irreducible.

Proof. Let $p \in R$ be a prime element we show p is irreducible. Assume $p = ab$. Then p/a or p/b. Suppose p/a hence $a = pr$ for some $r \in R$. Then $p = ab = prb \Longrightarrow p - prb = p(1-rb) = 0_R$. But $p \neq 0_R$, hence (since R is integral) $1 - rb = 0_R \Longleftrightarrow rb = 1 \Longrightarrow b$ is a unit. □

Remark 10.3.
The converse of Proposition 10.4 doesn't hold in general.

For example in $\mathbb{Z}[\sqrt{-5}]$ the element 3 is irreducible but is not a prime. Indeed suppose $3 = xy$ with $x, y \in \mathbb{Z}[\sqrt{-5}]$. Then $N(3) = 3^2 = N(x)N(y)$ hence $N(x) = 1, 3$, or 9. If $N(x) = 9$ then $N(y = c+\sqrt{-5})d) = c^2+5d^2 = 1$ and $y = 1$, or $y = -1$ is a unit, $N(x = a+b\sqrt{-5})) = a^2+5b^2 = 3$ is not possible, and if $N(x) = 1$ then x is a unit. Thus 3 is irreducible.

Now $3^2 = (2+\sqrt{-5})(2-\sqrt{-5}) \Longrightarrow 3/(2+\sqrt{-5})(2-\sqrt{-5})$ but 3 divides neither $2+\sqrt{-5}$ nor $2-\sqrt{-5}$. Indeed if say $3(a+b\sqrt{-5}) = 2+\sqrt{-5}$ then $N(3)N(a+b\sqrt{-5}) = N(2+\sqrt{-5})$ which implies; since $N(3) = N(2+\sqrt{-5}) = 9$, that $N(a+b\sqrt{-5}) = a^2+5b^2 = 1$ hence $a+b\sqrt{-5} = 1$ or $a+b\sqrt{-5} = -1$ which implies $3 = 2+\sqrt{-5}$ or $-3 = 2+\sqrt{-5}$, this is absurd. This show 3 is not prime in $\mathbb{Z}[\sqrt{-5}]$.

Definition 10.3.
A **unique factorisation domain (UFD)** is an integral domain R in which every element $r \in R \setminus \{0_R, \text{units}\}$ (non zero and non unit) has the following properties: **(i)** $r = p_1 p_2 \dots p_n$ is a product of irreducible elements $p_1, p_2, \dots, p_n$.

(ii) The above factorisation is unique up to associates, if $r = q_1 q_2 \dots q_m$ is another factorisation of r as product of irreducible elements $q_1, q_2, \dots, q_m$ then $n = m$ and after possibly renumbering the factors $p_i \sim q_i$; i.e. $p_i = u_i q_i$ with $u_i \in R^\times$, for all $1 \leq i \leq n$.

Example 10.4.
(1) $\mathbb{Z}$ is a UFD.

(2) $\mathbb{Z}[\sqrt{-5}]$ is NOT a UFD. Indeed in $\mathbb{Z}[\sqrt{-5}]$ we have

$$6 = 2 \times 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$

and 2, 3, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are irreducible.

We show 2 is irreducible: if $2 = (a+b\sqrt{-5})(c+d\sqrt{-5})$ then $2^2 = N(2) = N(a+b\sqrt{-5})N(c+d\sqrt{-5})$ hence $N(a+b\sqrt{-5}) = a^2+5b^2 = 1, 2$ or 4. Further $N(a+b\sqrt{-5}) = a^2+5b^2 = 2$ is not possible. Hence either $N(a+b\sqrt{-5}) = a^2+5b^2 = 1$ in which case $a+b\sqrt{-5} = -1, 1$ and $a+b\sqrt{-5}$ is a unit, or $N(a+b\sqrt{-5}) = a^2+5b^2 = 4$ hence $N(c+d\sqrt{-5}) = c^2+5d^2 = 1$ and $c+d\sqrt{-5}$ is a unit. Similarly one shows that 3, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are irreducible.

We have $(\mathbb{Z}[\sqrt{-5}])^\times = \{1, -1\}$, and 2, $1+\sqrt{-5}$, and $1-\sqrt{-5}$ are not associate. Thus the factorisation of 6 as a product of irreducible elements in $\mathbb{Z}[\sqrt{-5}]$ is not unique.

Proposition 10.5.
In a unique factorisation domain R an element is prime if and only if it is irreducible.

Proof. We only need to prove that an irreducible element is prime (see Proposition 10.4). Let $p \in R$ be an irreducible element and suppose p/ab, thus $ab = pr$ for some $r \in R$, we must show that p/a or p/b. Write a, b and r as a product of irreducible elements $a = p_1 p_2 \dots p_n$, $b = q_1 q_2 \dots q_m$, and $r = r_1 r_2 \dots r_t$. Thus we have

$$(p_1 p_2 \dots p_n)(q_1 q_2 \dots q_m) = p(r_1 r_2 \dots r_t).$$

The unique factorisation property implies $\exists 1 \leq i \leq n$ such that $p \sim p_i$, or $\exists 1 \leq j \leq m$ such that $p \sim q_j$. Suppose $p \sim p_i \iff p_i = up, u$ unit $\implies a = p_1 \dots up \dots p_n \implies p/a$. If $p \sim q_j \iff q_j = vp, v$ unit $\implies b = q_1 \dots vp \dots q_m \implies p/b$. □

Proposition 10.6.
Let a, b be nonzero elements of a unique factorisation domain R and suppose $a = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n}$, $b = p_1^{f_1} p_2^{f_2} \dots p_n^{f_n}$ are prime factorisations for a and b, the primes $p_1, p_2, \dots, p_n$ are distinct, and the exponents e_i, f_i are ≥ 0. Then

$$d = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \dots p_n^{\min(e_n, f_n)}$$

(where $d = 1$ if all exponents $\min(e_i, f_i)$, for $1 \leq i \leq n$, are 0) is a gcd *of a and b.*

Proof. Since the exponents of each of the primes occurring in the factorisation of d are no larger than the exponents occuring in the factorisation of both a and b, d is a divisor of both a and b. Let $c \in R$, c/a, c/b, and $c = q_1^{r_1} q_2^{r_2} \dots q_m^{r_m}$ the prime factorisation of c. A prime q_j dividing c divides a and b hence equals p_i for some $1 \leq i \leq n$, up to associates. Further in this case $q_j^{r_j}/p_i^{e_i}/a$, and $q_j^{r_j}/p_i^{f_i}/b$, which implies $q_j^{r_j}/p_i^{\min(e_i, f_i)}/d$. This shows c/d hence $d = \gcd(a, b)$. □

We state without proof the following result.

Theorem 10.7.
A PID is a UFD. In particular an ED (which is a PID by Theorem 10.1) is a UFD.

Summary

In summary we have the following strict containments of the various types of rings we encountered in this section

$$\Big\{(ED)\Big\} \subsetneq \Big\{(PID)\Big\} \subsetneq \Big\{(UFD)\Big\}.$$

Example 10.5.
(1) $\mathbb{Z}$ is a UFD.

(2) $\mathbb{Q}[X]$, and more generally $F[X]$ where F is a field, is a UFD since it is an ED.

(3) $\mathbb{Z}[i]$ is a UFD since it is an ED.

(4) $R = \left\{ a + b(\frac{1+\sqrt{-19}}{2}) : a, b \in \mathbb{Z} \right\}$ is a PID but is NOT an ED.

(5) one can show that $\mathbb{Z}[X]$ is a UFD which is not a PID.